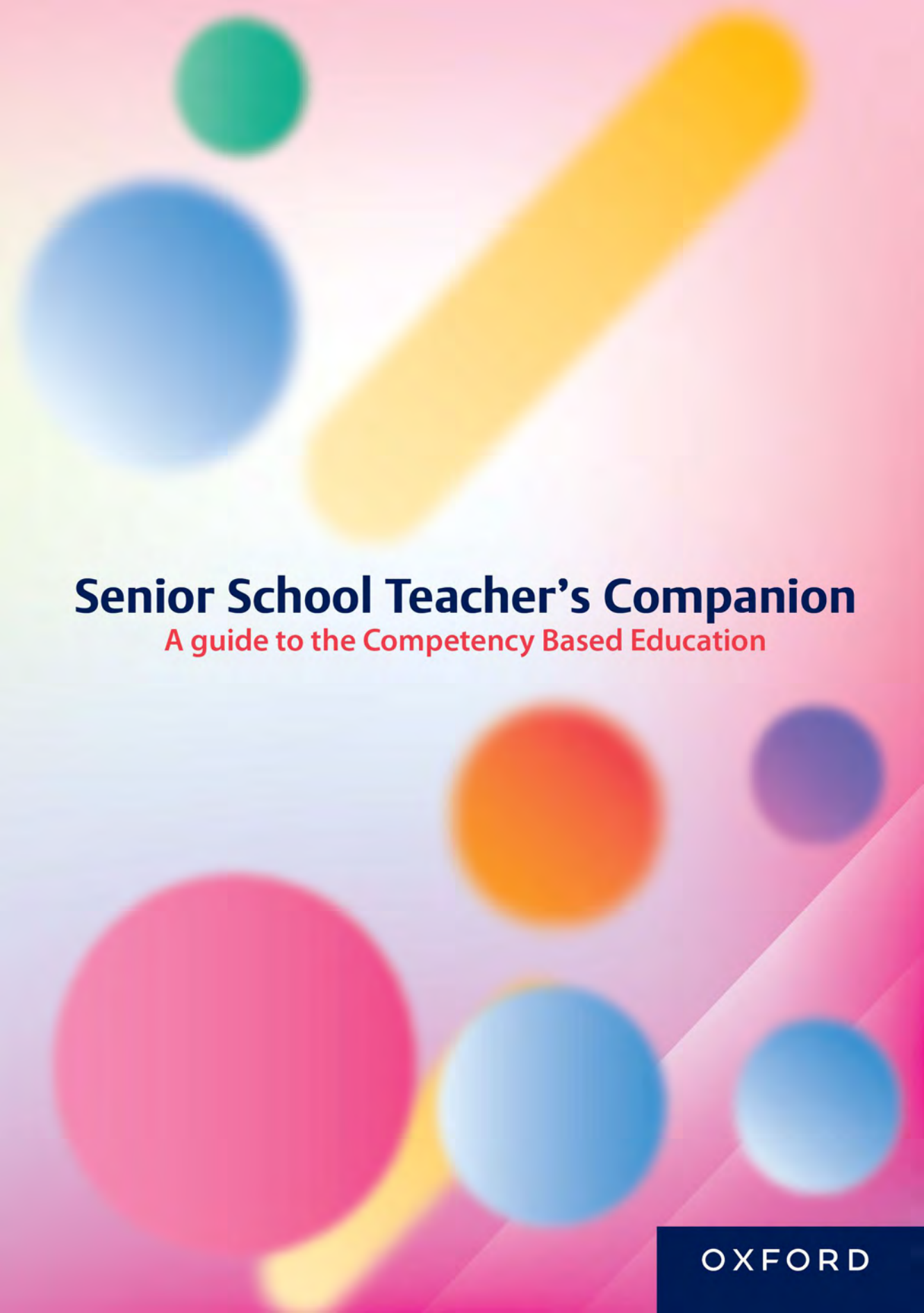


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Senior School Teacher's Companion

A guide to the Competency Based Education

OXFORD

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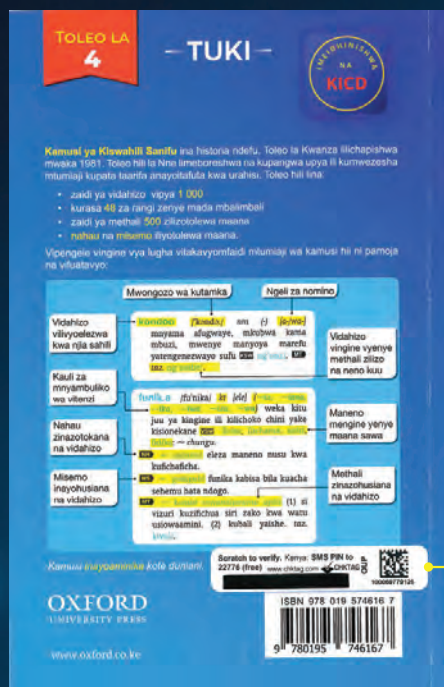
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kwenye maduka
yote ya vitabu
nchini!**



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Introduction to the Competency-Based Education (CBE)

1

What is competency-based education?

Competency-based education (CBE) is a product of the 2012 report of the Task Force formed by the then President Mwai Kibaki to review the education sector in Kenya. The word ‘competency’ is defined as ‘the ability to apply appropriate knowledge and skills to successfully perform a task’. CBE is therefore designed to emphasise the importance of developing skills and knowledge and using them in solving problems in real life.

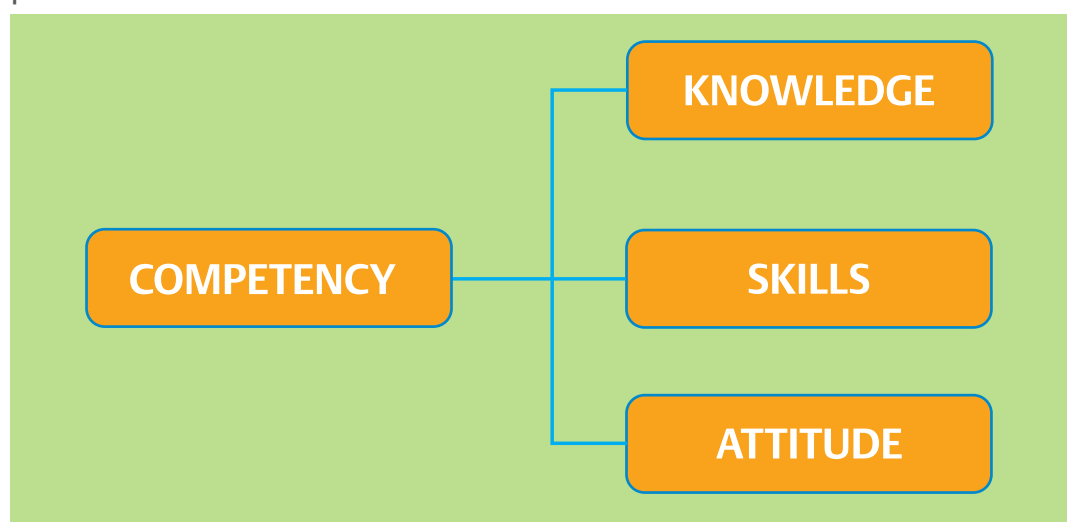


Figure 1.1: What comprises competency

Competency-based education therefore refers to a system of instruction, assessment, grading and academic reporting that is based on students demonstrating that they have learnt the knowledge and skills they are expected to learn as they progress through their education.

The main objective of the reforms was to improve the quality of education at all levels and to nurture every learner’s potential. In order to achieve this, teachers must be empowered with effective, efficient and sound instructional strategies, methodology and techniques that facilitate and assess competency-based learning. This also requires that pedagogical approaches are enhanced to support creativity, critical thinking and innovation.

Main differences between 8-4-4 and CBE

Table 1.1 gives a summary of how CBE differs from the previous 8-4-4 system of education.

Table 1.1: Summary of differences between CBE and 8-4-4 system of education

8-4-4 Curriculum	Competency-based education
Content focus Teachers disseminate information to students. Students are recipients of knowledge (passive learning).	Focuses on the acquisition of competencies Teachers have a dialogue with students, helping students construct their own knowledge (active learning).
Rigid and prescriptive curriculum with limited flexibility.	Learners' interests and special abilities should be considered in the learning process. Flexible with opportunities for specialisation.
Teaching where the role of the teacher is directing and authoritative.	Learning where learners take charge of their learning and teachers role is facilitating, guiding and providing resources.
Teacher centred.	Learner centred.
Emphasis on summative assessment and competition among the learners.	Emphasis on formative assessment and collaborative learning where learners share knowledge and experiences.
Strict adherence to a fixed curriculum.	Learners' interests are valued, which leads to identification and nurturing of talents.
Learning is based on repetition.	Learning is interactive and builds on what the learner already knows.

Overview of the Basic Education Framework

Basic education is organised into three levels in CBE:

- Early years
- Middle school
- Senior school

Figure 1.2 Summarises the different levels and the main focus for each.

Senior SchoolFocuses on pre-career skills (*Specialization*)**Junior School**

Focuses on exploration

Primary School

Focuses on socialization skills

Pre-Primary School

Focuses on interaction skills

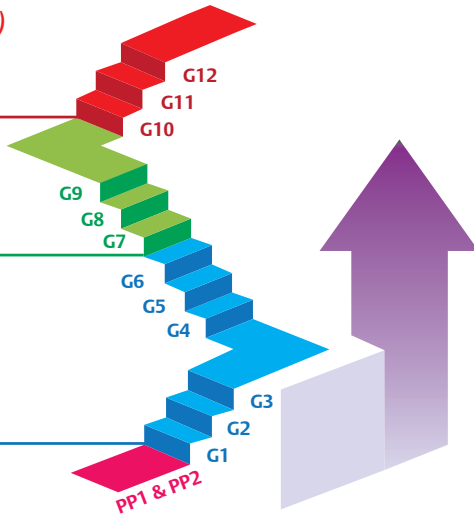


Figure 1.2: Different levels of education in CBE | **Source:** Revised KICD curriculum framework

(a) Early years education

This comprises two years of pre-primary and three years of lower primary school education.

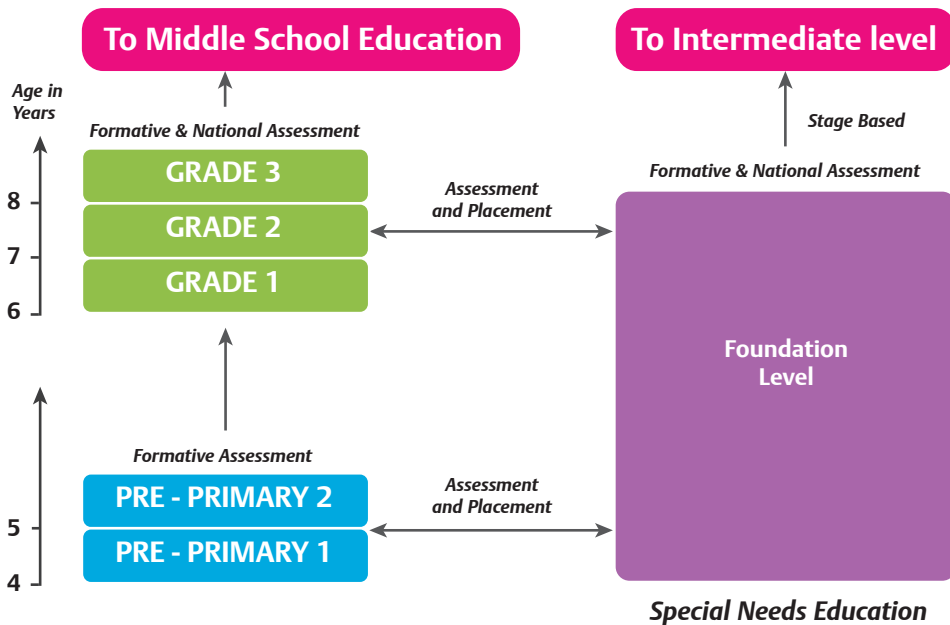


Figure 1.3: Early years education structural model

After the early years in school, the learners proceed to middle school education.

(b) Middle school education

This level comprises three years of **Upper Primary** education and three years of **Junior School** education. The upper primary part of middle school comprises a three-year programme where learners are exposed to a broad curriculum and given an opportunity for exploration and experimentation.

Secondary Education

Secondary education is organised into **two levels** namely, Junior School (Grades 7, 8 and 9) and Senior School (Grades 10, 11 and 12).

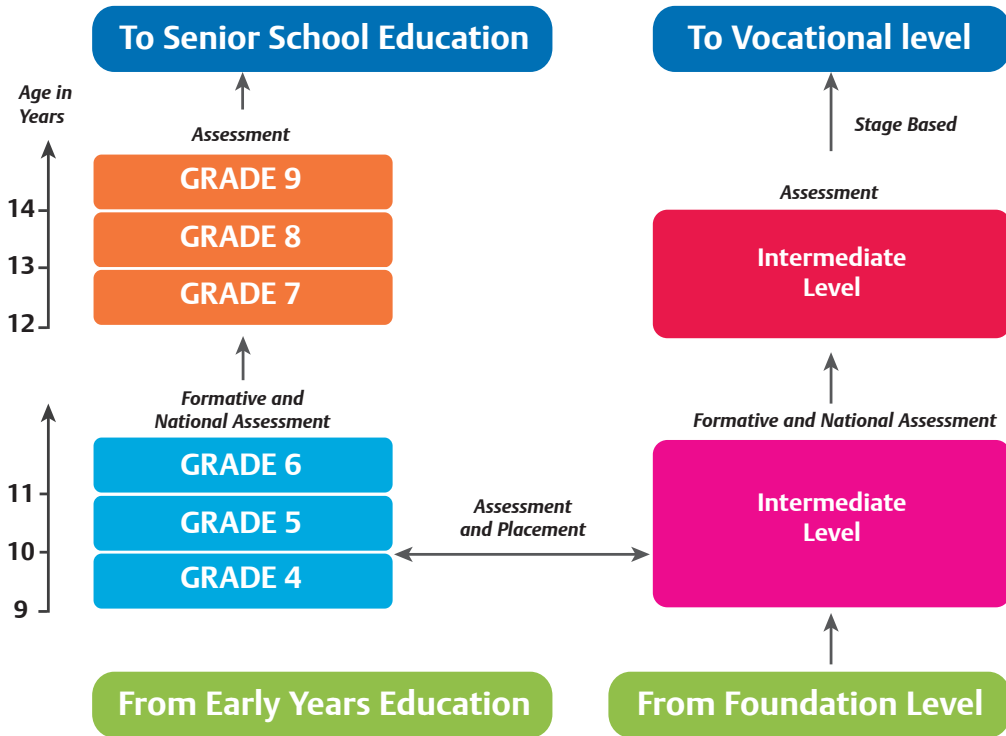


Figure 1.4: Middle school education structural model | [Source: Revised KICD curriculum framework](#)

(i) Junior School

Graduates of primary school after Grade 6 join Junior School at Grade 7. Junior School exposes learners to a broad-based curriculum as well to enable them explore their abilities, personality and potential as a basis for choosing subjects according to career paths of interest at the Senior School. As such, learners in Junior School should undergo a rigorous career guidance programme to enable them to make informed choices as they transit to Senior School.

Learning areas at Junior School

Learning areas in Junior School are divided into core subjects and optional subjects as shown in the Table 1.2.

Table 1.2: Learning areas in Junior School

Core Subjects	Optional Subjects
1. English 2. Kiswahili 3. Mathematics 4. Integrated Science 5. Pre-Technical Studies 6. Social Studies 7. Religious Education – learners choose one of the following: (i) Christian Religious Education (ii) Islamic Religious Education 8. Creative arts and sports 9. Agriculture	1. Foreign Languages: (i) German (ii) French (iii) Mandarin (iv) Arabic 2. Kenyan Sign Language 3. Indigenous Language

(ii) Senior School

Senior school comprises three years of education for learners between 16 to 18 years. It lays the foundation for further education and training at the tertiary level and the world of work. It marks the end of Basic Education as defined in the Education Act, 2013.

Learners exiting this level are expected to be **“engaged, empowered and ethical citizens”** ready to participate in the socio-economic development of the nation.

Senior schools are intended to be institutions of specialisation that provide opportunities for learners to focus in a field of their choice to form a foundation for further education and training and/or gain employable skills.

At this level, learners are supposed to specialise in a career path of choice since they have had opportunities to explore their potential, interests and abilities. The specialisation entails choosing to pursue studies in one of the three pathways available in Senior School namely:

- Arts and Sports Science Pathway
- Social Sciences Pathway
- Science Technology Engineering and Mathematics (STEM) Pathway

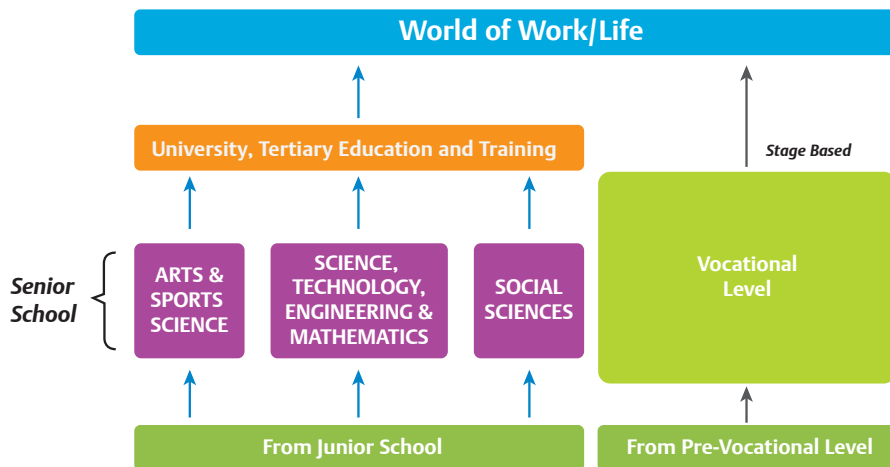


Figure 1.5: Pathways in Senior School | **Source:** Revised KICD curriculum framework

Below is a summary of the subjects in each of the pathways

Subjects in Senior School

Senior School learners (Grades 10-12) shall take **seven** subjects, four core and three elective from the pathways and tracks of choice as shown in the table below.

Table 1.3: Subjects and lesson distribution at Senior School

Subjects	Lessons per week
Core subjects	
1. English	5
2. Kiswahili/Kenya Sign Language (KSL)	5
3. Mathematics/Essential Mathematics	5
4. Physical Education (PE)	3
Elective subjects	
1. Elective 1	5
2. Elective 2	5
3. Elective 3	5
* Community Service Learning (CSL)	3
* Information Communication Technology (ICT)	2
* Pastoral Programme Initiative (PPI)	1
* Learner Personal/Group Study	1
Total	40 Lessons

*KEY

1. All learners will be provided with opportunities through CSL to integrate varied skills with instruction and reflection to enrich the learning experience, enhance

civic responsibility and strengthen community participation. This will be done through intentional learning experiences that allow them to apply knowledge, skills, values and competencies to address identified problems within the immediate environment.

2. ICT skills will be offered to all learners to facilitate learning and enjoyment. These skills include accessing, creating and caring for ICT devices and digital citizenship. The teacher is required to use ICT skills during lesson delivery.
3. Every school shall offer one lesson per week in Pastoral Programme Initiative (PPI) to enhance moral, spiritual and character development of learners.
4. One lesson shall be used for personal or group study to deepen knowledge and enhance skills in areas of choice by learners.

a) Arts and Sports pathway

This pathway is designed to nurture learners' creativity, physical abilities and cultural awareness, preparing them for successful careers and meaningful socio-economic contributions to society.

Core objectives include:

- **Foster creativity and expression** through arts; mainly performing arts and music.
- **Enhance physical fitness and sports skills** by building learners physical abilities, sportsmanship and understanding of sports science in training, nutrition and injury prevention.
- **Promote cultural awareness and appreciation** by encouraging learners to explore and appreciate local and global diverse cultural heritages and artistic traditions.
- **Prepare for careers** in Arts and Sports by equipping them with skills and knowledge needed for careers in arts, sports, entertainment and related industries.

Table 1.4: Subjects under Arts and Sports pathway

Track	Subjects	Basic infrastructure required
Arts	1. Music & Dance 2. Fine Arts 3. Theatre & Film	ICT/digital devices, Audio and video recording devices, costumes, props, Music instruments, dance studios, Art studios, paint and thinners, classroom instruction area and demonstrator's desk, storage area, clean-up area and a lockable teacher prep area.
Sports Science	1. Physical Education 2. Sports & Recreation	Indoor and outdoor Play fields, Athletics tracks, Gymnasium, equipment, protective gear, a standard swimming pool, water safety equipment, Lane ropes and Pool attendant, a store, appropriate uniform, ICT/digital devices and a standard classroom.

Career opportunities for learners who complete this pathway include:

- **Arts:** Fine arts, graphic design, fashion design, animation, film and media production, theatre, music performance and arts management.
- **Sports:** Professional sports, sports coaching, physical education teaching, sports management, fitness training and sports medicine.
- **Entertainment:** Acting, music production, dance performance, choreography, and media production.
- **Further studies** in universities, colleges or conservatories specialising in arts, music, dance, sports or related fields.

b) Social Sciences pathway

The pathway is designed to develop learners' understanding of society, human behaviour and global systems.

Core objectives include:

- **Develop critical and analytical thinking** by equipping learners with the ability to critically analyse social issues, human behaviour and societal structures and develop evidence-based arguments.
- **Promote civic responsibility and social awareness** by fostering an understanding of civic duties, rights and responsibilities, and active participation in community and societal development.
- **Enhance global awareness and cultural sensitivity** by cultivating an understanding of global issues, cultural diversity and interconnectedness of societies.
- **Prepare for careers and higher education in social sciences** by providing a strong foundation for further studies and careers in social sciences.

Table 1.5: Subjects under Social Sciences pathway

Track	Subjects	Basic infrastructure required
Languages and Literature	1. Literature in English 2. Indigenous Languages 3. Kenya Sign Language (KSL) 4. Fasihi ya Kiswahili 5. Sign Language 6. Arabic 7. French 8. German 9. Mandarin (Chinese)	• Tuition facilities • Libraries • Theatres • ICT/digital devices • Standard classroom • Language room
Humanities and business studies	1. Christian Religious Education 2. Islamic Religious Education 3. Hindu Religious Education 4. Community Service Learning (CSL) 5. Business Studies 6. History and Citizenship 7. Geography	• Subject rooms • Weather Station-Geography • ICT/digital devices

Career opportunities for learners who complete this pathway include:

- **Law:** Legal studies, law enforcement and judiciary roles.
- **Education:** Teaching, educational administration and curriculum development.
- **Public Administration:** Government service, public policy analysis and political science.
- **International Relations:** Diplomacy, international development and global affairs.
- **Economics:** Economic analysis, financial services and development economics.
- **Social Work:** Community development, counselling and social services.
- **Further studies** in universities and colleges offering social sciences for example in law, business and related fields.

c) STEM pathway

The STEM Pathway under the Competency-Based Education (CBE) is designed to equip learners with the knowledge, skills and competencies required to excel in scientific and technical disciplines. This pathway is ideal for learners who have a strong interest and aptitude in areas related to science, technology, engineering and mathematics.

Core objectives include:

- **Develop analytical and problem-solving skills** for learners to be able to think critically, analyse complex problems and develop innovative solutions using scientific and mathematical principles.
- **Foster technical competence** by providing hands-on experience in practical settings (laboratories, workshops etc) to build technical skills relevant to various STEM fields.
- **Encourage innovation and creativity** by inspiring learners to explore new ideas, technologies and methodologies in solving real-world challenges.
- **Prepare for advanced studies and careers** by ensuring that learners have a strong foundation in relevant subjects.

Table 1.6: Subjects under STEM pathway

Track	Subjects	Basic infrastructure required
Pure sciences	1. Mathematics 2. Essential Mathematics 3. Biology 4. Chemistry 5. Physics 6. General Science	• Fully equipped science labs • Standard classrooms • ICT/digital devices • Water points • Mathematics rooms

Track	Subjects	Basic infrastructure required
Applied Sciences	1. Agriculture 2. Computer Studies 3. Home Science 4. ICT	• Adequate land for crop and livestock production • Workrooms for Agriculture and Home science • Storage rooms for Agricultural and Home science equipment • Computer Lab • ICT/digital devices
Technical Studies	1. Metal Technology 2. Wood Technology 3. Aviation Technology 4. Building Construction 5. Electrical Technology 6. Power Mechanics 7. Media Technology 8. Marine & Fisheries Technology	• Classrooms • ICT devices • Multipurpose workshop(s) • Appropriate tools and equipment for the specific subjects

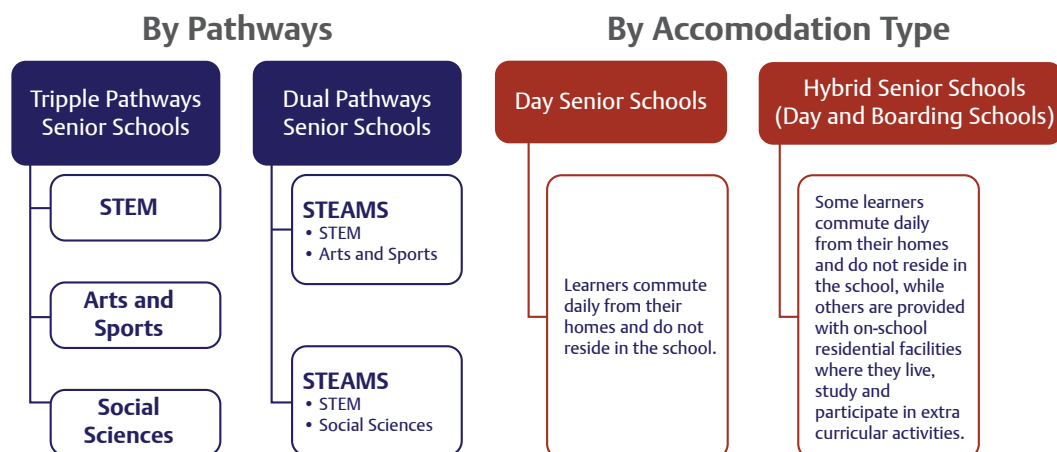
Career opportunities for learners who complete this pathway include:

- **Engineering:** Civil, Mechanical, Electrical, etc.
- **Information Technology** and Computer Science.
- **Medicine** and Health Sciences.
- **Environmental Science** and Sustainability.
- **Data Science** and Analytics.
- **Research and Development** in various scientific disciplines.
- **Advanced studies** in universities and other tertiary education institutions to specialise further in their chosen STEM fields.

Categorisation of schools in Senior school

Grade 9 learners will choose senior schools based on their career choices and academic performance.

Figure 1.6 shows the various categories of schools.



- **Triple Pathway Day Schools**

These are schools that offer three pathways with learners attending school during the day and returning home.

- **Dual Pathway Day Schools**

These are schools that offer two pathways with learners attending school during the day and returning home.

- **Triple Pathway Hybrid Schools**

These are schools that offer the three pathways with some learners attending school during the day and returning home while others reside in the school.

- **Dual Pathway Hybrid Schools**

These are schools that offer two pathways with some learners attending school during the day and returning home while others reside in the school.

- **Special Needs Schools**

These are schools that offer education to learners who experience difficulties within the mainstream education system (Hearing Impairment, Visual Impairment and/or Physical impairment).

- **Vocational schools**

These are schools that offer vocational education to learners who are following the stage-based pathway.

Placement of learners in Grade 10

The Ministry of Education has given comprehensive guidelines on how learners will be facilitated to choose senior schools at Grade 9. Learners will be guided on how to choose career pathways and subject combinations

through an automated selection process which is inclusive, fair and transparent. Principals of public senior schools are required to upload the subject combinations that they will offer on the system for the Grade 9 learners in Junior School to select. Key points to note about the selection process are listed below.

- Placement will be determined by learners' choice of pathways, tracks and schools, school category and capacity, merit and equity.
- Learners not placed in schools of their choice, will be placed in available slots in schools offering the pathway and tracks of their choice.
- No learner will be denied the opportunity to join senior school.
- Selection and placement will be fully computerised.
- Selection and placement will be restricted to learners who have successfully completed the requirement for Grade 9.

Cluster of learning areas in Junior School for placement

Learning areas at Junior School will be clustered to match the curriculum with students' interests, skills and future career goals. Further, performance in cluster learning areas will be used in placement of learners in appropriate pathways in senior school.

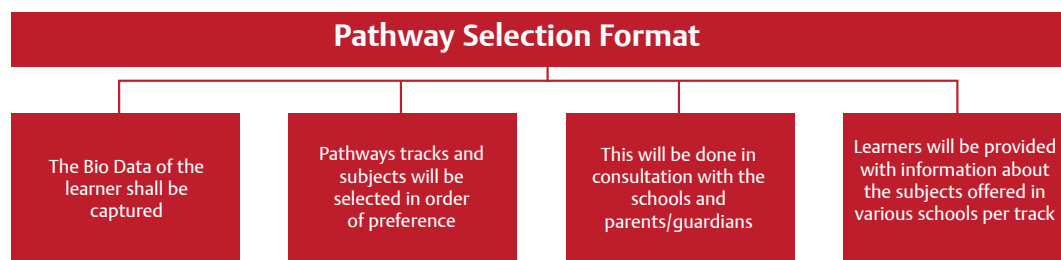
The cluster learning areas adopted for each track clearly relate to subjects in the pathways that learners would like to pursue in senior school as well as the careers they would like to pursue thereafter. The criteria highlighted in the following table will be used during placement.

Table 1.7: Cluster of Learning Areas for placement

Pathway	Track	Elective Subject at Senior School	Cluster of Junior School Learning Areas	Criteria
STEM	Pure Sciences	1. Mathematics 2. Chemistry 3. Physics 4. Biology	1. Integrated Science 2. Mathematics 3. English/Kiswahili/KSL 4. Pretechnical studies	At least 3 learning areas forming the cluster will be considered
	Applied Science	1. Agriculture 2. Computer Studies 3. Home science	1. Agriculture 2. Integrated Science 3. Mathematics 4. Pre-technical studies	
	Technical Studies	1. Metal Technology 2. Wood Technology 3. Aviation Technology 4. Building Construction 5. Electrical Technology 6. Power Mechanics 7. Media Technology 8. Marine & Fisheries Technology	1. Pre-technical Studies 2. Integrated Science 3. Mathematics 4. Creative Arts & Sports	
Social Science	Languages and Literature	1. Literature in English 2. Indigenous Languages 3. Kenya Sign Language (KSL) 4. Fasihi ya Kiswahili 5. Arabic 6. French 7. German 8. Mandarin (Chinese)	1. English 2. Kiswahili/KSL 3. Religious Education 4. Social Studies	At least 3 learning areas forming the cluster will be considered
	Humanities and Business Studies	1. Religious Education (CRE, IRE,HRE) 2. Business Studies 3. History and Citizenship 4. Geography	1. English, Kiswahili/KSL 2. Religious Education 3. Social Studies 4. Pre-technical studies	
Arts and Sports Science	Arts	1. Music & Dance 2. Fine Arts 3. Theatre & Film	1. Creative arts and sports 2. Kiswahili/ English/KSL 3. Social Studies 4. Pre-technical studies	At least 3 learning areas forming the cluster will be considered
	Sports Science	1. Physical Education 2. Sports & Recreation	1. Creative arts and sports 2. Kiswahili/ English/KSL 3. Integrated Science 4. Mathematics	

Placement will also take into account learners' interest and aptitude. Learners are expected to meet some minimum requirements in their cluster for consideration for the track of choice. Learner's personality, interest and assessment outcomes will be used to calculate the cluster weight for the tracks.

The selection of the pathway shall take the format shown below.



A learner will select a total of twelve (12) schools divided as shown below.

The learner will select:

- seven (7) schools for their first-choice pathway and track.
- three (3) schools for their second-choice pathway and track.
- two (2) schools for their third-choice pathway and track.

Nine (9) of these schools will be boarding schools, four (4) from the learners' home county, five (5) from outside their home county/county of residence and three (3) day schools in their home sub county/county of residence.

Data validation and verification will be done by regional supervisors who will ensure accuracy, consistency and reliability of the data.

Admission and replacement process

Admission process shall be automated, from access to joining instructions to reporting and replacements.

Points to note

1. Letters of admission shall be accessed online using the learner's assessment number.
2. Admission at the school level shall be through KEMIS.
3. All schools, both public and private, shall admit Grade 10 through KEMIS.
4. Lists of learners selected to a school shall be uploaded to the school's KEMIS account.
5. No principal shall enter a learner into KEMIS before he/she reports to the school.
6. Daily online reporting shall be monitored through KEMIS.
7. Senior schools with undeclared vacancies at the time of placement should declare them through the CDEs who will be responsible for putting them on KEMIS.

8. Schools with vacancies at the end of admission due to newly completed infrastructure and non-reporting of learners will be filled as follows;
 - Parents/guardians who wish to change schools and those wishing to be considered for replacement will make the requests through the Heads of Institution Junior School (JS) on KEMIS at least 2 weeks before the official Grade 10 reporting date.
 - Priority shall be given to those who had selected the schools requested.
 - MoE shall control the number of requests to be approved based on documented capacity.
 9. Replacement approvals shall be done through NEMIS as follows:

Boarding Schools: HDSE (assign and supervise)

Day Schools: CDEs will have system rights to authorise replacement.
 10. Upon approval, the joining instructions shall be accessed online.
 11. Replacements once approved shall be irreversible and done once.
 12. For multiple requests, one will be approved and candidate locked out from all the others.
- N/B** - At no time shall a school issue printed letters for replacement cases.

Role of principals of senior schools

Principals of Senior Schools will:

1. Ensure that accurate information on available vacancies is provided to MoE.
2. Prepare Grade 10 joining instructions in compliance with the MoE guidelines.
3. Receive and rank replacement requests.
4. Invite selected learners to their schools.
5. Create a conducive learning environment for students upon admission.
6. Ensure effective implementation of the MoE guidelines on selection and fees.
7. Make request for replacement through NEMIS.

Getting ready for CBE

To execute your mandate effectively in the wake of CBE, consider the following:

1. Understand the basics

Every teacher must understand the basics of the education framework in order to effectively implement the new curriculum. Refer to the notes above.

2. Know your policy documents

Teachers must ensure that they have the required policy documents to find the correct footing and develop good foundation in CBE. Below is a checklist of some of these documents:

- The Constitution of Kenya 2010
- The Basic Education Framework
- Sustainable Development Goals
- EAC Curriculum Education Harmonization Framework

- KICD Curriculum Designs
- CBE materials/coursebooks.

In addition to these documents, the teacher needs to be equipped with the 21st century learning/teaching skills and approaches.

3. Understand pedagogical approaches necessary for CBE

There are a variety of approaches teachers are encouraged to employ to enable them deliver the competency-based education. It should be remembered that learner-centred approaches to teaching are central to this curriculum. Table 1.7 gives some suggestions and explanations to some pedagogical approaches that a teacher can use.

Table 1.7: Suggested pedagogical approaches

Pedagogical approach	How to use the approach meaningfully
1) Open-ended instruction	Questions are structured in a way that multiple answers are possible so that learners are not directed towards one 'right' answer but rather multiple answers that open up room for further questions and answers.
2) Integrated learning	Content and skills from one learning area are used in other learning areas. This means content from the various related learning areas should be infused in the learning process.
3) Enquiry-based learning	Learning is focused through questions, problems or challenges that learners work to address. Learners are provided with questions/problems to solve and some guidance by the teacher on how to arrive at solutions given. The teacher is required to clarify the question(s) and skilfully lead the learners to think through the process to follow in order to arrive at solutions. Guided enquiry approach is appropriate for the younger learners while for older learners, the self-directed approach where learners generate the questions and assume much of the responsibility for how to solve them is more applicable.
4) Differentiated instruction	The teacher consciously designs activities and learning experiences that address a range of learning styles, abilities and readiness. There may be need to consider the gifted and talented learners as well as the time takers. This means that the teacher is tasked to develop a variety of instructional approaches that address the needs of different learners including those with learning difficulties.

Pedagogical approach	How to use the approach meaningfully
5) Experiential learning	Experiential learning brings real life experiences into the learning process, for example manipulating real objects. Coupled with simulation and mentorship, this approach provides experience beyond the classroom thus making it more likely to address real world issues and problems.
6) Co-operative learning	Group and co-operative learning strategies focus on learning in small groups. The teacher is however required to explicitly understand some of the co-operative learning strategies and try them out in the classroom for both teaching and assessment of learning.
7) Discussion	The teacher together with the learners form groups to discuss a specific problem. The group leaders need to be varied for different discussions in order to provide an opportunity for every member to participate. At the end, the group may make presentations to the rest of the class through their group leader.
8) Peer teaching	Learners learn best from each other! It would be useful to have teachers guide learners to learn from their peers. This approach provides opportunities for learners to actively present their knowledge and skills to peers and/or act as teachers and mentors of each other. Peer teaching promotes incidental learning from presentations and peer engagement. Peer teaching also creates an intentional opportunity to empower learners to teach others and for the teacher to master the concepts being taught.
9) Case studies	Case studies are thorough descriptions of real events from real situations. The teacher can narrate a story or let learners read stories on real life situations and allow learners to reflect on.
10) Role play, dramatization and storytelling	Learners internalize learning much better if it is done through role play, dramatization and storytelling. This helps the learners to associate with the characters, empathise with them and later practise the lessons learnt. For creativity and innovation, the teacher can ask the learners to construct their own brief stories, drama and role plays.

4. Understand the paradigm shift

Competency-based education emphasises development of competencies as opposed to knowledge acquisition. Figure 1.7 illustrates some of the changes in the new curriculum.

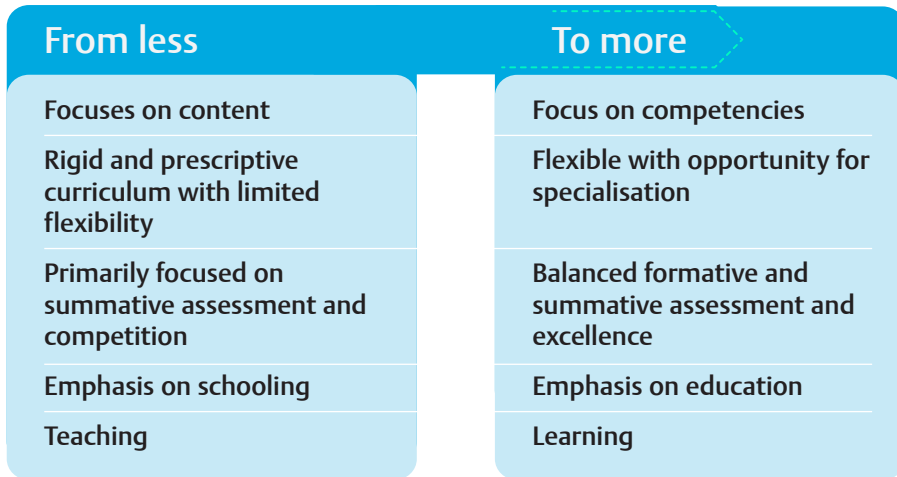


Figure 1.7: The paradigm shift

5. Get resourced

Appropriate resources with which to facilitate learning are critical for this curriculum. You need a good **Teacher's Guide (TG)** and **Learner's Course Book** with requisite competencies, stimulating visuals and relevant content as a priority to guide you in the acquisition of other relevant and suitable lesson resources. On the other hand, learners need a good course book that not only provides them with the required content, but also gives them visual stimulation to keep them interested in the learning process. It is critical that you choose this carefully.

6. Plan effectively

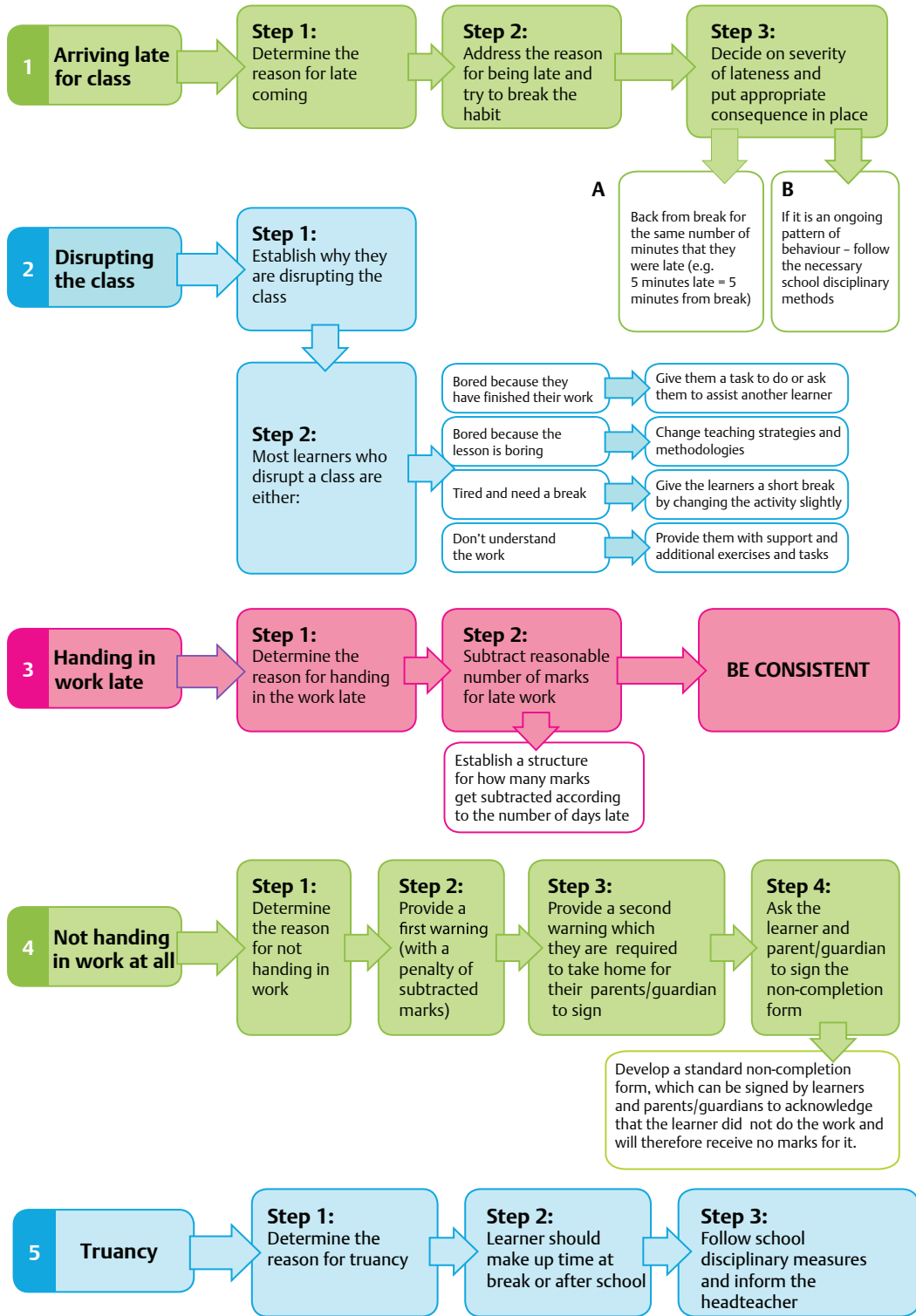
A successful CBE lesson is determined by the planning and the amount of thought put into the lesson planning. The lesson plans should have:

- a variety of activities and learning experiences,
- strategies and approaches to teaching CBE,
- strategies that cater for differentiation,
- attainable learning outcomes,
- key enquiry question(s) to trigger the learners' curiosity,
- learning resources,
- assessment items/tools,
- opportunity for reflection and evaluation.

Remember!

Competency based education is a very new concept to all stakeholders including parents, learners, teachers and school managements. An inclusive approach will be useful in enhancing greater buy-in by all stakeholders and for effective learning.

Step-by-step solutions to classroom problems



Teacher wellness

As an educator, your health and well-being are the keys to a productive and joyous learning experience for your learners and for yourself. To most teachers, however, their learners come first and they come second or not at all. The good news is that it only takes a few minutes a day to destress, energize and revitalize your body and mind.

There are many self-care techniques that can be practised in the classroom with your learners, so everyone benefits.

Some of the simple strategies you can use to guarantee you wellness include:

1. **Breathe.** Did you know you can trick your brain into thinking everything is fine with slow, deliberate breathing?
2. **Connect.** Reach out and connect (or vent!) with your fellow teachers and co-workers and hold each other accountable for your self-care contract.
3. **Take a timeout.** Schedule your timeouts during the day when you have no interruptions. Soon you'll look forward to this mini-recharge.
4. **Move more, sit less.** Incorporate some physical activity into your day – before, during and after school. Get started today by doing simple physical activities that take only two minutes or less.
5. **Enjoy the great outdoors.** Teachers don't have a lot of time for long hikes or adventure trips, but being with nature can be as simple as taking a leisurely walk during the lunch break or after work.
6. **Step away from the device.** Constant digital stimulation can increase your anxiety or sense of 'not doing enough'. Put down your phone and schedule some no-tech time each week.
7. **Reflect.** Think about all the things for which you are grateful. Even when life is stormy, you can always find one bright moment or thing to be grateful for.
8. **Get creative.** Do more inventive, original or artistic activities
9. **Create a comfort kit.** For those days when you're feeling particularly defeated or discouraged, a comfort kit reminds you that everything will be okay. It can include anything for example, your expression is one of the most therapeutic techniques for relieving stress. You might try drawing, painting, photography or crafting. It can include anything you love, from your favourite tea to a note from your best friend forever (BFF).
10. **Pamper yourself.** Set aside some time during the school term to do something special for yourself. It could be a nice dinner, a manicure or a weekend trip – experiences that will help you refuel and recharge for the school days ahead.

Additionally:

- **Rest.** Every teacher needs some downtime in the evening – even if it is only for an hour and a good night's sleep is a must. Teaching is one of the few professions that never lets you breeze through the day – even when you're not feeling well.
- **Eat right.** What you eat can have a big effect on how you feel, how you look and how much energy

you have for teaching. A simple diet of smaller meals, whole meal foods instead of processed foods and a variety of fresh fruits and vegetables is the most appropriate. Try to cut back on (or eliminate) your consumption of sugar, white flour and foods high in saturated fats. You will not believe how much energy you will have or how incredible you will feel.

- **Leave school work at school.** As soon as you leave the school compound, your workday is over. Leave it there. Although sometimes you may have to do marking or catch up at home, you should limit those instances as much as possible. Try not to carry your work frustrations or disappointments home.
- **Work out!** Nothing will clear your head faster or more completely than physical exercise. Go for a brisk walk or dance to your heart's content. Join a swim club. Whatever you do, make sure it's something you enjoy doing.
- **Create a culture of support.** Use your relationships to celebrate success and collaborate effectively. Ask yourself how often you share your highlights with others.

Assessment in Competency-Based Education (CBE)

2

What is assessment?

Assessment is the process of obtaining information that is used to make educational decisions about students, to give feedback to the student about their progress, strengths and weaknesses, to judge instructional effectiveness and curricular adequacy and to inform policy.

National Assessment System

According to this definition, assessment information can be used at different levels of the education system (i.e. national, county, sub county, school, classroom and at home) to make decisions about learners, about teaching and learning, about the curriculum as well as to make the relevant education policies. In practice, this definition is applicable to the entire assessment system that impacts on the education system in any country.

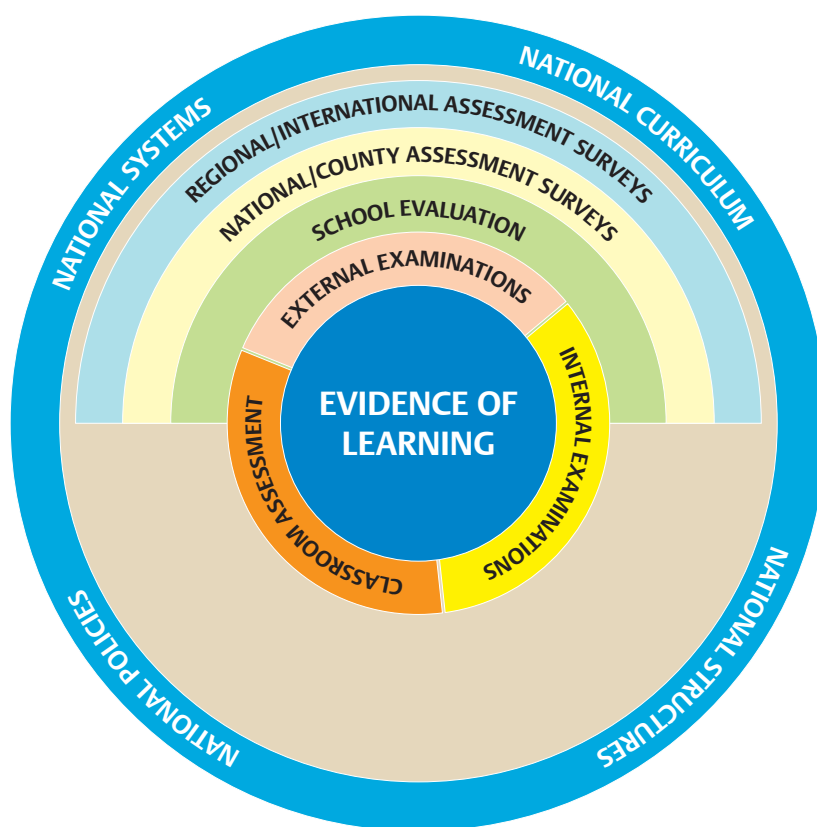


Figure 2.1: Components of a National Assessment System (Kanjee, 2008)

Classroom assessment versus examinations

(a) Classroom assessment

Classroom assessment refers to the process of obtaining evidence on the knowledge, understanding and skills of learners that can be used by teachers to improve learning and teaching and by learners to improve their own learning. The primary purpose of classroom assessment is to obtain evidence for use to improve learning. Typically, assessment activities are led by the teacher in the classroom that comprise class tests, classwork, homework, peer- and self-assessment, assignments, projects, questionnaires and observations. An integral aspect of all classroom assessments is feedback, which is used to improve both learning and teaching.

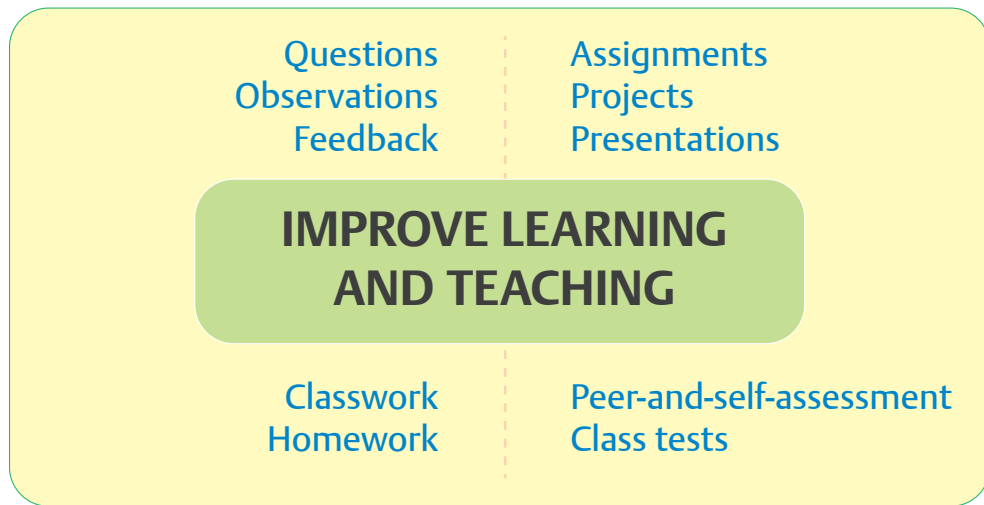


Figure 2.2: Overview of a Classroom Assessment System

(b) Examinations

Examinations refer to the process undertaken for determining a learners' mastery of relevant knowledge and skills taught. Examinations are generally the most common and most established component of the assessment system. The purpose of examinations is selection, usually based on specific, predetermined criteria. By definition, the function of examinations is to differentiate learners who are deemed to have or have not mastered the content being examined. Within the education sector, two types of examinations are usually conducted: internal and external.

- **Internal examinations** are conducted by the school. They are usually developed and administered by teachers and are conducted on a quarterly, half-yearly and/or annual basis. In addition, many countries also conduct school-based assessments (SBA) for obtaining learner scores that are used in calculating the final grade for specific external examinations. In practice, SBA may include similar tasks as those used for internal exams and classroom assessments. The key difference, however, is that these results

contribute to calculating the final grade of an external examination. This is the approach CBE has adopted.

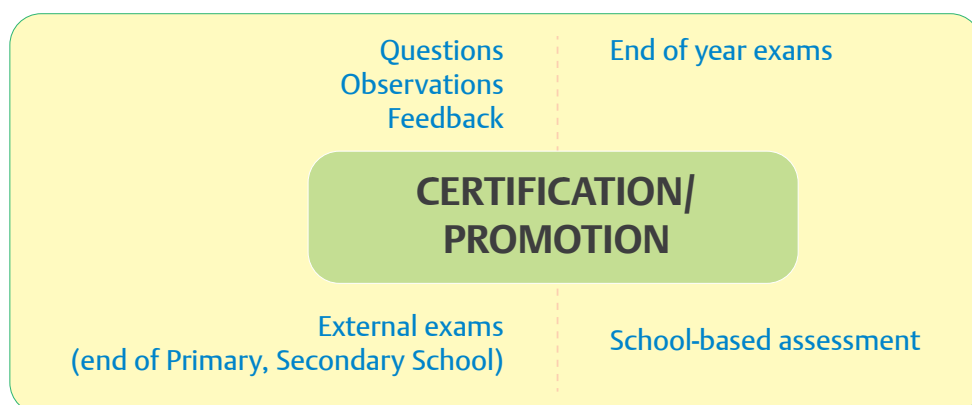


Figure 2.3: Overview of examinations

- **External examinations** are conducted by bodies outside the school to certify completion of a specific phase of education. In those countries where examinations are a feature of the education system, exams are always conducted at the end of schooling by an external agency, usually a national examination board. In addition, in many countries, especially those in Anglophone Africa, national examinations are also conducted at the end of primary schooling and/or end of lower secondary. Historically, examinations were introduced to ensure equitable distribution of educational and vocational benefits (Greaney and Kellaghan, 1995).

However, given the high stakes nature of external examinations, the use of exam results for accountability purposes and the context of limited resources and facilities, the form and format of modern day examinations have taken a very different nature that has had a profound impact on equity within the education system in a number of different ways.

Assessment surveys

Assessment surveys refer to the process of obtaining evidence from an education system (or part thereof) on the performance of learners and other role-players (such as teachers, principals, education officials, parents among others) as well as on the functioning of structures and programmes within that system (Kanjee, 2007). The primary purpose of assessment surveys is to obtain information to evaluate various aspects of the education system; to make decisions about the need for interventions and for resource allocation; for enhancing public awareness and for accountability purposes (Braun & Kanjee, 2006). Assessment surveys are generally conducted at the county, national, regional or international level and comprises the administration of standardised tests and questionnaires to learners and/or teachers and may include questionnaires, site visits and interviews with key education officials, including the school principal, ministry officials and other role-players in education. These surveys are usually conducted on samples of schools and learners. However, a number of countries have also conducted census-based

national surveys. In this instance, all learners within a particular grade level are surveyed in selected subject areas, thus information on performance levels is available to every single learner tested.

School evaluations

School evaluations refer to the process of gathering evidence about the effectiveness and efficiency of a school to develop and implement specific set of actions to improve learning and teaching (Kanjee, 2008). School evaluations comprise on-site visits, usually conducted over several days by education officials or their representatives, to review school policies and practices as well as the performance of learners, teachers, school leaders and where available, school governing bodies.

Within the context of most education systems, the use of assessment evidence to determine the performance of learners comprises a critical aspect of the school evaluation process. Thus, in practice, school evaluation may also involve the testing of learners, especially where learner performance data is not available. However, in most cases, available assessment evidence from internal and/or external examinations are used as a key aspect of the school evaluation exercise.

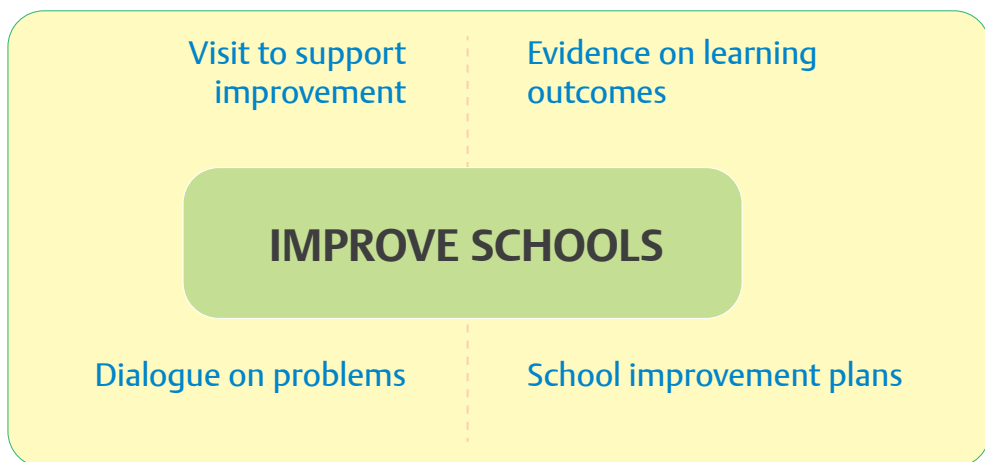


Figure 2.4: School evaluation activities

More importantly, findings and recommendations emanating from school evaluations can have a significant impact on how schools function. Specifically, school evaluations can impact on different assessment practices in schools and how assessment evidence is used by school leaders, teachers and learners to improve learning and teaching. This includes the amount of funding provided as well as the nature, content and target audience of training and support within schools. In essence, the school evaluation system as currently implemented can serve as a key lever for enhancing assessment systems and practices within the school system.

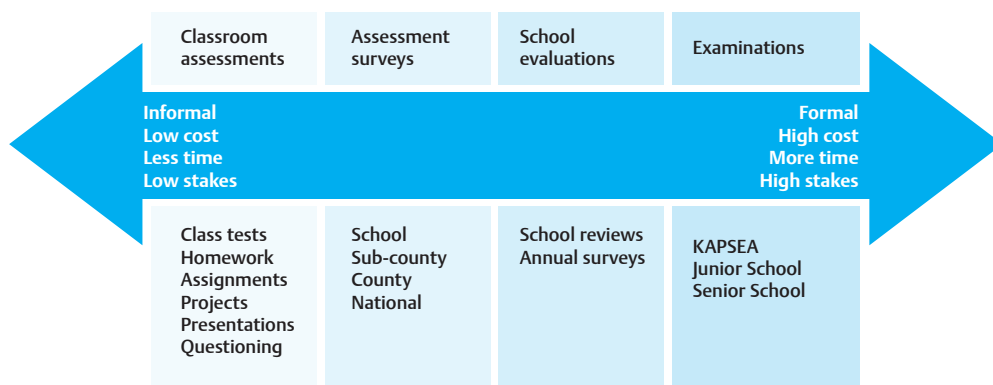


Figure 2.5: Assessment systems continuum

As noted in Figure 2.5, the practical implications for implementing different components of an assessment system vary widely depending on the purpose and use of the assessment evidence.

Competency-based assessment (CBA)

In any classroom context, it is important to note that assessing learning forms an important aspect of the teaching and learning process and that assessment is integrated into teacher's pedagogical practice. The primary purpose for conducting any assessment is to obtain evidence about learning. Teachers need evidence on what learners know, understand and can do, in order to adjust their teaching to better support learners or to make decisions regarding learners' mastery of the content covered.

In CBE, there are three primary types of assessment:

- Assessment **OF** Learning
- Assessment **FOR** Learning
- Assessment **AS** Learning

The primary purpose of assessment is to obtain evidence of learning.

(a) Assessment OF Learning

Assessment OF Learning (AoL) is also referred to as summative assessment. It is defined as the:

“process by which teachers gather evidence in a planned and systematic way in order to draw inferences about their students' learning, based on their professional judgment and to report at a particular time on their students' achievements” (Assessment Reform Group (ARG), 2002).

Summative assessment is carried out at the end of the course to find out the sum total of the learning that has or has not taken place. Summative assessment is separated from teaching and is carried out when achievement has to be summarised and reported.

It includes oral, practical and written activities in as well as tests, examinations, assignments and projects.

Most summative assessments are conducted under standardised conditions so that the assessment instruments and conditions are the same for every learner. In practice, this requires the application of a formal process, hence AoL is sometimes referred to as formal assessment. Christodoulou (2016) notes that the purpose of summative assessment is to produce a shared meaning among all users.

In other words, teachers, learners, parents and school leaders will have the same understanding of what a specific score on a test or an assignment means. For example, it is understood that a score of 70% is a high score and that the learner performed well on the test. It also means that this learner knows most of the work on which they were tested. However, a critical limitation is that test scores **do not provide any information on what learners know, understand and can do**. Thus has limited value for assisting teachers and learners to identify specific learning needs of learners.

Within the context of schools, the results from summative assessments can be used to:

*The primary purpose of summative assessment is to produce a **SHARED UNDERSTANDING**.*

- determine what learners know, understand and can do,
- monitor progress of learners,
- report on learners' achievement,
- select learners for progression purposes or awards,
- identify content areas that need to be revised or re-taught.

In the CBE system of learning, there will be an overall summative assessment at the end of each level. There will be early years exams which will take place at the end of Grade 3 to enable learners transit to Grade 4. Another exam will be done at Grade 6 to enable learners transit to Junior School and a final one at Grade 9 to enable their transition to Senior School.

The aim of the summative assessment is to find out if learners have acquired the expected competencies for their level before transition.

(b) Assessment FOR Learning

Research on Assessment FOR Learning (AFL) shows that teachers who effectively use this approach see considerable learning gains among all learners, irrespective of their socio-economic backgrounds (Kaliniec-Craig, 2017). However, teachers are required to have high levels of expertise, experience and appropriate content knowledge to successfully apply Assessment FOR Learning in the classroom. Assessment FOR Learning (AFL) is defined as:

“the process of seeking and interpreting evidence for use by learners and their teachers to decide where the learners are in their learning, where they need to go and how best to get there” (Assessment Reform Group, 2002).

In the context of classrooms, Assessment FOR Learning is implemented in two ways:

- (i) as formative assessment,
- (ii) as the formative use of summative assessments.

Assessment FOR Learning =

1. *Formative assessment +*
2. *formative use of summative assessment.*

What is formative assessment?

Formative assessment refers to all assessments undertaken by learners and teachers during the lesson to:

- (a) identify what learners know and can do,
- (b) take appropriate steps to support learners to improve their knowledge, understanding or skills.

The primary purpose of formative assessment is to produce a consequence for teachers and learners (Christodoulou, 2017). That is, formative assessment information must lead to actionable steps to improve what learners know, understand or can do. The characteristics that define formative assessment are that it:

- involves BOTH learners and teachers,
- is conducted DURING the lesson,
- is based on INFORMAL EVIDENCE which is obtained using planned and/or spontaneous activities,
- requires the teacher to provide FEEDBACK that learners can apply and leads to specific ACTIONS that address learners' learning needs.

The purpose of formative assessment is to produce ACTIONABLE next steps for learners AND teachers.

In practice, formative assessment is mainly based on teacher judgment. This process may be different for each lesson and also different for each learner, depending on their needs. Hence, formative assessment is also sometimes referred to as *informal assessment*.

NOTE

Formative assessment involves both teachers and learners, occurs during the lesson and leads to actionable steps to improve learning.

Formative use of summative assessments

The formative use of summative assessments refers to the use of summative assessment results to improve the teaching and learning process. In practice, this can only occur after the summative assessment results are available. For example, after marking a class test, the final exams or even a project, a teacher analyses the results to identify areas in which learners performed well or poorly and then uses this evidence to:

- identify what steps to take to improve their teaching,
- take specific action to improve on learners' knowledge, understanding and skills.

(c) Assessment AS Learning

Assessment as Learning (AaL) refers to the process where learners monitor their own learning to improve their knowledge, understanding and skills. In practice, Assessment As Learning is related to formative assessment. The main difference is that learners take greater responsibility of their OWN learning.

With the support of the teacher, learners determine for themselves where their strengths and weaknesses lie, plan for and implement actionable steps to address any gaps in their knowledge, understanding and skills. Assessment As Learning is defined as:

“a process where learners personally monitor what they are learning and use the feedback from this monitoring to make adjustments, adaptation and even major changes in what they understand” (Earl, 2012).

Assessment AS Learning
is where learners take primary responsibility to monitor and improve their OWN learning.

During this process, teachers support learners by:

- creating opportunities to review their work,
- providing appropriate and specific feedback,
- offering ideas and options to consider,
- ensuring learners focus on the lesson objectives.

REMEMBER

In the classroom context, all types of assessment should be seen as complementary. Use formative assessment to support the learning process during lessons and to determine the extent to which learning has taken place. Enhanced and effective use of formative assessment can lead to improved performance on summative assessments.

Formative assessment strategies

The primary function of formative assessment is to obtain and use evidence of learners' learning in order to improve teaching practices that better support learners and address their learning needs.

Wiliam and Thompson (2007) note that the use of formative assessment is based on answering three key questions:

- Where are learners in their learning? – Teachers get this evidence during lessons.
- Where are learners going? – Teachers get this information from curriculum designs.
- What should be done to get learners there? – Teachers facilitate and support learning.

To implement formative assessment during the lesson, the use of the following formative assessment strategies is proposed that teachers must know and be able to apply to better support learners to improve their learning:

- Planning and preparing lessons that focus on learning.
- Clarifying and sharing learning intentions and success criteria.
- Using questions, discussions and activities to obtain evidence of learning.
- Providing effective feedback to improve learning.
- Guiding learners to support each other's learning (peer assessment).
- Guiding learners to improve their own learning (self-assessment).
- Using assessment evidence to improve teaching.

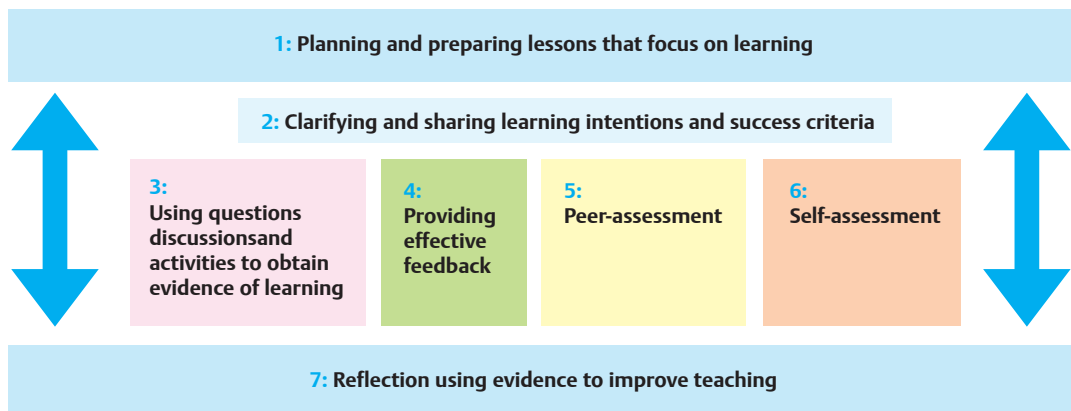


Figure 2.6: The relationship between key formative assessment strategies

Assessment tools

Competency-based assessment (CBA) requires the use of a variety of assessment tools to address the different learning styles of the learners.

Assessment tools are used to:

- Collect information on acquisition of competencies by a learner.
- Determine the level of acquired competencies.
- Provide feedback to the learner and other stake holders.
- Inform improvements on instruction strategies.
- Offer relevant and appropriate interventions.

*An **assessment tool** is an instrument used to determine the extent to which a learner has achieved specified learning outcomes.*

Examples of assessment tools which can be used in CBC are given below.



Figure 2.7: Examples of assessment tools

Below are explanations and samples of the above assessment tools for your review.

- **Written tests** – These can be generated by the teacher based on the content learnt in class or the teacher can use revision exercises and/or questions in the course books.
- **A portfolio** is a collection of samples of a learner's work over a period of time. It shows the effort, progress and mastery of the competencies. Portfolios can be physical or digital.

(a) Physical



(b) Digital



Figure 2.8: Portfolios

- Learner **journals** are learners' individual written accounts of their experiences during a lesson or a learning activity.

School:	
Learner's name:	Grade: 10
Teacher's name:	Date of assessment:
Strand: Citizenship	Sub strand: Concept of Community Service Learning (CSL)
Targeted competency	Communication and collaboration
Targeted learning outcome	Examining own civic identity as a member of the community.
What did I learn?	I learnt my rights and responsibilities as a member of my school community.
My thoughts and feelings on what I learnt	I felt glad because the election was conducted in an ethical manner. This made me realise the importance of participating in voting for decision making.
Teacher's comments: Thank you for exercising your right to vote and expressing your feelings. Reach out to your classmates and help them to understand their civic identity. Encourage them to participate in CSL projects.	

Figure 2.9: Journal

- An **observation schedule** is an outline of the characteristics and behaviour to be observed as a learner performs specific tasks, individually and/or in a group.

School: Freto Senior School	Learner's name: Angela Koli	Grade: 10
Teacher's name: Fred Atib	Date: 20 th January 2026	
Strand: Business and money management	Sub strand: Banking	
Indicators of knowledge, skills, attitudes and values being assessed		
1. Explaining the meaning and importance of banking in an economy. 2. Analysing types of accounts offered by banks. 3. Exploring ethical and unethical practices in banking. 4. Describing the trends in banking in Kenya.		
Observations		
1. The learner is always ready to volunteer to answer questions. 2. The learner asks questions during learning. 3. The learner assists other learners who have difficulties in identifying types of accounts offered by banks.		
Teacher's comments:		
1. The learner needs more guidance in explaining the importance of banking. 2. The learner needs more guidance in describing trends in banking in Kenya.		
Teacher's signature:	Learner's signature:	

Figure 2.10: Observation schedule

- A **checklist** is a list of characteristics of a learners' behaviour which is used to monitor the development of specific aspects.

Learning task: Meaning of budgeting in business					
Core competencies assessed Tick (✓) appropriately					
	Self-efficacy		Critical thinking and Problem-solving		Teacher's remarks
	Was the learner able to give the meaning of budgeting in business?		Was the learner able to share their own definitions with the class?		
Learner 1	✓		✓		The learner has acquired all the desired competencies.
Learner 2	✓			✓	The learner needs some assistance to acquire some of the desired competencies.
Learner 3		✓		✓	The learner has not acquired any of the desired competencies hence needs a lot of assistance.

Figure 2.11: Checklist

- A **project** is a set of activities implemented within a given timeframe with a clearly stated purpose and objectives. It gives an opportunity to the learners to apply their acquired knowledge and higher-order thinking skills to real-life situations.
- A **rating scale** is a tool that a teacher can use to record observations against a list of criteria, such as skills. The teacher selects the option that describes the degree of the skills displayed by the learner.

School: Daraji Senior School	
Learner's name: Alim Ketis	Date: 22nd January 2026
Teacher's name: Teacher Pecha	Grade: 10
Strand: Citizenship	Subject: Community Service Learning (CSL)
Sub strand: Leadership development	

Competency (knowledge, skills, attitudes and values assessed)	Always 4	Most of the time 3	Sometimes 2	Never 1
The learner contributes to the discussion as they think and share ideas on factors considered when setting business goals.	✓			
The learner shows responsibility by participating in the discussion on factors that are considered when setting business goals.		✓		
The learner can formulate SMART short term business goals.			✓	
Teacher's comments on the learner's performance Alim is a diligent learner who contributes and collaborates with other learners. He needs to take more initiative in the practical aspects of learning.				
Teacher's signature:		Learner's signature:		

Figure 2.12: Rating scale

- An **assessment rubric** provides a criteria that the teacher can use to evaluate a learner's performance. It gives a detailed description of characteristics for each level of performance in terms of quality.

Indicator	Exceeds expectation	Meets expectation	Approaches expectation	Below expectation
Ability to explain the principles of Community Service Learning as a learning strategy.	The learner correctly and consistently explains all the principles of Community Service Learning as a learning strategy.	The learner explains five of the principles of Community Service Learning as a learning strategy.	The learner explains three principles of Community Service Learning as a learning strategy.	The learner explains at most two principles of Community Service Learning as a learning strategy with help.

Figure 2.13: Assessment rubric

- A **learner's profile** gives a summary of the teacher's opinion on mastery of competencies by a learner. The information is derived from the other assessment tools such as learner journal, portfolio, written tests and projects.

School: Robert Kaida	
Learner's name: Queen Ampani	Date: 26 th January 2026
Teacher's name: Martin Otindo	Grade: 10
Strand: Business and money management	Sub strand: Money

Criteria	Learner's strengths	Learner's area of improvement	Learner's preferences or interests
Presenting the meaning and importance of budgeting in Business.	The learner can give the appropriate meaning of budgeting in business.	The learner needs to enhance her knowledge on the importance of budgeting in business.	Explaining how a business can be useful to a business.
Analysing the types of business budgets for financial planning.	The learner can identify the types of business budgets.	The learner needs to learn how to differentiate the different types of business budgets.	Identifying the types of business budgets.
Apply knowledge on types of business budgets to create and present a business budget.	The learner can create and present a business budget.	The learner needs to learn how to prepare a more comprehensive business budget.	Preparing a business budget.

Figure 2.14: Learner's profile

- A **questionnaire** is a list of questions used by the teacher to gather information from the learners.

School: Nadrib Senior School	Grade: 10
Learners name: Justin Ambeh	Date: 28 th January 2026
Teacher's name: Pete Kabim	Subject: Business Studies
Strand: Business and money management	Sub strand: Banking
1. Are you able to describe the steps followed when setting business goals? 2. If yes, (a) identify the steps and (b) describe the steps.	
Teacher's general remarks/ comments (strengths, areas that require improvements and recommendations)	
The learner has acquired some knowledge on the steps followed when setting business goals.	
Area of improvement: The learner needs to take more initiative in the practical aspects of learning.	
Recommendation: The learner is to be assisted further to enable her master the steps that are followed when setting business goals.	

Figure 2.15: Questionnaire

- **Oral or aural questioning** is aimed at assessing learner's listening and observation skills. The teacher asks questions verbally and the learner responds with an appropriate mode of communication.
- An **anecdotal record** is a detailed description of a learner that is recorded after a specific behaviour occurs. The information is factual and objective. The teacher uses anecdotal notes to record their observations of individual learners.

School: Busara Senior School	Subject: Chemistry
Learner's name: Gladys Sharon	Grade: 10
Strand: Inorganic Chemistry	Sub strand: The atom
Observation date: 30/06/2025	Observation time: 10.30 am
Description of the location or setting Grade 10 West classroom.	
Teacher's comments Emmanuel showed a great sense of responsibility. During the next lesson, I will mention this to the rest of the members of group 5 so that they can also learn to be responsible.	
Teacher's Name: Mr. Bagra Odek	Teacher's signature: B.O

Figure 2.16: Anecdotal record

In the competency-based education, the teacher is expected to continuously carry out formative assessment. The assessment should mainly focus on the skills acquired and how effectively the learner is able to apply those skills in their day-to-day lives.

3

Giving feedback to learners the CBE way

The provision of effective feedback to learners is one of the most important activities that teachers engage in during any lesson. Effective feedback is the use of assessment evidence by teachers and learners to modify the activities in which they are engaged, so as to improve learners' achievement of the learning outcome. While several definitions for what constitutes feedback exist (Hattie & Timperley, 2007; Shute, 2008; Tunstall & Gipps, 1996), the following definition is used:

Effective feedback refers to any oral and/or written information provided by teachers and/or peers to support learners attain the learning intentions and success criteria for the lesson.

In the context of the classroom, the primary purpose of providing feedback to learners is to reduce the gap between the learner's current level of understanding and/or performance and the desired curriculum or learning goal to be attained (MARS, 2012).

NOTE: Effective feedback can also be obtained from other learners (peer feedback).

Providing effective feedback

This is evident when teachers and learners use assessment information to support improvements in learning and teaching. Feedback which is designed to help learners understand strengths and weaknesses of their current performance is central to the learning process. Successful feedback will indicate to learners how and where they might improve on the specific learning outcome for any lesson. During each lesson, teachers must therefore first identify learner's current level of understanding, determine what assistance, if any, the learner requires and thereafter provide appropriate oral or written feedback to improve the learner's level of understanding.

Wiliam (2011) notes that the most useful feedback for improving learning is that which provides learners with detailed information on how to improve performance. In seeking an explanation for the results on feedback, Wiliam (2011) found that learners who were given comments demonstrated high levels of task-involvement; that is, they focused specifically on the tasks they were required to do. However, those learners who were given grades and/or written praise, showed high level of ego-involvement. In other words, giving praise or grades has no effect on achievement; their only effect is to increase the learner's ego.

Other key findings relating to effective feedback include:

- Effective feedback should relate to the learning goals and success criteria identified for the task (MARS, 2012).

- Feedback can be oral or written, depending on the context it is provided for (James, et al., 2006).
- The impact of any feedback provided to learners will depend on the type, delivery and timing of the feedback (Hattie & Timperley, 2007).
- Feedback is more effective if it focuses on the task rather than the individual (James, et al., 2006).
- Feedback affects learners' motivation to learn and their perceptions about their intelligence and their ability to learn (Black & Wiliam 1998).
- Grades, marks, ticks, scores, etc have little effect on subsequent performance (James, et al., 2006).

Principles for providing effective feedback

The primary purpose of providing feedback to learners is to support learners to identify and address learning gaps. Thus, any feedback that teachers provide must allow the learner to focus on what to do next rather than focusing on how well or how badly he/she has done.

Wiliam (2011) proposes three key principles for providing effective feedback;

1. *All feedback should relate to the success criteria.*
2. *All feedback must provide a recipe for future action.*
3. *All feedback should be focused on one or two key aspects of the work.*

Factors that impact an effective feedback

When providing learners with feedback, it is important for teachers to consider the following questions (Brookhart, 2008; Clarke, 2008; Hattie & Timperley, 2007):

- When was the feedback provided?
- How was the feedback provided?
- How much feedback should be provided?

Brookhart (2008) notes that four factors need to be considered when providing feedback. These factors are discussed in Tables 3.1 and 3.2.

Table 3.1: Four feedback strategies and recommendations to consider

Feedback can vary by	In these ways	Recommendations for good feedback
Amount	• How many points made? • How much about each point?	• Prioritise: Pick the most important points. • Choose points that relate to major learning goals. • Consider the learner's developmental level.

Feedback can vary by	In these ways	Recommendations for good feedback
Timing	- When given? - How often?	- Provide immediate feedback for knowledge of facts (right/ wrong). - Delay feedback slightly for more comprehensive reviews of the learner's thinking and processing. - Never delay feedback beyond when it would make a difference to learners. - Provide feedback as often as it is practical, for all major assignments.
Mode	- Oral - Written - Visual or demonstration	- Select the best mode for the message. Would a comment in passing the learner's desk suffice? Is a conference needed? - Interactive feedback (talking with the learner) is best when possible. - Give written feedback on written work or on assignment cover sheets. - Use demonstration if "how to do something" is an issue or if the learner needs an example.
Audience	- Individual - Group/class	- Individual feedback says, "The teacher values my learning." - Group/class works if most of the class missed the same concept on an assignment, which presents an opportunity for re-teaching.

Table 3.2: Questions to consider during feedback strategies and recommendations

	Feedback	Oral	Written
1.	When?	Classroom discussions, tasks and activities	After written work handed in e.g. individual or group tasks, projects, assignment, tests
2.	How?	In whole class discussion, groups or individually	Written comments and/or symbols
3.	How much?	Focus on at least one positive aspect and one area that needs improvement	

Types of feedback

Feedback can be oral, written, descriptive or evaluative.

(a) Oral feedback

As previously discussed, one of the modes of feedback is oral feedback. Oral feedback refers to comments given verbally to learners in a lesson by teachers, regarding specific tasks or activities. According to Smith (2014), feedback is **"the lifeblood of learning"**. He states that if learners were asked what would help

them most to learn better, the vast majority would say more immediate feedback from the teacher as they are learning. He believes that teachers must find time for oral feedback.

Practical guidelines for providing effective oral feedback to learners

Smith (2014) suggests a number of ways of providing oral feedback to learners. Always note that feedback should always help the learner progress in his or her learning.

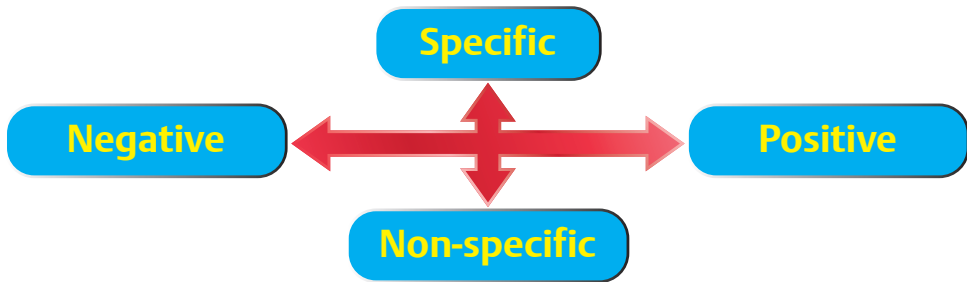


Figure 3.1: Types of oral feedback (Adapted from DfES, 2004)

NOTE

As you read these guidelines, think about the way you provide oral feedback to your learners.

(i) Focus on the task and not the learner

Making the person feel incapable of doing the task can be discouraging. Often teachers would say: “You’re not coping, are you?” It is advisable for teachers to say something about the task being challenging, for example “This exercise is challenging, isn’t it?” and then offer a suggestion on approaching it.

(ii) Recognise the learner’s effort

When learners are criticised for being dull or bright, they are often made to believe that there is nothing that can be done about it. Intelligence is not “fixed at birth”. Learners of all abilities will come across tasks that they will find challenging. Some learners will be frustrated because they now confront a task they are struggling to understand. Other learners will give up because they have already been conditioned to believe that they don’t have the mental capacity for it. By focusing feedback on the effort they are making and the strategies they are using, it can help learners reinforce the view that their ability can improve. Encourage them to thrive on challenging tasks. So instead of saying: “You’re a clever girl, aren’t you?” your comment could be: “I can see you put a lot of effort into your project.”

(iii) Make positive and specific comments

Learners need to be told what to improve in but not shown how to do it. Encourage learners to find their own solutions to problems by making positive

but specific comments. In a comment such as: “You need to give more detail on the conclusion of the experiment you conducted” – give guidance as to what detail is required. An example of good feedback could be: “I liked how neatly your graph was drawn. Perhaps you need to think whether a line graph was suitable for this experiment. What about attempting a bar graph?”

(iv) Link feedback to the learning outcome and success criteria

Feedback linked to learning outcomes and success criteria keeps the learners focused on what they need to learn in that lesson. If the learning outcome was: We are learning about the different types of road transport. A comment such as “You really need to improve your writing” isn’t helpful when the success criteria clearly states: “I can name the different types of vehicles that travel on land.” An example of a good comment could be: “I see you have been able to list at least four types of vehicles that travel on land.”

(v) Don’t overpraise or give it when it is unwarranted

While giving positive feedback is important, beware of its drawbacks. Praise should be commensurate with the effort and should allow learners to reflect critically on their work. If the teacher gives gold stars often because they think everything the learner does is wonderful, it can do more harm than good. Praise what has been accomplished rather than the person. If learners believe they have achieved something because of who they are, it can lead to complacency. Persuade learners to work hard AND smart. Encourage them to overcome their challenges. Smith (2014) notes that “praise often comes at the end of a task”, has to be earned and can be vague. Praise may lead to learners doing the work to please the teacher. However, according to Smith (2014) “encouragement can be given at any time”. It is given for effort put into a task and is therefore clear and specific. A comment such as “I want you to not be afraid of trying a new idea” can encourage creativity. Genuine appreciation may be better to use than praise. It is based on fact and is not evaluative. So rather say “I appreciate how you keep putting effort into your work” instead of “You are the best in this class.”

(vi) Judgments vs opinions

Teachers should guard against making judgments about learners too soon or based on past efforts. Don’t judge the learner by suggesting that she/he is good or bad. It is important to value the effort that had been made. When giving an opinion use more “I” statements than “you” statements. Instead of saying: “Your spelling is really terrible. When are you going to get it right?” the teacher could say: “I am a bit disappointed with this essay. I know you have the potential to do better.”

(vii) Respect the diverse abilities of the learners

No two learners are the same. Saying “Why can’t you be more like Kamau?” has a negative effect on the self-esteem of the learner. Learners have a combination of different experiences and abilities.

Appreciate their endeavours to overcome challenges: “I know it was difficult for you to do this, but I am glad you made the attempt.” Beware of gender-stereotyping learners by expecting girls to perform or behave one particular way and boys another. Girls can do as well in mathematics as boys.

(b) Written feedback

Written feedback is the teacher’s written comments provided to a learner in response to tasks or activities completed. The type of feedback provided also impacts on whether and how learning takes place: specifically, whether the feedback is procedural, evaluative, descriptive (showing) or descriptive (guiding).

Below are several teacher comments and some feedback criteria (A–C) to help you judge the helpfulness of the comments to the learners. Decide whether the comments:

A: Highlight one or more successes.

B: Identify an area of improvement.

C: Make a specific suggestion for improvement.

Table 3.3: Examples of written feedback

No	Written feedback	A	B	C
1.	You really did well with the first few questions but was challenged by the rest. Try the box method for multiplication.	✓	✓	✓
2.	Your rap song about poverty was profound. What do you think causes poverty?			
3.	3 for effort, 1 for spelling. Neat work and improve your spelling.			
4.	Again, you have not completed your work.			
5.	Good work - 7/10			
6.	Too many errors. Re-do!			
7.	You’ve identified all the 2D shapes. Excellent.			
8.	Your description uses a number of adjectives to make the topic interesting. Try writing more complex sentences.			
9.	This is a great project. Your presentation using the diagrams showed clearly why conservation is important.			
10.	Nice work. Your presentation grabbed our attention. What more do you think could be done to increase employment for the youth?			

(c) Descriptive feedback

Feedback that is descriptive is linked to the learning that is expected, specifically the success criteria and highlights key issues like faulty interpretations and lack of understanding (Earl, 2012).

This type of feedback is intended to help learners to improve learning by providing information about their current achievement (Where am I now?) with respect to a goal (Where am I going?) and identifying appropriate next steps (How can I close the gap?) (Sadler, 1989). In addition to providing learners with visible and manageable next steps, Earl (2012) notes that this type of feedback should also provide examples of what good work looks like.

The difference between these two types of feedback is that for **descriptive-showing** feedback, the teacher shows or tells learners how to go about improving their work. That is, the teacher decides on the next steps that should be followed and informs learners on what to do next. For **descriptive-guiding**, the teacher guides and supports the learner with suggestions or ideas for the learner to decide that the next steps should be for their work.

(d) Evaluative feedback

Evaluative feedback may provide some information about learning but does not convey the information that learners can use to improve. For example, grades, marks and comments such as “well done”, “excellent”, “untidy”. No information is provided about how to improve the work. It is important to note that evaluative feedback can have a negative impact on learning and motivation (Black & Wiliam, 1998), especially when this type of feedback focuses on characteristics of the learner rather than on the characteristics of the work (Dweck, 2007).

Other comments that focus on routine and/or irrelevant (in terms of the success criteria) practical aspects of the learners’ work, for example, “Incomplete!”, “Where are your answers?” “Not again”, “Pull up your socks” are not necessary.

NOTE

Remember, ALL learners have the potential to learn and your role as a teacher is to use feedback positively to improve the way all learners learn.

Planning and preparing lessons in CBE

For teachers to plan and prepare effective lessons so that learners acquire the knowledge and skills specified in the curriculum, teachers must be clear on what learners need to learn and what they need to teach. Therefore, teachers must be fully conversant with their subject content area(s) as well as the specifications of the national curriculum. With this understanding, teachers will be able to plan and prepare lessons that engage all learners to develop the important knowledge, understanding and skills on key concepts in the curriculum.

Research shows that many teachers spend limited time to plan and prepare their lessons (Kanjee, 2018). While some only use documents such as the Annual Teaching Plans, others use structured lesson plans provided by some publishers, while others teach directly from the textbook. Also some teachers use one page monthly or weekly schedules that list topics to be taught and present lessons without any lesson plans.

The limitation of these practices is that teachers are unable to clearly determine what learners need to learn, and thus the lessons they present are mainly focused on their teaching instead of learning. In practice, this means that only the teacher is aware of the learning outcomes for the lesson, the tasks to be completed during the lesson, and the assessment criteria required to demonstrate that learners have attained the lesson outcomes. Moreover, many teachers do not share this information with learners before the completion of the lesson. Thus, most learners participate in lessons with no prior knowledge about what they are going to learn, nor what evidence they have to show that they had learnt the concepts being taught.

The value of planning and lesson preparation

It is vital that teachers plan and prepare for all lessons because “planning provides structure and context for both teacher and learners, as well as a framework for reflection and evaluation” (Spencer, 2003). Preparation means spending time reading about and understanding the formative assessment strategies and techniques, how these can be used, and the resources required to effectively implement them during the lesson. Planning and preparation is also contextual, it is not done in isolation. Teachers need to consider their learners, the context of the school, the availability of resources, as well as how one lesson links to the next and if the different lesson activities are related.

There are distinct advantages to planning and preparation. These include:

- **It will improve teaching skills.** Teachers cast themselves as researchers. Teachers learn educational theory – in this instance the theory of formative assessment and understand how it will impact on teaching and learning. They then learn new practices that will change the way they think about teaching.

Proper planning and preparation also requires a clear understanding of the content that needs to be taught.

- **It saves time.** Planning and preparation, especially when using formative assessment strategies and techniques, allows for efficient and effective delivery of the lesson.
- **Learners will become more disciplined.** When used, formative assessment strategies and techniques allow for greater learner participation, whether in answering questions, contributing to discussions or in ownership of learning. This means that the teachers need to plan and prepare interesting lessons.
- **The confidence of the teachers are boosted.** Planning and preparation creates a positive attitude because teachers know the content of their subject area and are increasingly improving their pedagogy, in this instance the use of formative assessment. Confidence comes from teachers constantly reflecting on the way they teach, but more significantly on how learners are learning.
- **Teachers will earn respect.** Other stakeholders, but particularly the learners, will know which teachers are putting in an effort to make learning interesting and helping learners to perform better.
- **Learner performance will improve.** In addition to teachers mastering the content of their subject area, effective implementation of formative assessment makes learning interesting and helps to engage learners more effectively during lessons. Additionally, it enhances their understanding of the content presented. Using formative assessment in a classroom requires careful planning and preparation. In practice, this will mean:
 - (i) Examining the revised curriculum to determine exactly what learners need to learn.
 - (ii) Identifying the objectives and assessment criteria for the topic (this information may not always be stated).
 - (iii) Specifying activities that will be used and how these will be applied.
 - (iv) Writing the higher-order thinking questions that will be used in the lesson.
 - (v) Listing the resources required and how these will be applied.
 - (vi) Indicating which techniques and strategies will or will not be used for the lesson.

Lesson plans should be written with learners in mind. Therefore, the plans must first address the question:

- *What is it that learners must attain?*

And only thereafter, teachers ask the question:

- *What must the teacher do to ensure that learners meet the lesson objectives?*

Below is an example of a lesson planning and preparation template to support teachers to plan and prepare for their lesson.

Sample lesson plans

Below are sample lesson plans for the various learning areas in Senior School.

Table 4.1 Sample **Kiswahili** lesson plan

Taarifa za Kiutawala		
Jina la Shule: Shule ya Wajibika	Tarehe: 11/02/2026	Saa: 8.00 – 8.40 asubuhi
Somo: Kiswahili	Mwaka: 2026	Gredi: 10
Muhula: 1	Idadi: 35 Wavulana: 20 Wasichana: 22 Jumla: 42	
Taarifa za Mwalimu.		
Jina: Imani Pendo	Namba ya TSC: 1000005	Jinsia: kike
Mada Kuu: Kusoma		
Mada Ndogo: Kusoma kwa ufahamu: Kifungu simulizi		
Maswali Dadisi: Je, unazingatia nini ili kupata ujumbe katika kifungu simulizi?		
Nyenzo: kifaa cha kidijitali, nakala ya kamusi		
Mpangilio wa Ujifunzaji		
Utangulizi: Kujadili swali dadisi kama darasa		
Maendelezo ya Somo:		
Hatua ya 1: Kusoma kifungu simulizi kuhusu kukabiliana na shinikizorika.		
Hatua ya 2: Kudondoa habari mahususi katika kifungu simulizi.		
Hatua ya 3: Kupanga matukio yanavyofuatana katika kifungu simulizi.		
Shughuli zaidi: Kufanya utabiri na ufasiri kutokana na kifungu simulizi.		
Hitimisho: Wanafunzi kufanya zoezi la kupata maana za maneno na kuyatungia sentensi kama kazi ya ziada.		
Tafakuri kuhusu Kipindi: Malengo ya kipindi yaliafikwa.		

Table 4.2 Sample **Community Service Learning** lesson plan

Administrative details		
School:	Date: 10 th June, 2026	Time: 11:00 – 11.40 am
Subject: Community Service Learning	Year: 2026	Grade: 10
Term: 1	Roll: 35	
Teacher details. Name: Mr. Albert Jatich		TSC NO: 822442 Gender: Male
Strand: Citizenship	Sub strand: Intercultural competence	
Lesson learning outcomes: By the end of the sub strand, the learner should be able to: - Name intercultural activities in their community. - Explain the importance of participating in intercultural activities. - Select and participate in an intercultural activity. - Develop genuine interest in participating in intercultural activities.		
Key inquiry question Why do we participate in intercultural activities?		
Learning resources: Learner's Book, resource person, lesson notes and digital devices.		
Organisation of learning: Learning will take place in the classroom. Learners will work in pairs and also in groups.		
Introduction The teacher introduces the lesson by narrating to learners a brief experience he had while attending a recent intercultural activity. This will arouse their interest. The teacher asks the learners to talk about intercultural activities they know about and share with their deskmate.		
Lesson development Step 1: Learners are asked to discuss and share with the class some intercultural activities that they have participated in or attended in their school or home communities. The teacher observes as they collaborate and communicate with each other respectfully. The teacher assesses the relevance of their responses. Step 2: Learners are asked to study the pictures on page 46 of the Learner's Book and to answer the questions. Learners are guided to orally give answers to question 1 while answers to question 2 are written on their exercise books. The teacher marks some of the answers written and suggests corrections where necessary.		
Step 3: In pairs, learners are guided to discuss the meaning of the proverbs on page 47 of the Learner's Book and to answer the questions asked. The teacher assesses correctness of responses and offers necessary guidance.		
Step 4: In groups, learners are guided to select one intercultural activity from the ones they listed in their responses to question 3. Learners are allowed time to briefly rehearse and role play the intercultural activity of their choice. As they present the skits, a learner uses a digital device to record the role play. The teacher assesses the learners' creativity and imagination as well as their ability to work together harmoniously, for social cohesion.		
Extended activity: Learners are asked to seek the support of a parent/guardian or any other resource person and learn the importance of participating in intercultural activities in the community. They are asked to prepare the points they will present in class in the next CSL lesson.		
Conclusion: The teacher uses oral questions to sum up the main points of the lesson on the importance of participating in intercultural activities for social cohesion.		
Reflection on the lesson:		

Table 4.3 Sample **Business Studies** lesson plan

Administrative details		
School: Ufanisi Senior School	Date: 12/1/2026	Time:
Subject: Business Studies	Year: 2026	Grade: 10
Term: 1	Roll:	
Teacher Name:	TSC NO:	Gender:
Strand: Business and money management		Sub strand: Money
Lesson learning outcomes: By the end of the sub strand, the learner should be able to: - Define money. - Identify and explain the security features found on the Kenyan currency. - Appreciate the importance of the security features on the Kenyan currency.		
Key inquiry question; What are the security features of the Kenyan currency?		
Learning resources: Teacher's Guide, Learner's Book, lesson notes, print resources, currency coins and notes.		
Organisation of learning:		
Introduction (5 minutes)		
Begin with a brief question and answer session on the meaning of money and the various denominations of the Kenyan currency.		
Engage learners by asking questions about the reasons why it is necessary for money to have security features.		
Lesson development (30 minutes)		
Step 1: Observing pictures of Kenyan coins. In groups, let the learners go through the questions in the Learner's Book in Activity 1 page 1. Allow them time to observe the pictures and then answer the questions that follow. Use oral questions to correct those that have challenges when answering the questions. In groups, allow learners to observe and touch Kenyan coins that you have made available to them. Guide them as they identify the security features on the notes that they are handling. Control the class so as to minimise the chorus answers from the learners. Give learners with challenges enough time to complete the task.		
Step 2: Observing pictures of a Kenyan note. In groups, direct the learners to undertake Activity 2 in the Learner's Book pages 2 - 3. Allow them to observe the pictures and then answer the questions that follow. Guide them as they identify the security features on the notes that they are handling. In groups, allow learners to observe and touch the Kenyan notes that you have made available to them. Guide them as they identify the security features on the coins that they are handling. Control the class so as to minimise the chorus answers from the learners. Give learners with challenges enough time to complete the task.		
Step 3: Filling in a table. Allow the learners to perform the tasks in Activity 3 in the Learner's Book page 3. Give the learners adequate time to copy the table in their exercise books and then fill in the appropriate information on security features that are unique to some Kenyan currency notes. Let them peer assess their answers. Ensure use of correct grammar in communication.		
Step 4: Reading a chart. Direct the learners to carry out the tasks in Activity 4 in the Learner's Book pages 3 - 4. Allow them time to answer the questions that come after the chart. Use oral questions to guide the learners that have challenges in coming up with appropriate answers to the questions in this activity.		
Extended activities		
Learners will discuss the learning spot points to reinforce what they have learnt on the security features of the Kenyan currency. Let them give their views as this develops the core competencies on communication and collaboration as well as learning to learn.		
Conclusion (5 minutes)		
Allow the learner(s) to summarise key points and the learning outcome achieved during the lesson. Identify key information by conducting a brief interactive activity where learners can ask each other questions on the security features on Kenyan currency. Prepare learners for the next lesson by previewing the upcoming topics or posing questions for them to consider.		
Reflection on the lesson:		

Table 4.4 Sample **Geography** lesson plan

School: Fanaka Senior School	Date: 12/1/2026	Time:
Learning area: Geography	Year: 2026	Grade: 10
Term: 1	Roll: Boys21	Girls.....24 Total 45
Teacher details: Name:	TSC NO:	Gender:
Strand: Practical Geography	Sub strand: Introduction to Geography	
Lesson learning outcome By the end of the lesson, the learner should be able to analyse the branches of Geography for in-depth understanding of the subject.		
Key inquiry question What are the branches of Geography?		
Learning resources: Teachers' Guide, Learners' Book, lesson notes, pictures showing natural and human-made features and immediate environments.		
Organisation of learning: Introduction (5 minutes) Welcome learners to Grade 10 and explain that Geography will now be studied as a separate subject, unlike in previous grades where it was integrated with Social Studies. Briefly discuss what Geography is and ask learners to share their thoughts. Guide them to understand that Geography studies the physical and human environment.		
Lesson development (30 minutes) Step 1: Explain that Geography is the study of the earth as the home of humankind, focusing on both the physical and human features. Highlight that the subject explores how people interact with their surroundings.		
Step 2: Guide learners to discuss how physical features influence human activities and how human activities can, in turn, affect physical features. Ask the learners to do Activity 1 on page 1 of the Learner's Book. This activity helps to develop the competence of communication and collaboration and as they share their answers.		
Step 3: Guide learners to brainstorm on the meaning of Geography based on their observations from Activity 1. Lead a class discussion to agree on a comprehensive definition. As they brainstorm and discuss, learners develop critical thinking and imagination. Listening to each other's definitions and sharing ideas nurtures the values of respect and unity.		
Step 4: Use Activity 2 on page 2 of the Learner's Book to guide the learners to understand that Geography has two main branches: Physical Geography and Human Geography.		
Step 5: Let learners work in pairs to sort various aspects of Geography into physical Geography and human Geography as guided in Activity 3 on page 3 of the Learner's Book. Allow different groups in pairs to share their completed tables and explain their choices.		
Conclusion (5 minutes) Recap the key points covered in the lesson by asking learners to summarise the two main branches of Geography and their characteristics. Reinforce the importance of understanding the relationship between physical features and human features. Prepare them for the next lesson by mentioning that they will learn about the importance of Geography in promoting sustainable development.		
Reflection on the lesson Learners engaged well in group and pair activities, demonstrating an understanding of the branches of Geography. A few struggled to differentiate between physical features and human features. I will reinforce this concept in the next lesson.		

Table 4.5 Sample *History and Citizenship* lesson plan

Administrative details		
School:	Date: 12/1/2026	Time:
Learning area/subject History and Citizenship	Year: 2026	Grade: 10
Term: 1	Roll: Boys21 Girls.....24 Total 45	
Teacher details: Name:	TSC NO:	Gender:
Strand: Themes in Kenyan History and Citizenship Sub strand: Introduction to History and Citizenship Specific learning outcomes By the end of the sub strand, the learner should be able to: 1. classify sources of information on History and Citizenship in Kenya, 2. examine how the past historical information has shaped the modern History and Citizenship, 3. justify the importance of studying History and Citizenship, 4. value the importance of studying History and Citizenship for posterity. Key inquiry questions 1. What are the sources of History and Citizenship in Kenya? 2. How do primary and secondary sources differ in their usefulness and limitations when studying historical events? 3. How can digital tools and archives be strategically used to enhance access to diverse historical sources? Learning resources: Teacher's Guide, Learner's Book, lesson notes, print resources, digital device, resource person, poems and charts. Organisation of learning: Introduction (5 minutes) Begin with a brief recap of what was covered in the previous grades about career development to create a link to this lesson. Engage the learners by asking questions about key concepts discussed previously in the sub strand mentioned. Ask the learners what they remember about pathway choices and pre-career systems. Link this to History and Citizenship. Lesson development (30 minutes). Step 1: Classifying the sources of information on History and Citizenship in Kenya In their pairs let them undertake Activity 2 in the Learner's Book pages 1 – 2. Let them read the stories and identify the sources of History and Citizenship discussed. Ensure class control and minimise chorus answers from the learners. Give learners with challenges enough time to complete the task. Step 2: Categorising the sources Allow the learners to categorise the sources from the stories on accuracy and the one containing first-hand experience. Let them peer assess other answers. Ensure use of correct grammar in communication. Step 3: Sharing their feedback Peers and teachers will provide constructive feedback, highlighting aspects that are well done or could be improved. Extended activities Learners will discuss the learning spot points to reinforce what they have learnt. Let them give their views as this develops the core competence of communication and collaboration. Conclusion (5 minutes) Allow a learner(s) to summarise the key points and learning outcomes achieved during the lesson. Identifying key information – Conduct a brief interactive activity where the learners can ask each other questions based on classifying the sources of information on History and Citizenship in Kenya. Prepare the learners for the next session by previewing upcoming sub strands or posing questions for them to consider. Reflection on the lesson:		

Table 4.6 Sample **Computer Studies** lesson plan

Administrative details		
School: Baraka Senior School	Date: 12/1/2026	Time: 10.30 a.m.
Learning area: Computer Studies	Year: 2026	Grade: 10
Term: 1	Roll: Boys: 25 Girls: 23 Total: 48	
Teacher details: Name: John Thomas	TSC NO:	Gender: Male
Strand: Foundation of Computer Studies Sub strand: Evolution and development of computers Specific learning outcome By the end of the lesson, the learner should be able to identify early computing devices and state how they relate to evolution of electronic computers. Key inquiry question What were the early computing devices and how do they relate to evolution of electronic computers? Learning resources: The Grade 10 Learner's Book, the teacher's guide, digital devices connected to the internet, reference textbooks, computer magazines and modern abacus. Organisation of learning: Learning takes place inside the classroom where the learners are organised into groups. Groups are well-balanced in terms of gender and ability. Introduction (Engagement) 5 minutes The learners are probed to identify the items in the get started activity on page 1 of the Learner's book and give their opinion about what would happen today if the items were not invented. Lesson development - 30 minutes Step 1: Exploration Learners to watch a video on evolution of computers as a class through the guidance of the teacher. Step 2: Explanation Learners to do research using textbooks or the internet about evolution of computers guided by the questions in Activity 1 on page 1 of the Learner's book. Step 3: Elaboration Learners to use the knowledge acquired so far to draw a chart on evolution of computers. Step 4: Evaluation Through question and answer method, the teacher finds out if the concept has been understood. Conclusion (5 minutes) The teacher guides the learners through the summary notes on pages 3-4 of the Learner's book making clarifications where necessary. Reflection on the lesson: The teacher reflects on whether the learning outcome was achieved, whether the learner activities were appropriate and effectiveness of the resources used so as to thoughtfully modify the approach and resources and if necessary, organise individualised education programmes for particular learners.		

Table 4.7 Sample **Biology** lesson plan

Administrative details		
School:	Date:	Time:
Learning area: Biology	Year: 2026	Grade: 10
Term: 1	Roll: Boys: 25 Girls: 23 Total: 48	
Teacher details: Name: John Thomas	TSC NO:	Gender: Male
Strand: Cell Biology and Biodiversity Sub strand: Introduction to Biology Lesson learning outcomes By the end of the lesson, the learner should be able to: 1. explain the meaning of Biology as a field of science. 2. describe the applications of Biology in everyday life. Key inquiry question 1. What is Biology as a field of science? 2. How is the knowledge gained from the study of Biology applied in our day-to-day activities? Learning resources: - Learner's Book - Lesson notes - Print resources - Digital devices Organisation of learning: Introduction (5 minutes) Begin with a brief recap of what was covered in Grade 7 on the branches of science. Engage learners by asking questions about key concepts discussed previously in Junior School. • Direct learners to relevant sections in the Learner's Book and other resources. • Encourage a short discussion about what they know concerning Biology. Lesson development - 30 minutes Step 1: Defining Biology In groups, learners to brainstorm the meaning of "Biology" as a field of science. Learners to use digital or print resources to research and write a concise definition. Step 2: Application of Biology in everyday life Learners to use digital or print resources to research on applications of Biology in everyday life. Learners to engage in a group discussion on the applications of Biology in everyday life. This will develop the core competency of communication and collaboration as the learners engage in group discussion and promote the value of respect as learners appreciate the opinions of their peers during the discussion. Step 3: Presentation and feedback Groups to present their findings to the class. Peers and teachers to provide constructive feedback, highlighting aspects that are well done or could be improved. Extended activities 1. Learners to make posters showing various applications of Biology in everyday life. 2. Interactive web: Learners to explore educational websites or applications that provide more information on the applications of Biology in everyday life. This will develop the core competency of digital literacy. Conclusion (5 minutes) • Summarise the key points. • Recap the definition of Biology as a field of science. • Reinforce the significance of learning Biology. • Conduct a question-and-answer session on the meaning of Biology and the applications of Biology in everyday life. • Brief learners on the next concept which is the fields of study in Biology. Reflection on the lesson: Reflect on whether the learning outcome has been achieved and whether the learning activities were appropriate. Also, check whether the learning resources were effective. Organise individualised education programmes, if necessary.		

Table 4.8 Sample **Chemistry** lesson plan

School: Fanaka Senior School	Date: 12/1/2026	Time:
Learning area: Chemistry	Year: 2026	Grade: 10
Term: 1	Roll: Boys21 Girls.....24 Total 45	
Teacher details: Name:	TSC NO:	Gender:
Strand: Inorganic chemistry		Sub strand: Introduction to Chemistry
Lesson learning outcome		
By the end of the lesson, the learner should be able to:		
1. explain the meaning of Chemistry as a field of science. 2. describe the branches of Chemistry.		
Key inquiry questions		
1. What is Chemistry as a field of science? 2. Which are the branches of Chemistry?		
Learning resources: Grade 10 Chemistry Learner's book, lesson notes, print resources and digital devices.		
Organisation of learning:		
Introduction (5 minutes)		
• Begin with a brief recap of what was covered in Grade 7 on the branches of science. • Engage learners by asking questions about key concepts discussed previously in Junior School. • Direct learners to relevant sections in the Learner's Book and other resources. • Encourage a short discussion about what they know concerning Chemistry.		
Lesson development (30 minutes)		
Step 1: Defining Chemistry		
• In groups, learners to brainstorm the meaning of "Chemistry" as a field of science. • Learners to use digital or print resources to research and write a concise definition of chemistry.		
Step 2: Exploring branches of Chemistry		
• Learners to use digital or print resources to research the branches of Chemistry. • Learners to engage in a group discussion to identify branches of Chemistry and their scope. This will develop the core competency of communication and collaboration as the learners discuss the branches of Chemistry and promote the value of respect as learners appreciate the opinions of their peers during the discussion.		
Step 3: Presentation and feedback		
• Groups to present their findings to the class. • Peers and teachers to provide constructive feedback, highlighting aspects that are well done or could be improved.		
Extended activities		
• Learners to make posters showing various branches of Chemistry. • Interactive web: Learners to explore educational websites or applications that provide more information on branches of Chemistry. This will develop the core competency of digital literacy.		
Conclusion (5 minutes)		
• Summarise the key points. • Recap the definition of Chemistry as a field of science. • Reinforce the significance of learning Chemistry. • Conduct a question-and-answer session on the meaning of Chemistry and the branches of Chemistry. • Brief learners on the next concept which is the role of Chemistry in day-to-day life.		
Reflection on the lesson:		
Reflect on whether the learning outcome has been achieved and whether the learning activities were appropriate. Also, check whether the learning resources were effective. Organise individualised education programmes, if necessary.		

Table 4.9 Sample **General Science** lesson plan

Administrative details		
School: Tumaini Senior School	Date: 7 th January 2026	Time: 9.20 a.m.
Learning area: General Science	Year: 2026	Grade: 10
Term: 1	Roll: Boys: 20 Girls: 20 Total: 40	
Teacher details: Name: Mary Jane	TSC NO: 884293	Gender: Female
Strand: Life Science Sub strand: Introduction to General Science Lesson learning outcome By the end of the lesson, the learners should be able to explain the meaning of General Science as a learning area. Key inquiry question What is the meaning of General Science? Learning resources: Relevant textbooks, digital devices such as laptops or smartphones, a chart on learning areas in General Science, Manila papers and marker pens. Organisation of learning: Introduction (5 minutes) Ask learners to brainstorm on what they know about General Science. Lesson development - 30 minutes Step 1: Research (10 minutes) 1. Provide learners with available resources for research. Encourage them to share. This promotes the value of love. 2. Ask learners to search for information on the meaning of General Science. Instruct learners to write down their findings. Step 2: Group discussion (10 minutes) 1. Guide a class discussion on the meaning of General Science as a learning area. 2. Encourage peer-to-peer learning. Step 3: Class sharing (10 minutes) 1. Allow each group to presents their findings in class. 2. Encourage other learners to ask questions. Extended activities 1. Instruct learners to create a poster showing the meaning of General Science and its related learning areas. 2. Invite a resource person to present on the meaning of General Science and its related learning areas. Conclusion (5 minutes) Conclude the lesson by referring learners to the ' Key Points ' section as outlined in the Learner's Book. Instruct learners to write short notes. Reflection on the lesson: 1. Ask learners what worked well and what challenges they faced when carrying out the activities. 2. Ask learners to mention what improvements they need when carrying out the activity.		

Table 4.10 Sample **Mathematics** lesson plan

Administrative details		
School: Hekima Senior School	Date: 12 th January 2026	Time: 3:20 – 4:00 p.m.
Subject: Mathematics	Year: 2026	Grade: 10
Term: 1	Roll: Boys22 Girls.....23 Total 45	
Teacher Name: Mr Hassan	TSC NO: 456789	Gender: Male
Strand: Numbers and algebra Sub strand: Whole numbers Lesson learning outcomes By the end of the sub strand, the learner should be able to: - state the meaning of odd and even numbers in different situations, - classify whole numbers as odd and even in different situations, - promote the use of odd and even numbers in day-to-day activities. Learning resources: Number charts and number cards. Organisation of learning: Learning takes place inside the classroom and learners work in groups. Introduction: (5 minutes) Learners to discuss the divisibility test of 2 as they learnt in Grade 7. Lesson development (30 minutes) Engage learners to classify their ages as either even or odd numbers in Activity 1 on page 1 of the Learner's Book. Reinforce their understanding by guiding them through Example 1 on page 1 of the Learner's Book. Ask them to individually attempt the questions in Assessment 1 on page 1 to 2 of the Learner's Book. Check the correctness of their responses. Learners will develop <i>communication and collaboration</i> as they discuss how to classify whole numbers as odd and even numbers. <i>Self-esteem</i> will be enhanced as the learners contribute to the discussion. Extended activity Learners to identify numbers in real life and classify them as odd or even. Conclusion Learners to discuss how they classified numbers as odd and even. Reflection on the lesson Learners were able to classify numbers as odd and even. However, they need to do more classification of numbers.		

Choosing the best CBE Course Book

There are two important aspects to consider when choosing the best Course Book to use:

- The content of the Learner's Book and the Teacher's Guide.
- The information and guidelines on planning, methodologies, assessment and subject-related tools.

The following checklist below can be helpful in selecting the best CBE Course Book for your class.

Table 4.11 Checklist for the best CBE Course Book

Learner's Book	Teacher's Guide
(a) Activities and learning experiences • Are the activities clear? • Are the activities sufficient for differentiation and inclusion for all learning styles? • Are there enough activities for each lesson for each day? • Do the activities show progression beyond the classroom? • Is there a variety of activities? • Are there activities designed for the time takers? • Are the learning experiences appropriate for the grade? • Do the activities exclude any bias? • Do the activities stimulate critical thinking and problem-solving skills?	(a) Activities and learning experiences • Are there sufficient additional activities for those learners that struggle with difficult concepts? • Do the teaching guidelines give you new ideas for your teaching? • Are there additional information that can help to reinforce learning?
(b) Structure and layout • Does it have a contents page? • Is it visually attractive? • Is it appealing to the learners? • Is the font size appropriate for the grade? • Is it easy to find things in the book? • Does it have headings and sub-headings?	(b) Structure and layout • Does it have a contents page? • Is it easy to reference with the Learner's Book? • Is it easy to find things in the book? • Does it have headings and sub-headings?
(c) Language and Content • Is the level of language used appropriate? • Can the weaker learners understand the language? • Is the level of language appropriate for the gifted learners? • Does the language support the learning process? • Are all of the main content, concepts and skills included? • Is the content structured from the known to the unknown? • Is the content written in an interesting way? • Does the content encourage learners to apply concepts and skills in different situations?	(c) Content • Are there guidelines on teaching methodologies? • Are there planning guidelines? • Is there ease in progression from one term to the next? • Are there sufficient guidelines for the teacher on how to complete the assessment tasks? • Are there sufficient guidelines for assessment? • Is the assessment aligned to the CBE? • Are there model or suggested answers for all the questions?

Learner's Book	Teacher's Guide
(d) Illustrations - Are the illustrations clear? - Do they enhance understanding of the concepts covered? - Do the illustrations support the text?	(d) Extras - Are there additional ideas that can be used in your teaching? - Are there additional formal assessment tasks included that could save you time setting your own? - Is the planning comprehensive?

Checklists for CBE preparedness

For an effective learning process, the teacher is required to develop a **'to do list'** that will help them to decide on what is priority, urgent and important or even important but not urgent items. This will give the teacher confidence and direction on what to prioritise. Some items to include on the list are:

- Design strategies to guide the process and achievement of the goals.
- Understand your focus.
- Create a planner for your students.
- Develop your expectations.
- Make sure all the equipment in your classroom work properly.
- Develop a list of classroom management solutions.
- Create your class rules and procedures.
- Put together materials in case you need a substitute resource.

Competence is not simple addition of knowledge, skills and attitude. It is the proven ability to use them in context to achieve the desired results. A competency-based education teacher ought to desire to go beyond knowledge delivery to proficiency.

Checklist 1.

Teacher's proficiency in modelling competency

A needs assessment checklist will help you to know where you are. An example is given below.

Teacher individual competency skill area with learners	Competency rating score				Training required		
	Excellent	Good	Average	Below average	Yes	No	Not sure
Communication and collaboration							
Creativity and imagination							
Critical thinking and problem solving							

Teacher individual competency skill area with learners	Competency rating score				Training required		
	Excellent	Good	Average	Below average	Yes	No	Not sure
Learning to learn							
Self-efficacy							
Citizenship							
Digital literacy							

Checklist 2.**Ask yourself: Have I created a warm learning environment?**

A successful classroom invites students to question and to explore. Below are some of the questions you may wish to ask yourself about the kind of learning environment you wish to create:

- How will I arrange the desks to foster classroom discussion (communication) and collaboration?
- How have I selected the groups and in-group roles and responsibilities?
- Where will I display my learners' work?
- Are important announcements and notices clearly displayed?
- Are there books on display to stimulate student interest and curiosity?
- Where do I store other learning materials?
- How do I maintain a clean hygienic environment that is learner-friendly?

Checklist 3.**Planning and delivery: What items will I need in order to plan a successful lesson?****I will need to have:**

- The curriculum designs
- A scheme of work
- A monitoring tool to cross-check a balanced delivery of the competency, PCI and values
- Reference materials
- Learner's workbooks
- Teacher's Guide
- Learning activities and learning experiences
- Lesson plan
- Attainable learning outcomes
- Other teaching resources
- Templates for assessment
- Class rules of engagement
- Journal for reflections

Checklist 4.**Audit tool to check quality**

Quality evidence to embed competencies, PCIs and values. An example is given below.

Lesson	Competencies, PCIs and values	What does the teacher do?	What does the learner do?	What does the learner's work look like?
Lesson 3: Spelling	Competencies: Communication & collaboration PCI: Learning to live together Values: Responsibility and unity	• Distributes materials • Guides the learners • Assesses to gauge understanding • Probes • Provides feedback	• Learners in their groups select and spell out words from the materials given. • Learners pair up for peer assessment and support.	• Word spell displayed in groups or classroom walls. • Presented to the rest. • Entered in the portfolio of learning.

Checklist 5.**Assessment processes**

- Know what assessment to give at every level.
- Develop and design a rich toolkit with a variety of assessment
- Understand the modes of assessment to use for:
 - assessment AS learning,
 - assessment FOR learning,
 - assessment OF learning.

Use of generative AI in lesson planning

What is generative AI?

Generative AI refers to artificial intelligence systems that can create content such as texts, images and videos. These systems can be powerful allies in lesson planning, offering innovative ways to create engaging and personalised learning experiences by using prompts to generate content.

A prompt is a specific instruction or question given to an AI tool (AI Chatbot) to generate the desired content. It guides the AI chatbot in producing relevant and accurate outputs.

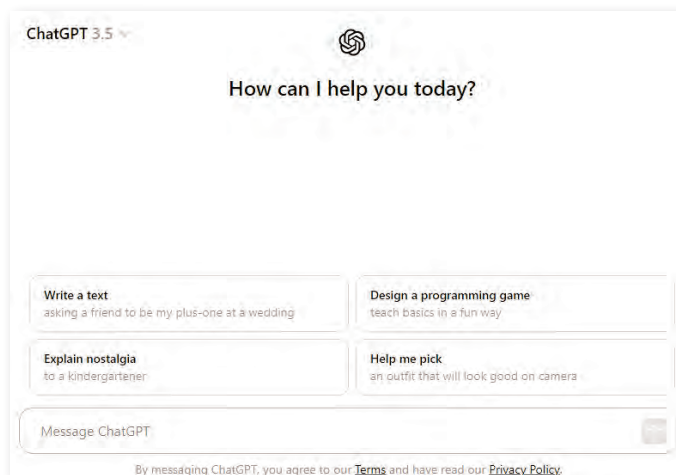
Examples of AI Chatbots

Some useful AI chatbots include microsoft copilot, ChatGPT and Gemini.

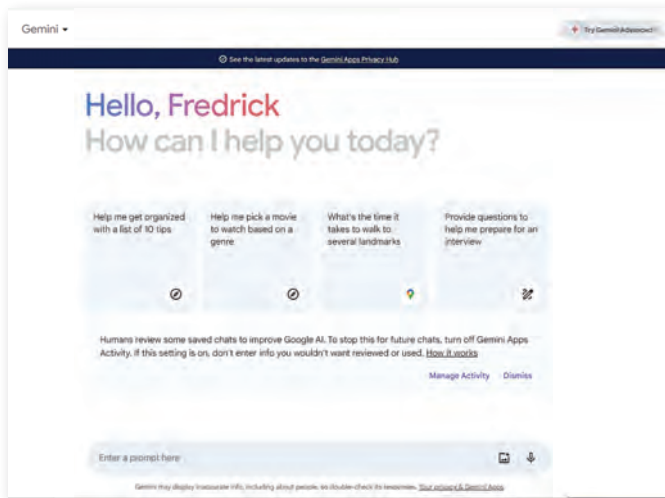
- Microsoft Copilot



- ChatGPT



- Gemini



Tips for generating prompts for use in Chatbots when developing lesson plans

Consider the following factors:

1. **Clarity:** Briefly explain your goal in your prompt.
2. **Context:** Specify the subject, grade level and desired learning outcomes in your prompt.
3. **Specificity:** Mention the type of curriculum support needed in your prompt.
4. **Role:** Define the role the chatbot will act as in your prompt.
5. **Format and Style:** Indicate any preferences for format or style of output in your prompt.

Demo prompt

I'm developing a lesson plan for my [Grade Level] [Subject] class. The topic will be [Specific Topic]. My goal is to create an engaging and interactive learning experience that fosters [Desired Learning Outcome]. The target audience for this lesson is [Target Audience]. I'm particularly interested in incorporating [Specific Activity Type]. Can you act as a [Desired Role] and help me achieve this goal? Ideally, the output would be [Desired Format].

Sample prompt

I am developing a lesson plan for my Grade 6 Science class on the topic of the human circulatory system. My goal is to create an engaging and interactive lesson that fosters critical thinking skills. The target audience for this lesson is a mix of visual and kinesthetic learners. I'm particularly interested in incorporating simulations and group activities that allow students to explore the circulatory system in a hands-on way. Can you act as a Curriculum Assistant and help me brainstorm engaging activities for this lesson? The output should be a list of different activity ideas that

cater to both visual and kinesthetic learning styles. List three learning outcomes, describe the suggested activities in detail, outline instructions for the teacher and indicate the time for each activity.

The above prompt, when typed in the various chatbots, the outputs looks as shown below.

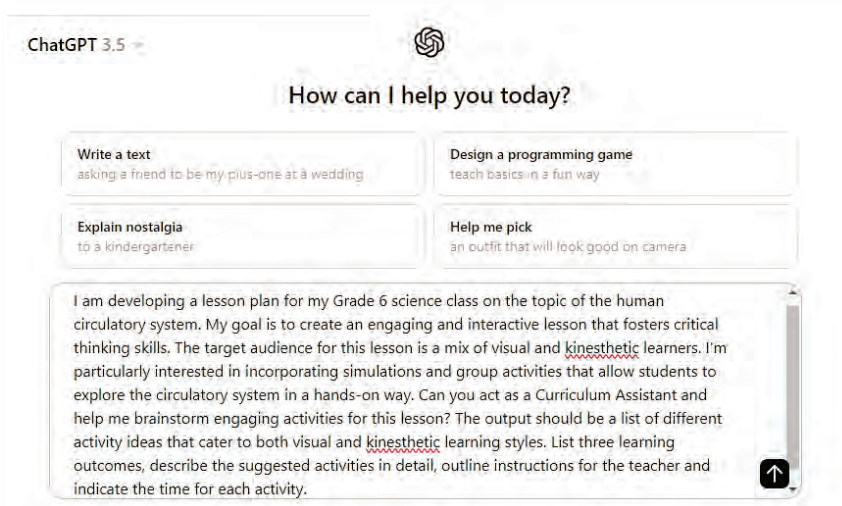


Figure 2.8: ChatGPT screen

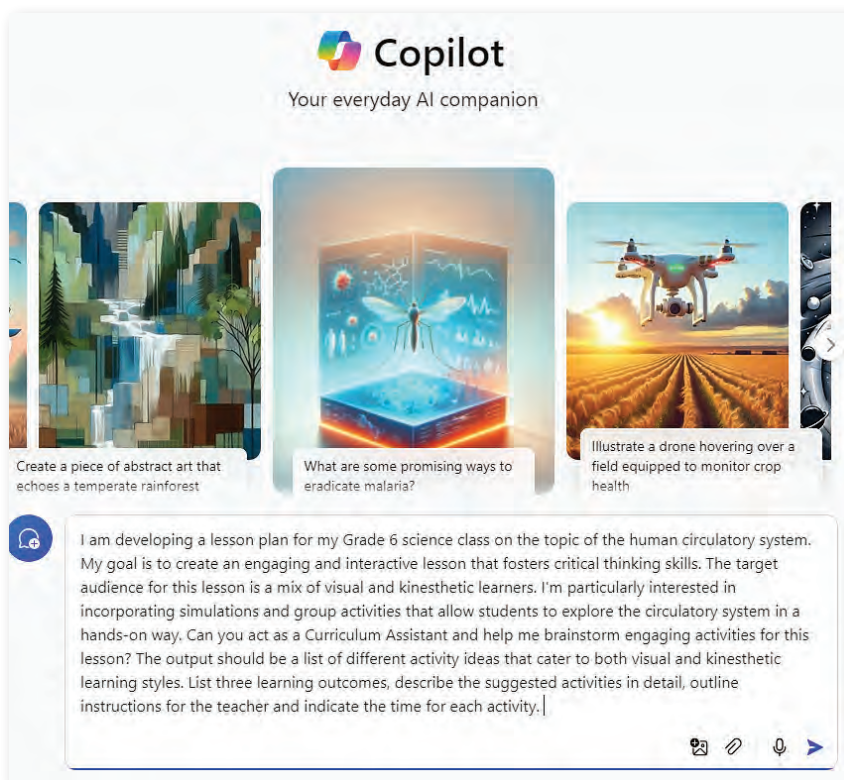


Figure 2.8: Copilot screen

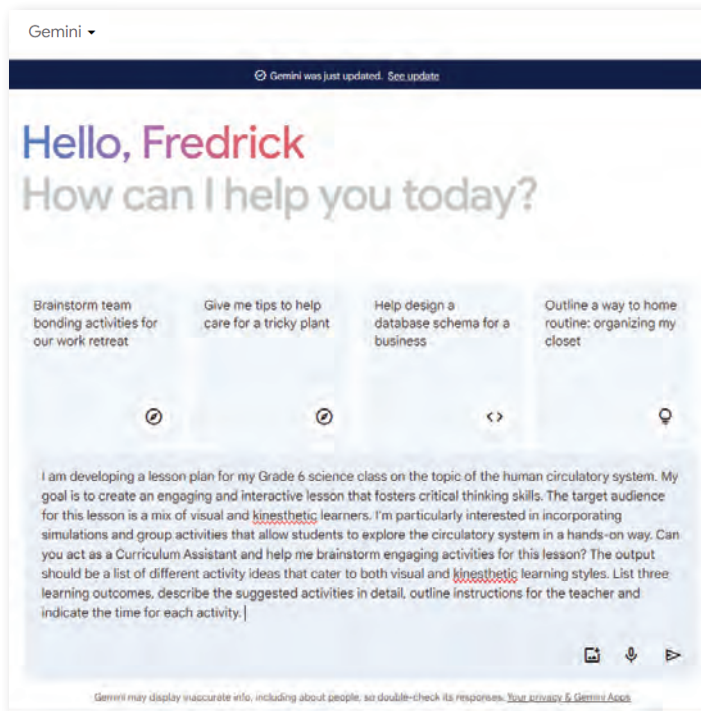


Figure 2.8: Gemini screen

Sample output (Chat GPT)

Learning outcomes:

- 1. Identify parts and functions:** Students will be able to identify and explain the functions of major components of the human circulatory system.
- 2. Understand circulation pathway:** Students will understand the pathway of blood through the heart, lungs and the body.
- 3. Apply knowledge:** Students will apply their understanding of the circulatory system to explain its importance to overall health.

ChatGPT

Learning Outcomes:

- 1. Identify and Describe Components:** Students will be able to identify and describe the main components of the human circulatory system.
- 2. Understand Circulatory Pathway:** Students will understand the pathway of blood through the heart, lungs, and body, including the functions of each part.
- 3. Apply Knowledge:** Students will apply their understanding of the circulatory system to explain how nutrients and oxygen are transported throughout the body and how waste products are removed.

Activity Ideas:

Activity 1: Circulatory System Role-Play (Kinesthetic)

Time: 20-30 minutes

Description: In this activity, students will simulate the journey of blood through the circulatory system using their bodies to represent different parts of the system.

Activity ideas:

Activity 1: Circulatory System Model

Learning Outcome: Identify parts and functions of the human circulatory system

Description:

- **Materials Needed:** Large poster paper, markers, sticky notes, scissors and a tape.
- **Activity:** Divide students into small groups. Each group will create a large-scale model of the human circulatory system on the poster paper using markers and sticky notes. They should label and color-code the heart, arteries, veins and capillaries. Each group will present their model to the class, explaining the function of each part.

Instructions for Teacher:

1. Preparation:

- Create cards or labels for each student that represent different components of the circulatory system (e.g., heart, lungs, arteries, veins, cells).
- Assign each student a role (component) and give them their card/label.

2. Activity Steps:

- **Introduction (5 minutes):** Briefly review the main components of the circulatory system.
- **Role-Play (15-20 minutes):**
 - Ask students to stand in a large circle.
 - Start with a student representing the heart and guide them to pump their arms slowly.
 - Direct the flow of students (representing blood) through the circulatory system by calling out the next component (e.g., lungs for oxygenation).
 - As each component is called, students representing that component should move accordingly (e.g., oxygen exchange in lungs).
 - Discuss the functions of each part as they are represented in the role-play.
- **Discussion (5 minutes):** Facilitate a discussion about the process and functions observed during the role-play.



Instructions for the teacher:

Time: 30 minutes

Instructions:

1. Introduce the activity by briefly reviewing the major components of the circulatory system.
2. Divide students into groups of 4 - 5.
3. Provide each group with the materials and explain the task: create a model that represents the circulatory system.
4. Encourage students to discuss and assign roles within their groups (e.g., artist, labeler, presenter).
5. Allow 20-25 minutes for group work, followed by 5 - 10 minutes for each group to present their model to the class.

Activity 2: Blood Flow Simulation

Learning Outcome: Understand circulation pathways

Description:

- **Materials Needed:** Large open space, cones or markers to designate stations, red and blue yarn or string, small balls (representing blood cells).

- **Activity:** Set up a simulation of blood flow through the heart, lungs and body. Use cones or markers to represent different stations (heart, lungs, body parts). Assign students roles such as heart, lungs, arteries, veins and body cells. Students will pass the “blood cells” (small balls) through the stations, following the pathway of oxygenated and deoxygenated blood.

Activity 2: Circulatory System Model Building (Visual and Kinesthetic)

Time: 40-50 minutes

Description: Students will work in small groups to construct a 3D model of the human circulatory system using simple materials.

Instructions for Teacher:

1. Preparation:

- Prepare materials such as colored clay, straws, cotton balls, and cardboard.
- Have diagrams or pictures of the circulatory system available for reference.

2. Activity Steps:

- **Introduction (5 minutes):** Provide an overview of the circulatory system and its main components.
- **Model Building (30-40 minutes):**
 - Divide students into small groups.
 - Assign each group specific components of the circulatory system to focus on (e.g., heart, blood vessels).



Instructions for the teacher:

Time: 40 minutes

Instructions:

1. Set up the stations in a large open space (e.g., gym or playground).
2. Assign roles to students and explain the flow of blood through each station.
3. Demonstrate the flow with a few students before allowing the whole class to participate.
4. Start the simulation and monitor to ensure students understand the sequence and pathway of blood circulation.
5. Debrief after the activity, discussing any challenges or clarifications needed.

2. Activity Steps:

- **Introduction (5 minutes):** Provide an overview of the simulation and its purpose.
- **Simulation Exploration (20-30 minutes):**
 - Allow students to explore the simulation individually or in pairs.
 - Guide them to follow the pathway of blood, identify different components, and understand their functions.
 - Encourage students to take notes or answer specific questions provided.
- **Discussion (5-10 minutes):** Lead a discussion based on their exploration:
 - Ask students to share their observations and insights.
 - Clarify any misconceptions and reinforce key concepts related to the circulatory system.

Conclusion:

These activities are designed to cater to both visual and kinesthetic learners by providing hands-on experiences, visual representations, and interactive simulations. Each activity fosters critical thinking by encouraging students to explore, create, and apply their knowledge of the human circulatory system in engaging ways. Adjust the time allocated for each activity based on your class dynamics and schedule, ensuring enough time for thorough exploration and discussion.

Activity 3: Health effects debate

Learning Outcome: Apply Knowledge

Description:

- **Materials Needed:** Debate topic cards (e.g., “Is regular exercise crucial for heart health?”, “Does diet impact blood circulation?”), timer, whiteboard or flip chart.
- **Activity:** Divide students into two groups for a debate on health effects related to the circulatory system. Each group will prepare arguments supporting their assigned stance using scientific evidence. After preparation, facilitate a structured debate where students present arguments and counterarguments.

Instructions for the teacher:

Time: 50 minutes

Instructions:

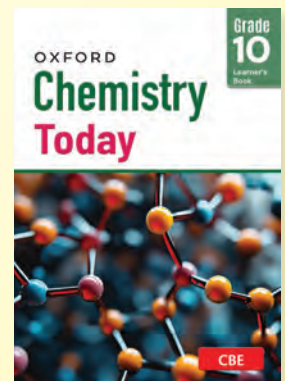
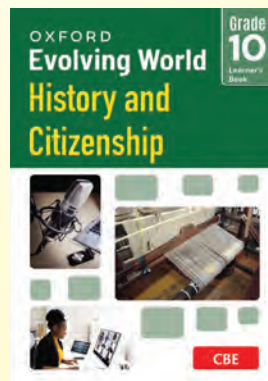
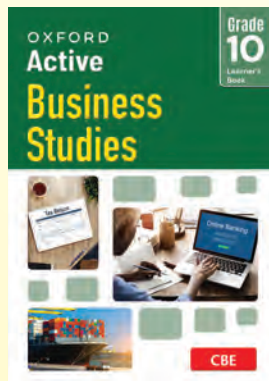
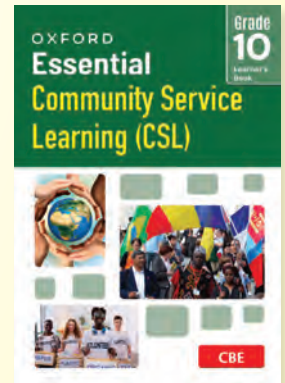
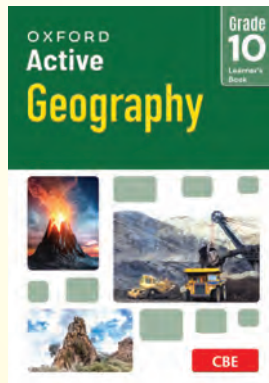
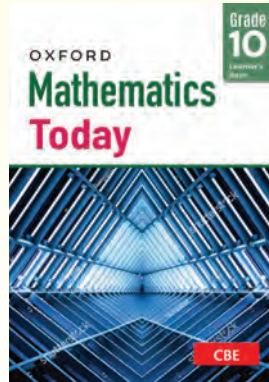
1. Introduce the debate topic and assign groups.
2. Provide time for each group to research and prepare their arguments (20-25 minutes).
3. Outline debate rules (e.g., speaking time limits, rebuttals).
4. Conduct the debate, allowing each group to present their arguments and respond to opposing arguments.
5. Facilitate a class discussion post-debate to summarise key points and encourage critical thinking about the circulatory system’s importance.

Conclusion:

- These activities are designed to engage both visual and kinesthetic learners through hands-on modelling, physical simulations and interactive debates.
- Adjust the complexity and detail of explanations based on the students’ prior knowledge and understanding.
- Ensure to integrate safety guidelines, especially for physical activities like the blood flow simulation.

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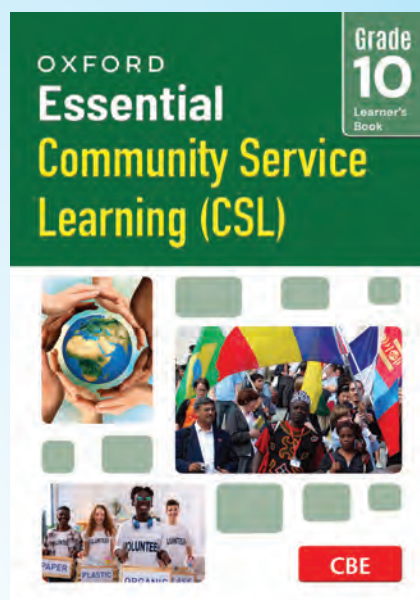
Oxford CBE Senior School course books features



Essential Community Service Learning (CSL)

Grade 10 Community Service Learning (CSL) Learner's Book offers comprehensive curriculum coverage with engaging activities and visually appealing illustrations. Through the activities, it nurtures core values, addresses PCIs, and builds essential life skills that not only lead to personal growth and development but also prepare learners for the 21st century challenges. A dedicated digital component further enhances the learning experience. The Grade Ten Community Service Learning Learner's Book is therefore more than an academic resource. It is a guide to lifelong learning and meaningful community engagement.

The key features of Essential Community Service Learning are explained below:



Introduction

Learners are introduced to each sub strand through engaging activities that spark curiosity on and lay a foundation for the sub-strand. These pre-assessment activities encourage learners to appreciate the sub strand, thereby fostering enthusiasm for the concepts to be covered.

Introduction

You will need a mirror.

- (a) **Self-examination:** Look at yourself in the mirror. If you do not have a mirror, close your eyes and think about who you are. Consider your strengths, emotions, values, weaknesses, physical and spiritual aspects of your life. Who do you see?
(b) Imagine someone else is looking at you in the mirror. Who do they see? Do they see the same person you saw in (a)?
You can draw two mirrors and write down your thoughts. At least two for each.
- Share your thoughts:** Pair up with a classmate and share what you thought about yourself and how you think others see you.
- Give positive feedback:** Tell your partner one positive thing you see in them.
- Self-reflection and discovery:** What did you learn about yourself from the activity?



Wouldn't living life knowing who we are and what we care about be great? How about awareness of how other people perceive or see us? Could this help us improve our interactions and relationships?

Our communities play a key role in shaping who we are. Understanding ourselves affects how we connect with and interact with people in our communities. In this sub strand, we will explore factors that influence our public self-awareness, analyse the importance of positive public self-image and apply public consciousness in our day-to-day life.

Learning Spot

Concise summaries are presented in a variety of ways to reinforce concepts covered in each sub strand.

Learning spot

Action research is cyclical, meaning it is continuous. Each study cycle entails planning, action, observation and reflection phases or stages.

1. Planning phase

- Identify the problem: Clearly define the issue you want to address.
- Set objectives: Determine what you aim to achieve with your research.
- Develop a plan to address the issue: Outline the steps and methods you will use to collect information (data collection), analyse it, interpret and share the results and intervention action(s) determined.

4. Reflect

- Evaluate effectiveness: Assess whether your objectives were met and what worked well.
- Plan the next steps by deciding on any further actions or improvements based on your reflection.
- Action research is continuous, so use your findings to start the cycle again, for continuous improvement.

2. Action phase

- Implement the plan by carrying out the actions you outlined.
- Keep track of your activities and ensure everyone involved in the research knows their roles.
- Be flexible and ready to make adjustments if things do not go as planned.

3. Observation phase

- Monitor your progress regularly to check how the implementation is going.
- Gather information on the outcomes and any changes or impact observed. Use various methods including observation and interviews.
- Analyse the data collected to understand the impact of your actions.
- Keep detailed records of what you observe for analysis.
- Share the results with stakeholders.

Cycle of action research

Summary

It has clear, structured overviews that highlight key takeaways from each sub-strand.

Summary

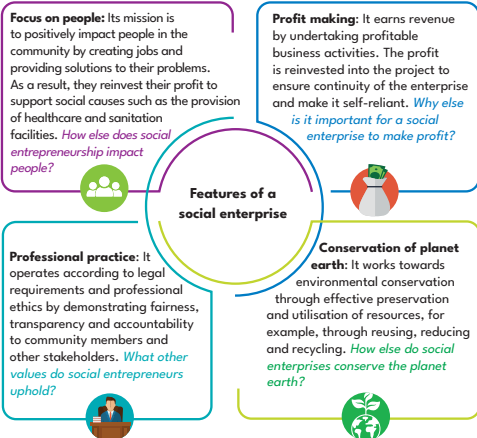
- Effective leadership attributes include integrity, honesty, kindness and responsibility.
- Effective leadership skills include empathy, effective communication, peaceful conflict resolution, decision making, emotional intelligence and critical thinking.
- Leadership styles include transformative, democratic, situational and servant leadership.

Activities

The book comprises well researched and designed tasks and examples that are creative, fun and intellectually stimulating, thereby promoting evidence based learning. Whether working individually, in pairs, in groups or as a class, learners engage in tasks that encourage them to think critically, collaborate and apply their knowledge to develop essential competencies, nurture values and address PCIs.

Activity 2 Class discussion

- Study the information below and answer the questions asked.

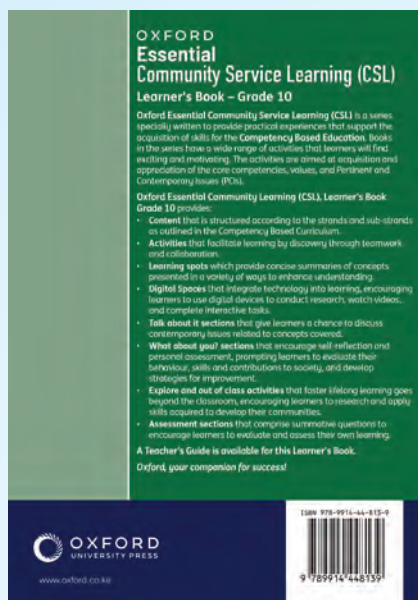


- (a) Identify one social enterprise in your community or any that you know about.
(b) How does it meet the features of a social enterprise?

Differences between social entrepreneurship and other enterprises

Activity 3 Group work

- Reflect on the features of a social enterprise. How is social entrepreneurship different from other enterprises? Explain your points using relevant examples.
- Write down your answers and present them to the class.



Reflection

A guided self-assessment where learners analyse their understanding of key outcomes in the sub-strand and identify areas where they may need clarification or support. This section helps learners take ownership of their learning journey.

Reflection

Are you able to:

- explain the principles of community service learning as a learning strategy?
- outline the rationale of community service learning in the learning process?
- examine your own civic identity as a member of the community?
- examine the purpose of community service learning in promoting responsible citizenry?

Seek help where you have challenges.

Out-of-Class Activities

Lifelong learning extends beyond the classroom. Through school, home and community projects, learners are guided to apply knowledge and skills to real-life situations, thereby making them responsible members of their communities.

Out of class activity

- Prepare a talk to create awareness on the needs of your community. You can also create skits or role plays.
- Give the talk or conduct the skits or role play during school gatherings, such as parents' and learners' meetings.

Digital Space

Integrating technology into learning, this section encourages learners to use digital devices to conduct research, watch videos and complete interactive tasks, making learning more relevant and engaging.



Digital space

Create and maintain a database of community needs, resources and stakeholders for your CSL activities using digital and print resources.

Talk about it

A thought-provoking conversation section where learners discuss pertinent and contemporary issues affecting them (related to the concept). They analyse different perspectives, assess challenges and collaborate on solutions, thereby acquiring essential life skills to navigate life's challenges effectively.

Talk about it

Biases, stereotypes and misconceptions about our cultures affect us. Reflect on the common stereotypes about your culture. How would you respond to them? Share with a peer. You can also ask them to guide you on how to respond.

Stereotype

I am afraid I cannot assign you this leadership role to avoid disappointment. People from your culture are known to be lazy.

Response

Thank you for sharing your concerns. I believe that judging someone's abilities based on their community is unfair and unlawful. I am confident in my skills and work ethic, and I would appreciate the opportunity to demonstrate my leadership capabilities.

Assessment

Summative questions help learners evaluate their understanding, test their ability to apply knowledge and measure their progress.



Assessment

- Explain your understanding of the following terms:
 - Community
 - Community learning
 - Community service
 - Community service learning.
- Identify four key principles of community service learning as a learning strategy.
- Why is reflection important in community service learning? Explain two reasons.
- Why is it important to examine our own civic identity as members of the community?
- How does community service learning contribute to the development of responsible citizenry? Explain three ways.
- Explain three benefits of community service learning to learners and the community.

What about you?

It encourages self-reflection and personal assessment as well as help learners evaluate their behaviour, skills and contributions to society and set goals /develop strategies for improvement.

What about you?

- Consider the entire process of implementing your CSL project. Identify the critical leadership skills needed for successful execution of a CSL project.
- Which skills, attributes or areas of leadership might be challenging for you during the implementation of the CSL project and other group projects in other learning areas? Give reasons for your response.
 - Create a detailed plan to address these challenges. What measures will you put in place to ensure they do not hinder your effectiveness as a leader?
 - Share your plan with your CSL teacher and a learner who is good in leadership to assess it and give constructive feedback.
 - Implement your plan. Monitor your progress after sometime and seek help in areas where you are experiencing challenges.

Explore

This section encourages learning from the community and mentorship as learners engage different stakeholders to deepen their understanding of concepts or develop skills.



Explore

Ask your teacher to guide you to do this activity.

- Identify members of the school or neighbouring community from different cultures.
- Engage them in a dialogue to learn about their culture.
- Find out whether they have any issues (needs or gaps) that affect cohesion and harmony between them and other cultures. List down your findings.
- Discuss how to overcome the issues raised in 3 to promote social cohesion. You may need to consult members of the communities, research on the internet or read print media.
- Write down an action plan and share it with the relevant stakeholders for approval, for example, community members and the school principal.
- Implement your plan, taking note of any improvements and challenges faced.
- Reflect on your activity. How effective was it? What challenges did you face? How can you improve on your action plan? What lessons did you learn?
- Write a report and share with members of your school community.

Remember/Did You Know?

Contains **insights** that enhance deeper understanding and appreciation of concepts.



Remember

It is not our cultural differences that divide us. It is our inability to recognise and respect these differences. Be kind to everyone.



Did you know?

It is a violation of the Kenyan constitution to discriminate against others based on negative stereotypes, biases, misconceptions or hostilities. Such actions can lead to severe consequences, including huge fines or imprisonment.

Mathematics Today

Mathematics Today is a series specially designed to provide practical experiences that support the acquisition of skills for the Competency based Education. Books in this series have diverse activities that are aimed at acquisition and appreciation of core competencies, values and Pertinent and Contemporary issues (PCIs). The series has the following key features:

Numerous **interactive activities** to enable learners to acquire practical skills to solve real-life problems.

Activity 1 Group work

- Write down the ages of learners in your group.
 - Classify each age as either even or odd number.
- Let each learner in your group write down a number.
 - Classify each number you have written as either even or odd number.

Activity 4 Group work

- Make a number spinner like the one shown on the right.
- Spin the spinner once.
 - Determine the probability of the arrow stopping at:
 - 3
 - 4
 - 6
 - Determine the probability of the arrow stopping at:
 - red
 - blue
 - yellow
 - green



Activity 6 Group work

- Determine the reciprocal of 1.56 using a calculator as follows:
 - Press the keys $1 \div 56$ in that order.
 - Press the keys $\times 100$ in that order.
 - What is the reciprocal of 1.56 from the calculator?
- Use a calculator to determine the reciprocal of other different numbers.

Learning point that provides explanations on concepts learnt through the interactive activities.

Learning point

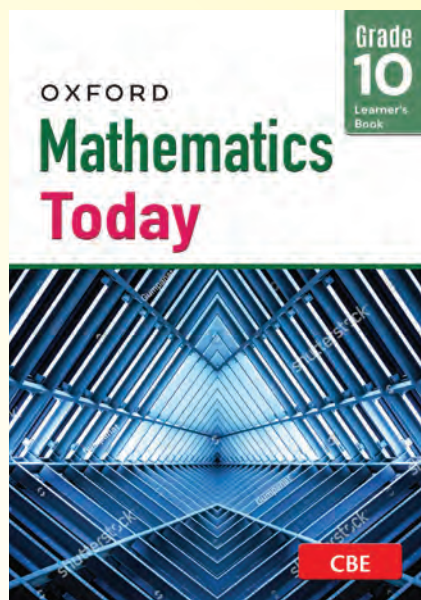
Figures are said to be similar if they have the same shape, irrespective of their sizes. Two figures are similar if:

- the corresponding angles are equal.
- the ratio of the corresponding sides is a constant.

Extended activity section allows learners to engage family in learning. It promotes parental engagement and empowerment in learning.

Extended activity

Using the knowledge of statistics, collect data on the number of days in a week your family eats proteins, carbohydrates and vitamins. Together with your peers, represent the data you have collected. You can also guide your family members on the importance of eating a balanced diet.



Digital activity that allows learners to apply digital technology in learning. It creates awareness in ICT.

Digital activity

- Connect the external storage device to a computer.
- Open the **Statistics and Probability** folder and then the **Probability** folder.
- Work out the questions.

QR code is provided at the end of each sub strand and at the back cover of the books. The code gives access to interactive exercises by scanning the code using a QR code scanner that can be installed on a smart phone.



Digital activity

- Scan the QR code.
Alternatively, search the following link:
<https://www.oxford.co.ke/grade9/mathematics/data%20handling%20and%20Probability>
- Select **Probability** and work out the questions.



Introduction that relates what learners will cover in a given sub strand to real life.

Introduction

To make house plans, to print pictures in photography and to make maps, we can use the knowledge of similarity and enlargement.

Biology Today

The following are the key features of Biology Today Learner's Book.

Key features of the Learner's Book

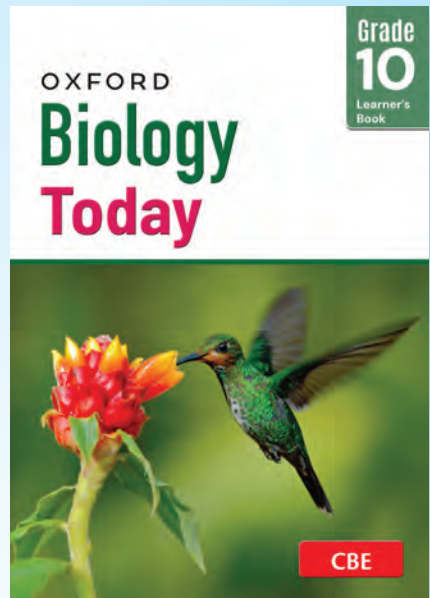
The book:

- has an **entry point activity** at the beginning of each strand to stimulate learner's interest.
- is **well designed** and the **illustrations are in full colour** to make the book learner-friendly.
- is written in a **simple and clear** language making it easy for learners' understanding of the concepts.
- has **summary notes** in form of **key points** aimed at enabling learners to quickly revise and reflect on the sub strand.
- has **self-check questions** at the end of each sub-strand to enable learners evaluate their understanding of the concepts covered.
- has **wrap-up activity questions** to prompt learners to think about real-life issues and find practical solutions to the problems.
- covers all the **core competencies, PCIs, skills and values** as required by the curriculum design.

Key features of the Teacher's Guide

The following are the key features of Biology Today Teacher's Guide. The book:

- comprehensively covers all the learning outcomes as guided by the curriculum design.
- covers all the core competencies, PCIs, skills and values as required by the curriculum design.
- has answers to self-check questions and wrap-up activities to enable teachers to teach more effectively.
- has guidelines on how to manage learners with special needs when conducting various activities.
- has a sample lesson plan, schemes of work and learner's progress record to enable the teacher to prepare effectively during teaching.



Read ahead

These are activities meant to enable learners to research concepts that are new before they are introduced in class. They infuse digital literacy as learners are required to use the internet and other digital resources and develop competencies of self-efficacy and learning to learn. They also take care of non-formal activities.

Read ahead!

Work individually

Using relevant textbooks and digital devices, search for information on the following mechanisms of opening and closing of the stomata:

- (a) The photosynthetic theory (b) The starch-sugar interconversion theory
(c) The potassium ions theory

Write short notes explaining how each mechanism takes place. Discuss your findings during the next lesson.

OXFORD
Biology
Today

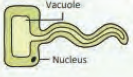
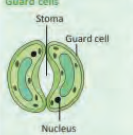
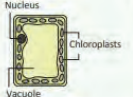
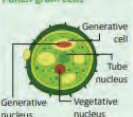


CBE

Notes

There are summary notes appearing after every activity. They are meant to explain the concept further and answer the key inquiry questions. They are also used to conclude the activity/concept.

Table 1.6: Specialised plant cells, their features and functions

Specialised plant cell	Function	Differentiated structures
Root hair cells 	They absorb water and mineral salts.	- They have thin walls for absorption of water and mineral salts. - They are elongated to increase the surface area for absorption of water and mineral salts. - They have large vacuoles that store sugars and salts. This increases the concentration of the cell sap.
Guard cells 	They open and close the stomata to allow gases to enter and leave the leaf.	- They are bean-shaped cells that occur in pairs. - The inner wall is thicker and less elastic than the outer wall. The outer wall therefore stretches more, making the guard cells bulge outwards and open the stomata. - They have chloroplasts that carry out photosynthesis.
Palisade cells 	They carry out photosynthesis.	- They contain numerous chloroplasts that carry out photosynthesis to make food for the plant. - Their cell wall is transparent to allow light to penetrate through to the chloroplasts.
Pollen grain cells 	It carries the male gametes to the ovules for fertilisation.	- It has a generative nucleus which divides to form the male gametes or sperm cells. - It has a vegetative nucleus that germinates and forms the pollen tube, which grows down through the style to reach the ovule and deliver the male gametes.

A closed transport system consists of a transport fluid which is confined within **vessels**, and a pumping mechanism which consists of the heart. The main type of closed transport system in animals is the **blood circulatory system**. Closed transport systems are found in all **vertebrates** such as fish, amphibians, reptiles, birds and mammals.

The closed transport system is further divided into **single circulation** and **double circulation**. In single circulation, blood flows through the heart once in every complete circuit. Organisms with this type of circulation include fish. Figure 3.15 illustrates single circulation.

In double circulation, blood flows through the heart twice in every complete circuit. Amphibians, reptiles, birds and mammals have double circulation. Figure 3.16 illustrates double circulation.

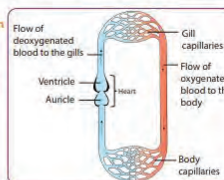


Figure 3.15: Single circulation in fish

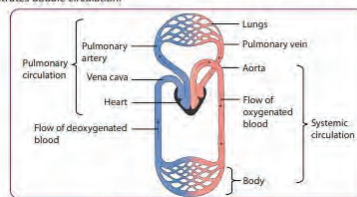


Figure 3.16: Double circulation in a mammal

Substrate concentration

An increase in substrate concentration increases the rate of enzyme activity. This is because more substrate molecules bind with the active sites of the enzyme to form products.

However, the rate of enzyme activity increases up to an optimum level until the enzyme concentration becomes the limiting factor. The graph in Figure 1.47 shows the effect of substrate concentration on the rate of enzyme activity.

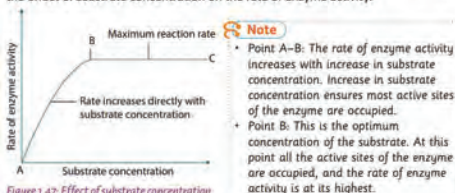


Figure 1.47: Effect of substrate concentration on the rate of enzyme activity

Revision exercise

These are activities that offer summative assessment of concepts. They appear at the end of every sub strand.

Revision Exercise 3.3

- Figure 3.45 shows the tracheal system of an insect. Study it and answer the following questions.
 - Name the parts labelled A, B, C and D.
 - State the function of the liquid found at the end of the structure labelled C.
 - Explain the importance of the spiral rings of chitin on the wall of the part labelled B.
- Explain how a moist respiratory surface helps in gaseous exchange.
- Figure 3.46 represents an organ obtained from a bony fish. Study it and answer the following questions.
 - Name the parts labelled A, B and C.
 - State the function of the parts labelled A.
 - Describe how the structures labelled C are adapted to their function.
- Figure 3.47 shows the structure of the mammalian breathing system. Study it and answer the following questions.
 - Name the structures labelled A, B, C and D.
 - State one adaptation of the part labelled D to its function.
 - Briefly describe how gaseous exchange takes place in the part labelled D.
- Explain how the following processes take place in the human breathing system.
 - Inhalation
 - Exhalation

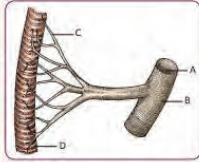


Figure 3.45

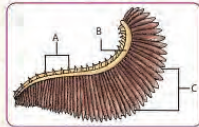


Figure 3.46

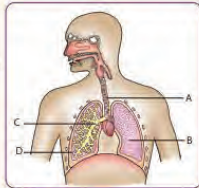


Figure 3.47

Project

This section allows learners to apply what they have learnt in class to solve real-life problems. It also builds higher order skills of synthesising, analysing and creating.

Project Model the structure of animal and plant cells as seen under the electron microscope

- Using suitable sources of information such as textbooks, relevant pictures or the internet, search for information on models of plant and animal cells as seen under the electron microscope.
- Study 3D models and take note of how the various cell organelles and structures have been modelled.
- Discuss and choose suitable models of plant and animal cells you can create. Your models can either be 2D or 3D.
- Draw a sketch of the models, showing the various cell components and structures.
- Discuss the locally available materials you can use to construct your models. Such materials can include cardboard, clay, leaves, bottle tops, small sticks, bottles and papers.
- Assemble the materials and use them to create each component of the cells as outlined in your sketch.
- Place each component on the different sections of the models according to your sketch. Create labels and assign each component or structure accordingly.
- Present your models in class. Explain the functions of the different organelles and structures of your models. You may take photographs and store them in your Biology portfolio.



Figure 1.35: Models of animal and plant cells

Diverse interactive activities

The activities are well researched, relevant, creative and fun. They enable learners acquire key competencies. The activities range from **group discussions, experience sharing, interpretation of pictures** and use of **resource persons** to enable learners learn from the community.

Activity 2 Find out the application of Biology in everyday life

Work in groups

Requirements: Pictures showing people engaged in activities connected to various careers related to Biology, dictionaries, textbooks, computing devices with reliable internet connectivity.

Instructions

- Think of what people in our communities engage in their day-to-day lives. Answer the following questions.
 - Are there some people who are employed in careers that are related to Biology? What do the careers involve?
 - Are there some people that do business? Which businesses apply knowledge and skills gained from the study of Biology?
 - How do farming activities make use of the knowledge and skills gained from the study of Biology?
- Discuss how learning Biology can enable us lead healthy lives and improve living standards in our communities.
- Find out how biological knowledge enables us to take care of our environment.
- Note down your ideas and share them with other groups in class.

Activity 5 Discuss with a resource person the careers opportunities in the fields of study in Biology and the factors that influence career choices

Work as a class

Requirements: A resource person, writing materials.

Instructions

- Visit an institution such as a hospital, pharmacy, agricultural centre, college, environmental office, forest reserve or veterinary centre.
- Identify the various careers of the people working in the institution and find out how the fields of study in Biology relate to their careers.
- Talk to a resource person working in a career related to a field of study in Biology about his or her career and other career opportunities related to the fields of study in Biology.

Note

You can also listen to the resource person through a video call.

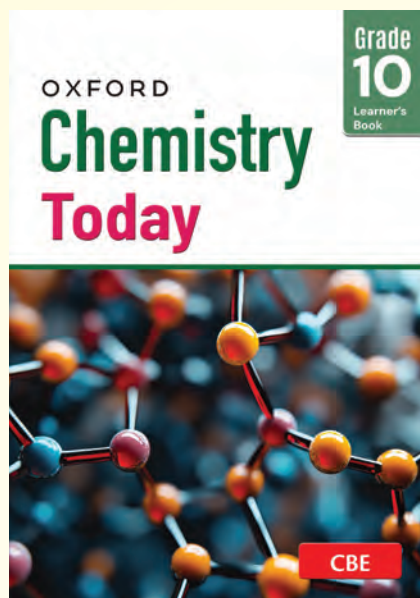
- Discuss with the resource person, the factors that led them to pursue their career. Discuss also the factors that should not influence career choices.
- Share, with the resource person, about a career related to the fields of study in Biology that interests you most. Talk about why that career interests you.
- Note down your findings and discuss them when you get back to class.

Chemistry Today

Key Features of Learner's Book

The following are the key features of Chemistry Today Learner's Book:

- Covers all the strands, sub-strands and learning outcomes as per the curriculum design.
- Develops the seven core competencies, addresses the PCIs and instils the core values.
- Encourages learning by discovery with the teacher or technician acting as a facilitator.
- Language used is simple, clear, scientific and precise.
- Experiments in the book have been practically done and hence, they are all workable. The procedures given are accurate.



Why learn about?

Explains the importance of the sub strands. It shows the application of the knowledge and skills acquired in a sub strand in day-to day life, for example in industries and environmental conservation.

Why learn about the atom?

Atoms can be compared to bricks or blocks in a building. Without the blocks, the building will not stand. Similarly, atoms are the building blocks of all matter. Learning about atoms will help you to understand the changes that occur in matter under different conditions. It will also help you to understand how different substances interact and combine to form new substances and materials that we use in day-to-day life.



Activities

Structured experiences to facilitate acquisition of knowledge and skill development. They include experiments and discussion activities of digital activities. All activities are designed to promote learning by discovery.

What is Chemistry?

Activity 1 To discuss the meaning of Chemistry

Requirements

Print media such as Chemistry textbooks that have information on the meaning of Chemistry, digital devices and internet access.

Instructions

1. Brainstorm the meaning of the word 'Chemistry'.
2. Find the meaning of the word 'Chemistry' from the relevant print media available. You can also search the meaning of the word from the internet.
3. Discuss and agree about the meaning you have found with other members of your group.
4. Write the meaning in your exercise book and present it to the rest of the class.
5. Listen to the presentations from the other groups.

Key points

Short notes at the end of every concept. They help the learners to grasp the main ideas quickly and effectively. They also help them to recall important ideas.

Key Points

Chemistry is the study of composition, structure and properties of matter and the changes that matter undergoes. **Matter** refers to everything around us including air, plants, animals, water, stones and soil.

Evaluation test

Tests at the end of every sub strand. They are designed to test the learner's understanding, skills and knowledge in the sub strand.

Evaluation test 1.1

1. We experience Chemistry in many aspects of our daily life. Explain this statement giving four examples.
2. Chemistry leads to very interesting and life-saving careers. State three such careers and explain what each entails.
3. James went to hospital and was given medicine with the following instructions.

Take 5 ml 4 times a day.

Study it and answer the questions that follow:

- (a) What is the dose?
 - (b) After how many hours should James take the medicine?
 - (c) What is the route of administration of the drug?
 - (d) What is likely to happen if James:
 - (i) takes 5 ml 2 times a day?
 - (ii) takes 10 ml 4 times a day?
4. Drug and substance use has health, social and economic effects. State:
 - (a) three effects of drug and substance use to a Grade 10 learner.
 - (b) three social effects of drug and substance use you have seen in your locality.
 - (c) three health effects of drug and substance use.

Teacher's Guide at a glance

- Provides a summary of core competencies, PCIs and values that are key pillars in CBE.
- Gives a summary of different methods of teaching and learning science.
- Guides the teacher on how to handle learners with special needs.
- Provides guidance on how to conduct both formative and summative assessment.
- Provides sample assessment tools at the end of every sub strand.
- Provides sample professional documents; lesson plan, scheme of work and learner's progress record.
- Provides clear guidelines on how to handle every activity in the Learner's Book.
- Guides the teacher on how to improvise various apparatus using locally available materials.
- Guides the teacher on how to approach Community Service Learning (CSL).

Quick assessment

Assessment that comes after every outcome. It provides instant feedback on whether the outcome has been achieved, enabling the teacher to find out what went well and what needs improvement right away.



Quick assessment 1

1. What do you understand by the term Chemistry?
2. Why is Chemistry organised into branches?
3. Identify three branches of Chemistry and explain what each of the branches entails.

Self-assessment

Section that enables learners to reflect on their understanding and progress. It also helps the teacher identify areas where individual learners need support, hence prepare individualised education programmes to assist them.

Self-assessment

1. Find out if you can do the following:
 - (a) State the meaning of Chemistry as a field of science.
 - (b) Describe the role of Chemistry in day-to-day life.
 - (c) Explain the effects of drug and substance use in day-to-day life.
 - (d) Explain the rights and responsibilities to a safe and healthy learning environment.
2. Consult the teacher in the areas where you have challenges.

Kiswahili Fasaha

Kiswahili Fasaha Gredi ya 10 kina masuala makuu **kumi (10)**. Masuala kuu hayo ndiyo yaliyotumika kama sura.

Vipengele muhimu vya Mtaala wa Kiumilisi:

1. **Umilisi** – Mtaala umebainisha aina saba za umilisi ambazo kama mwalimu unapaswa kuwasidia wanafunzi wako kukuza katika mchakato wa ujifunzaji. Umilisi huu ni **mawasiliano na ushirikiano, kujiamini na kujithamini, ubunifu, uwazaji kina na utatuzi wa matatizo, uraia, ujuzi wa kidijitali na kukuza hamu ya ujifunzaji.**
2. **Maadili** – Hakikisha shughuli unazowapa wanafunzi zinachangia ukuzaji wa maadili kama vile **upendo, ushirikiano, kujiamini, uzalendo, uadilifu na umoja.**
3. **Masuala mtambuko** – Mtaala umependekeza masuala mbalimbali ambayo yanahusiana na maisha yetu ya kila siku. Masuala haya ni kama vile **afya kwa jumla, afya ya akili, sayansi na teknolojia, utunzaji wa mazingira, haki za binadamu, haki za watoto, haki za wanyama na utunzaji wa maliasili.** Tafuta jinsi ambayo unaweza kuwapa wanafunzi wako nafasi ya kutangamana na masuala haya.

Tanbihi: Hupaswi kuwatajia wanafunzi kuwa wanajenga umilisi au maadili yoyote. Panga shughuli utakazowapa wanafunzi kiasi kwamba wanapoitekeleza shughuli fulani, mwishowe watakuwa wamekuza umilisi fulani au maadili fulani. Kwa mfano, unapowapa kazi za vikundi, bila shaka watahitajika kushirikiana ili kuzifanikisha kazi hizo. Katika kufanya hivyo, bila wewe kutaja, watakuza umilisi wa mawasiliano na ushirikiano na pia maadili ya heshima, ushirikiano, upendo na umoja.

Mtaala wa Kiswahili wa Daraja ya Juu una masuala kuu kumi. Kila suala kuu lina mada kuu nne, yaani **Kusikiliza na Kuzungumza, Kusoma, Kuandika na Matumizi ya Lugha.**

Nyaraka za kitaaluma/Vitendea kazi

Mwongozo wa Mwalimu umeonyesha mifano na sampuli za nyaraka za kitaaluma ambazo mwalimu anapaswa kujiundia mwenyewe ili kuweza kufanikisha shughuli nzima ya ujifunzaji. Mifano ya nyaraka hizi ni **rekodi ya maendeleo ya mwanafunzi, rekodi ya kazi ya mwalimu, azimio la kazi na andao la somo/mpangilio wa somo.**



Vipengele muhimu vilivyoangaziwa katika msururu wa vitabu vya Kiswahili Fasaha G10

Kiangaza ni sehemu iliyo na maswali yanayomdadisi mwanafunzi ili aweze kuwa tayari kuhusu dhana atakazokumbana nazo katika mada husika. Waongoze wanafunzi wayajadili maswali haya. Kumbuka si lazima wapate majibu pale mwanzoni lakini ni muhimu kuhakikisha kuwa wanaweza kuyajibu kikamilifu baada ya ujifunzaji wa mada husika. Watagusane na vifaa hivyo katika kufanya shughuli husika.



Kiangaza

Je, ni mambo gani ambayo unazingatia ili kutambua ujumbe katika matini simulizi unayoisikiliza? Jadilini kama darasa.

Tudijitike ni sehemu yenye shughuli zinazompa mwanafunzi nafasi ya kutumia vifaa vya kidijitali ili kujifunza dhana mbalimbali. Ili kukuza umilisi wa ujuzi wa kidijitali, wape wanafunzi nafasi ili watagusane na vifaa hivyo katika kufanya shughuli husika.



Tudijitike

- Mwalimu atawachezea rekodi ya matini. Isikilizeni kwa makini.
- Jadilini vipengele vya fani vifuatavyo kama vinavyojitokeza katika matini mliyoisikiliza katika nambari ya 1.
 - msuko
 - wahusika
 - mandhari
 - matumizi ya lugha

Zingatia ni sehemu yenye taarifa za kina kuhusu dhana zinazofunzwa. Chukua muda kuwaelekeza wanafunzi wasome yaliyo hapa ili wapate uelewa zaidi wa dhana husika.



Zingatia

Kusoma kwa ufasaha ni kusoma matini kwa namna ambayo wasikilizaji wanaweza kupata ujumbe jinsi ulivyokusudiwa na mwandishi. Kusoma kwa ufasaha hujumuisha vipengele vifuatavyo.

- Matamshi bora:** Msomaji ayatamke maneno yote kwa usahihi. Matamshi bora humsadia msikilizaji kupata ujumbe ulivyokusudiwa. Kwa mfano, iwapo msomaji atataka neno 'baba' badala ya 'papa' katika sentensi 'Mvuvi alimvua papa mkubwa baharini', ujumbe ulivyokusudiwa utakuwa umepotoka.
- Kiwango kifaacho cha sauti:** Msomaji asome kwa sauti inayosikika vyema na wasikilizaji. Sauti inayofaa ni ya wastani. Ni lazima sauti inayotumika iwafikie wote wanaolengwa. Sauti ikiwa ya juu sana itawakera wasikilizaji na ikiwa ya chini sana haitasikika na kila mtu anayelengwa.
- Kasi ifaayo:** Msomaji asome kwa kasi ya wastani. Kusoma kwa haraka sana kunaweza kumfanya msikilizaji asipate ujumbe unaokusudiwa. Aidha, kusoma polepole sana kunaweza kumchoshia msikilizaji. Katika kiwango hiki, kasi ya kusoma ifaayo ni ya maneno sahihi yasiyopungua mia moja na ishirini (120) kwa dakika moja.

Shughuli jumuishi inapatikana popote palipo na dhana zaidi ya moja katika mada fulani. Ni zoezi ambalo linashirikisha dhana zote kwa pamoja.

Shughuli ya 1

- Mkiwa wawiliwawili, someaneni sentensi zifuatazo kwa zamu.

A	B
(a) Majirani wale wanapendana kama chanda na pete.	(a) Majirani wale wanapendana sana.
(b) Mzee Kirior ni kobe safarini.	(b) Mzee Kirior hutembea polepole.

- Kati ya sentensi zilizo katika safu ya A na safu ya B, ni zipi ambazo zimetumia lugha yenye mvuto? Elezaneni huku mkitoa sababu za majibu yenu.
- Elezaneni mbinu za lugha zilizotumiwa katika sentensi zilizo katika nambari ya 1.
- Mkiwa katika vikundi, fanyeni mambo yafuatayo.
 - Elezaneni maana ya fani katika matini simulizi. Mnaweza kutafiti katika vitabu vya marejeleo au kwenye mtandao salama.
 - Tajaneni vipengele vya fani mnavyovujia.

Tathmini binafsi ni sehemu inayompa mwanafunzi nafasi ya kujitathmini iwapo amepata uelewa wa yale yaliyokusudiwa na pia anachostahili kufanya iwapo ana tatizo.



Tathmini binafsi

- Kwa makadirio yako, sema iwapo unaweza au bado unahitaji usaidizi kwa kurejelea vigezo vifuatavyo.
 - Ninaweza kueleza maana ya ujumbe na fani katika matini simulizi.
 - Ninaweza kutabiri ujumbe wa matini simulizi kwa kuzingatia anwani au picha.
 - Ninaweza kutambua ujumbe katika matini simulizi niliyoisikiliza.
 - Ninaweza kuchambua vipengele vya fani katika matini simulizi niliyoisikiliza.
- Kama bado kuna mambo usiyoweza kuyafanya kati ya 1(a) – (d), basi shauriana na mwalimu wako ili akuelekeze.

Computer Studies Today

Computer Studies is a series specially designed to provide practical experiences that support the acquisition of skills for the Competency Based Education. Books in this series have diverse activities that are aimed at acquisition and appreciation of core competencies, values and Pertinent and Contemporary issues (PCIs). The series has the following key features:



Key points section that provides a summary of the learning outcome that learners are expected to understand after completing an activity. It highlights the outcome achieved and reinforces the knowledge or skills developed through the activity. This section serves as a quick reference for revision and reflection.

Key Points

Remote storage refers to a type of data storage that allows users to save and access files from a location other than their local device or computer. This storage is typically hosted on external servers, data centers or cloud platforms and accessed via the internet or a network.

There are three types of cloud storage: **public**, **private** and **hybrid cloud**.

(a) Public cloud

This is an open cloud storage system where storage of data and information on servers are available for free up to a certain capacity. More storage space can be acquired by paying extra fees. Public cloud is accessible and shared by many organisations and individuals as shown in Figure 7. Examples of companies that offer public cloud services include Amazon Web Services (AWS), Google Cloud and Microsoft Azure.



Figure 7: Public cloud

Assessment task questions appear at the end of every sub strand and are specifically designed to evaluate whether learners have understood the key concepts and skills covered. These tasks provide both the teacher and the learner with feedback on the learning progress and highlights areas that may need reinforcement.

Assessment Task 1.2

1. Which part of the CPU fetches and decodes instructions?
2. What is the function of ALU?
3. Explain the purpose of the Control Unit (CU) in von Neumann architecture.
4. Describe the stored-program concept and why it's important.
5. What are registers and how do they help the CPU?
6. Why does the von Neumann architecture use a single bus for data and instructions?
7. Draw a simple diagram of von Neumann architecture and label its main components.
8. Identify one advantage and one disadvantage of von Neumann architecture. Explain your answer.
9. Name and briefly describe the three main functional units of a von Neumann architecture.
10. Explain two advantages of the fetch-execute cycle.
11. Using a well-labeled diagram describe the basic fetch-execute cycle within the CPU.

Extended/Further activity section offers additional tasks or exercises that learners can complete to extend, reinforce or deepen their understanding of the concepts introduced in the sub strand.

Further activity

Write simple programs that will accept input and display output in python to calculate the following.

- (a) Volume of a cylinder (b) Volume of a cone (c) Volume of a sphere

Activities in the book are designed to support learning by discovery. A learner-centered approach where learners actively explore, investigate and solve problems to construct their own understanding rather than receiving information passively, learners are guided through hands-on, inquiry-based experiences that allow them to uncover concepts, build skills and make meaningful connections.

Activity 2 Discussing categories of computer storage**Instructions**

1. Use the internet and print materials to search for the definition of computer storage and the categories of computer storage.
2. Discuss the categories of computer storage devices with your classmates.
3. Write down the points in your exercise books.
4. Share your findings with other groups.

Activity 13 Discuss the importance of using input and output devices in different areas**Instructions**

1. Use the internet to find out the input and output devices used in the following areas: home devices, roads, cars, ATMs, point of sale terminals, business, communication and safety systems.
2. Download the pictures from the internet and keep them in your portfolio.
3. Discuss the importance of the input and output devices in each area.
4. Write down and share your findings with other groups.

Reflection section provides learners with a structured opportunity for self-assessment and personal engagement with the content covered at the end of each sub strand. It encourages critical thinking, self-evaluation and deeper understanding by prompting learners to consider what they have learned, how they learned it and how they can apply the knowledge.

 Reflection

For each of the following statements, evaluate your ability and respond with **Yes** or **I have a challenge**. I can:

- a) describe the functional organisation and architecture of a computer.
- b) analyse the relationships among the functional elements of a computer.
- c) create a model of computer architecture that depicts the structural elements of a computer.

New word

Artificial intelligence (AI) is the science of making machines that are capable of performing tasks that usually need human intelligence such as learning, decision making and problem solving.

New words section serves as a vocabulary support tool for learners. It introduces and explains unfamiliar or subject-specific terms that appear in the key point section. This section helps learners deepen their understanding of key concepts.

Take away provides learners with a task they can complete outside the classroom to help them apply or extend what they have learned. It serves as a bridge between classroom learning and real-world application, encouraging learners to reflect and practice independently or with family and peer what they have learned.



Take away activity

Challenge 1: Use relational operators to compare two numbers and print a message indicating which number is greater or less than.

Challenge 2: Create a program using logical operators to determine if a user is eligible for a discount (for example, if they are a member **and** have a coupon).

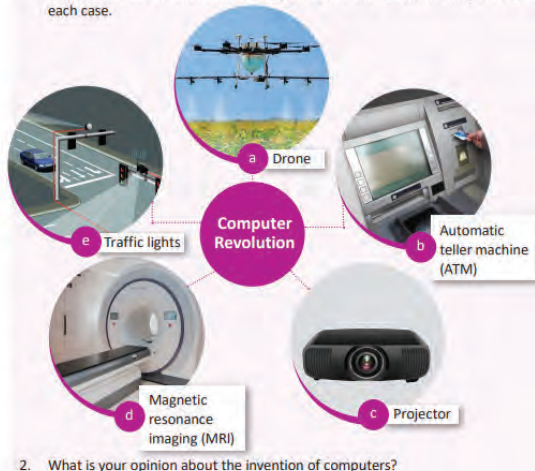
Get started section

appears at the beginning of each strand and serves as an introductory gateway to the strand. It uses a theme picture that give learners a glimpse of what they will explore in the upcoming sub strands to spark curiosity, activate prior knowledge and engage learners in exploring what they will learn in the strand.



Get started

1. The chart below shows some areas where computer technology is applied today. Think about what used to happen before the advent of computers for each case.



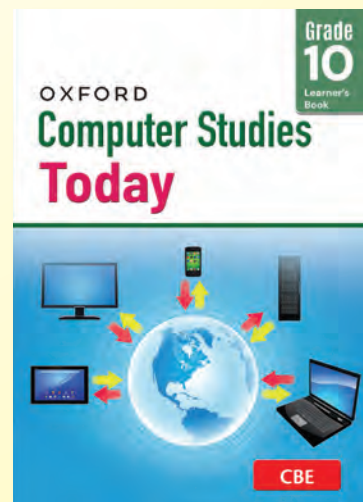
2. What is your opinion about the invention of computers?

Teacher's Guide has:

- Strand overview.
- Sub strand overview.
- Suggested learning resources.
- Preparation for the activity.
- Teaching instructions as per the activities in the LB. The instructions outline how the core competencies, PCIs, values and skills are developed in each activity.
- Answers to the activities in the Learners' Book.

Assessment

- Provides suggestions on competency-based assessment methods at the end of each sub strand.
- Different sample assessment tools are provided.
- Elaborate notes on competency-based assessment methods and tools are provided in the Appendix.



General Science Today

Key features of the Learner's Book

The General Science Learner's Book;

- Covers all the strands, sub strands and learning outcomes as outlined in the curriculum design.
- Develops all the core competencies, addresses PCIs and instills core values in the learners as stipulated by the curriculum design.
- Guides the learners on how to apply the knowledge and skills acquired in every day-to-day life.
- Has theme pictures and key questions at the beginning of each strand to stimulate learner's interest.
- Is written in a simple and clear language making it easy for learners to understand concepts.
- Is well designed with clear illustrations to make it learner friendly.



Organisation of the Learner's Book

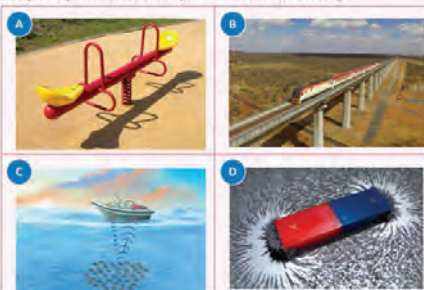
General Science Learner's Book is organised into three strands. Each strand having the following features.

The strand opener comes at the beginning of every strand. It is aimed at stimulating learners' interest in the strand.

3 Natural physical science

Get started

1. The pictures below show application of natural physical science in different fields in day-to-day life. Describe the importance of each area of study.



2. From your analysis, what do you expect to learn in this strand?

Introduction provides information on what will be covered in the strand.

Introduction

Natural physical science is the branch of science that studies non-living systems, natural phenomena and the fundamental laws of nature. It helps to explain how the universe functions by studying force, motion, waves and magnetism. By studying Natural Physical Science, we gain a deep understanding of the natural world and the physical laws that govern it.

Assessment tasks contains questions or tasks that help to evaluate what the learner has learnt in a sub strand. This section is aimed at helping the learner identify their weak areas and look for support.

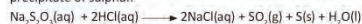
Assessment Task 3.2

1. Distinguish between the following terms as used in linear motion.
 - (a) Distance and displacement
 - (b) Speed and velocity
2. A car moves 100 m east and then 60 m south. Calculate the distance and displacement of the car.
3. A body moving at 10 m/s accelerates at a constant rate of 1.5 m/s^2 for 20 seconds. Calculate the total distance travelled by this body during this time.
4. Discuss the dangers of accelerating on sloping surfaces.
5. What is the significance of linear motion and free fall in day-to-day life?

Key points sections which have brief notes on the concept or learning outcome that is achieved after carrying out an activity. It summarises the concept developed in the activity and provides learners with an opportunity to quickly revise and reflect on the learning outcome.

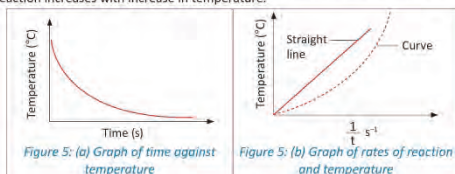
Key Points

The reaction between sodium thiosulphate and hydrochloric acid forms a yellow precipitate of sulphur.



An increase in temperature means that more energy is supplied to the particles which then move at greater speeds thus increasing the rate of reaction.

Figure 5 (a) shows a graph of temperature against time. From the graph, it is evident that at high temperatures, less time is required to produce the sulphur. The graph of temperature against $1/t$ is a straight line as shown in Figure 5(b). This shows the rate of reaction increases with increase in temperature.



Note: Graphs involving $1/t$ are not always straight lines. This depends on the type of reaction.

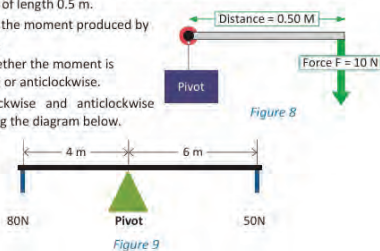
Practice Exercises which are numerous to enable learners acquire mathematical skills to solve problems.

Practice Exercise 1

- The diagram below shows a force of 10 N acting downward at the end of a horizontal bar of length 0.5 m.

- Calculate the moment produced by the force.
- State whether the moment is clockwise or anticlockwise.

- Calculate clockwise and anticlockwise moments using the diagram below.



Discussion questions, Did you discover and **Explanations** sections aimed at helping learners understand what they should have discovered during experiments.

Discussion questions

- Why is it necessary to cut the leafy shoot under water?
- Suggest the purpose of the petroleum jelly.
- Why is the water allowed to run out of the reservoir?
- Account for the results of the experiment.

Did you discover? As the leafy shoot take up water, the air bubble moves along the glass capillary tube. The speed of movement of the air bubble along the capillary tube corresponds to the uptake of water.

When the air bubble moves along distance within a short period of time, it indicates that the rate of water uptake by the leafy shoot is high. This also means that the rate of water loss through transpiration is high and vice versa.

Explanation: When setting up the potometer, the stem of the leafy shoot is cut while still under water to prevent an air bubble from entering the xylem vessels. The air bubble can block the xylem vessels and hence interfere with the uptake of water. A slanting cut is made on the stem to increase the surface area for water uptake.

Petroleum jelly is applied around the hole in the bung to keep the set up airtight. Before reusing the potometer, the reservoir is emptied to reposition the air bubble at the beginning of the scale in the capillary tube.

Further activity that provides additional activities learners will engage in to enhance their learning.

Further activity

During your science club meeting, organise a debate on the factors that affect rates of chemical reactions. Note the points presented.

Warm-up activity that comes at the beginning of each sub strand. It helps to jog the learner's mind on what they already know about the concepts in a sub strand. As the learners answer the questions or carry out the tasks in this activity, they are helped to create a mental picture of what they are expected to learn in the sub strand.

Warm-up activity

1. Which career would you like to pursue in future? Is the career related to the study of General Science?
2. Describe how the knowledge and skills you acquired in Junior School will support your career goals.

Worked examples to enable the learners to follow through as they work out the problems in the practice exercises.

Worked examples

1. A car starts from rest and accelerates at 3 m/s^2 for 6 seconds. Calculate its final velocity.

Solution

Write what you have: initial velocity $u = 0$ as the car is coming from rest, time = 6s, acceleration, $a = 3 \text{ m/s}^2$

Using the equation $v = u + at$

$$v = 0 + (3 \text{ m/s}^2) 6 \text{ s}$$

$$= 18 \text{ m/s}$$

2. A train moves with an initial velocity of 5 m/s and reaches a final velocity of 25 m/s after 10 seconds. Calculate the acceleration of the train?

Solution

Write what you have: $u = 5 \text{ m/s}$, $v = 25 \text{ m/s}$, $t = 10 \text{ s}$

Using the equation $v = u + at$, rearrange the equation to solve for a :

$$a = \frac{v - u}{t}$$

Substituting the known values:

$$a = \frac{25 - 5}{10} = \frac{20}{10} = 2 \text{ m/s}^2$$

Therefore, the acceleration of the train is 2 m/s^2 .

Know more section which provides a platform for interesting facts.

Know more

Auxin hormones in plants cause **apical dominance**. Apical dominance is the inhibition of the growth of lateral buds in the shoot tip. This phenomenon is employed in agriculture during pruning.

Activities are structured in a way that enables learning through discovery and hands-on approach. This approach of learning promotes learner engagement, motivation and independence. It also enables learners to develop their creativity, cognitive, psychomotor and affective skills.

Activity 9 To investigate the effect of pressure on the rate of reaction

Requirements: 50 cm^3 gas syringe with stopper, nitrogen (IV) oxide gas generator, a notebook and a pen.

Instructions: Your teacher will demonstrate this experiment.

Caution! Nitrogen (IV) oxide is poisonous.

Procedure

1. Set up the apparatus as shown in Figure 9.
2. Collect nitrogen (IV) oxide in 50 cm^3 gas syringe with stopper.
3. Allow the gas to attain room temperature. Record the colour change of the gas.
4. Slowly push the piston and note any change in the colour of the gas.

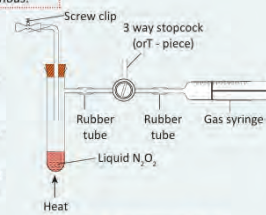


Figure 9: Experimental set-up

Self-evaluation section which provides an opportunity for learners to do self-assessment based on the content covered.

Self-Evaluation

For each of the following statements, evaluate your ability by responding with **Yes** or **I have a challenge**. I can:

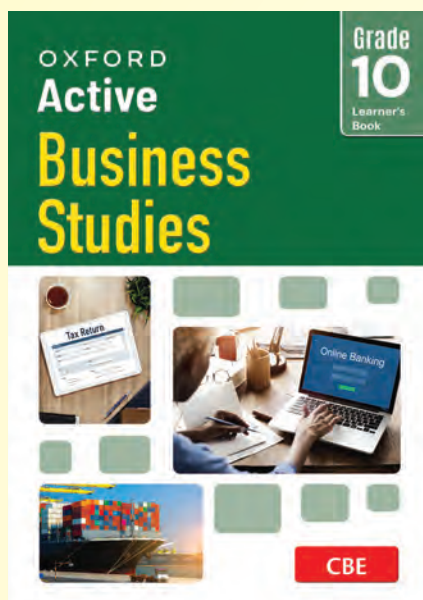
- (a) describe rate of chemical reactions.
- (b) perform experiments involving the rate of chemical reactions.
- (c) explain factors that influence the rate of reactions in biological, chemical and physical processes.
- (d) appreciate the importance of manipulating factors affecting the rate of chemical and biological processes in day-to-day life.

Seek the help of your teacher or classmates in areas where you have challenges. You can also look for more information from relevant textbooks or use digital devices with internet connection.

QR codes are provided in the book and at the back cover of the Learner's Book. It provides access to videos, animations and interactive digital exercises in the book. It enables learners develop their digital literacy skills.

Digital Interactive Exercise 2

1. Scan the QR code.
2. Open the folder named 'Matter and chemical reactions' and do the exercise.



Active Business Studies

Active Business Studies is a series that is designed to help the learner gain relevant experience and practical skills for the business world. The content in the book is strictly in line with the Competency-Based Education. The activities throughout this series are tailored to help the learner acquire and appreciate the core competencies, values and Pertinent and Contemporary Issues (PCIs). The series has the following key features:

Digital space which helps the learner to develop digital literacy competency. It involves either watching of video clips or doing research using digital devices connected to the internet.



Digital space

Discussing and presenting on the sources of public finance

1. Using digital devices, search for information on the sources of public finance.
2. Discuss the sources of public finance, providing full descriptions and examples in each case.
3. Using the digital devices, prepare a table to record your findings. (Alternatively, draw the table in your exercise books.) It should have the source of public finance, its description and examples.
4. Print your table and present it to the class. (Alternatively, present the tables you drew in your exercise books.)

Explore more section whereby learners are given an opportunity to discuss key issues in their environment related to the concepts learnt.



Explore more

1. Find out ways in which county governments are involved in public finance in Kenya.
2. Discuss the levies imposed by county governments in Kenya.

Different kinds of **activities** to enable the learners tackle the outcomes. These activities are presented as either group, pair or individual activities.

Activity 3 Group work

Discussing and presenting on the importance of public finance

The following is part of an article that appeared in a trade magazine. Read it and answer the questions that follow.

Did you know that the taxes you pay are only a small part of public finance? Public finance includes aspects such as revenue, expenditure, grants and aid and loans. Yet, taxes are the most known aspect because all income earners have to pay them.

Without your taxes, it would be impossible for the government to provide public services like education, security and health care. Do you know how expensive it is to provide education? In Kenya, one-third of the annual budget goes to the education sector. So, by contributing to public finance, you help educate millions of Kenyan children.

The government invests part of the funds it receives from various sources in development projects like roads and railways. I am sure you have seen construction workers on a road near you. The government utilises the resources you contribute to public finance to improve infrastructure. Hence, they make the business environment conducive enough for local and foreign investors.

Do you know where county government get their revenue from? The main source of funds for these governors is your taxes. They receive funds from the government equitably to promote balanced regional development, reduce poverty and enhance social cohesion.

Introduction section to each sub strand which gives information on the importance of the content to be covered. This helps the learner appreciate the need to learn the sub strand.

Public finance

Introduction

The government collects its revenue through various means, including taxes and fees for public services. Residents of the country, whether individuals or corporations, contribute to this revenue. The government in turn uses it to provide public goods and services.

Learning spot section which gives a short summary of the concepts learnt in a particular outcome.

Learning spot

Public finance refers to the managements of a governments revenue, debt and expenditure for purposes of development, provision of public services and economic stability. Public finance has components such as **public revenue**, **public expenditure**, **recurrent expenditure**, **development expenditure** and **public debt**.

Sub strand summary section which contains short notes in point form that summarise the content covered in the sub strand.

Sub strand summary

- Public finance is important because it enables the provision of public goods and services, promotes investment, enables the redistribution of resources, among others. The sources of public finance include tax revenue, non-tax revenue, grants and aid and loans (public debt).
- Taxation helps to raise revenue, control consumption, protect local industries, control inflation, reduce inequality among others.
- The principles of taxation include equity, diversity, simplicity, elasticity, certainty, economy, flexibility, convenience and non-oppression.
- The requirements for tax compliance in Kenya include registering with KRA, filing tax returns, keeping accurate records, payment of taxes, compliance with tax laws and regulations, among others.
- In Kenya, items exempted from taxation include equipment and machinery. The people exempted from taxation include persons with disability, diplomats and staff of foreign missions. Organisations exempted from tax include PBOs, Central Bank and cooperative societies.
- The types of customs duties in Kenya include import duty and export duty.
- The trends in taxation in Kenya include introduction of digital tax, expansion of the tax base, use of technology in tax administration, increase in excise duty, introduction of environmental taxes, among others.
- Ethical issues in taxation include tax evasion, tax avoidance, inequitable tax system, corrupt tax administrators, misallocation of tax revenue, manipulation of tax policy, double taxation, among others.

Self-assessment section which is a reflection meant to assist the learner assess themselves on what they have covered in the sub strand based on the outcomes.

Self assessment

- For each of the following statements, respond with **Yes** or **I have a challenge**.
 - I can explain the importance of public finance in Kenya.
 - I can assess the concept of taxation in Kenya.
 - I can analyse the types of customs duties in Kenya.
 - I can evaluate the trends in taxation in Kenya.
 - I can identify ethical issues in taxation.
 - I can write an article on the importance of taxation in Kenya to sensitise the community.
 - I can appreciate the role of public finance in Kenya.
- If you have a challenge with any of the above seek help from your teacher or classmates.

Out-of-class activity section which consists of school, home or community activities where learners research on something and apply the knowledge learnt to solve a problem or identify gaps and opportunities.

Out-of-class activity

School activity

- Discuss and formulate messages to sensitize your community on legal and ethical issues in taxation.
- Using manila paper, felt pens and other relevant materials, design posters with the messages you prepared in 1 above.
- Display your posters in prominent places within the school or the community to reach as many people as possible.

Assessment section which contains high order and low-order questions which test the learners' grasping of concepts taught in the sub strand.

Assessment

- KRA officials held outreach meetings to sensitise the residents of Masingo on newly-introduced taxes and levies. They explained the importance of public finance to the community and the country.
 - Explain the meaning of public finance.
 - Explain five ways in which public finance is important.
 - Describe four sources of public finance.
- Public finance plays a very important role in Kenya, enabling the government to provide and operate public goods and services. Describe the three components of public finance.
- The government of Kenya raises its revenues mainly through taxation. As such, individuals and businesses pay different types of taxes to the government.
 - Explain five principles of taxation.
 - Discuss the importance of taxation.
- One day on the 9.00 pm news bulletin, there was a story of a businessman who had been arrested for not paying taxes. KRA officials had raided his business premises, discovered untaxed goods and confiscated them.
 - Discuss five requirements for tax compliance in Kenya.
 - Discuss two ethical issues in taxation that affect the businessman above.
- Every year, the government makes changes to the tax system by altering tax rates or introducing new taxes or levies.
 - Describe two types of custom duties.
 - Discuss five trends in taxation in Kenya.
- Not everyone in Kenya pays taxes. Some individuals and organisations are exempted from taxation.
 - List the individuals exempted from taxation in Kenya.
 - Name the organisations exempted from taxation in Kenya.

Quick facts section which gives important information related to the concepts learnt for general knowledge.

Evolving World History and Citizenship

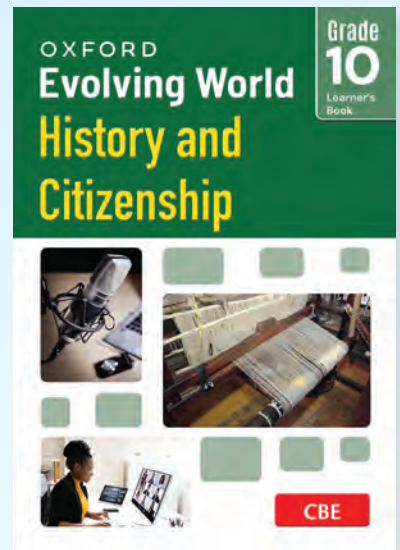
Learner's Book at a glance

The learners book has:

1. All the learning outcomes in the curriculum design covered.
2. Well-developed concepts.
3. Creatively presented activities.
4. Relevant and colourful illustrations.
5. Accompanying digital component.

Key features of the Learner's Book

The History and Citizenship Learner's Book has the following key features:

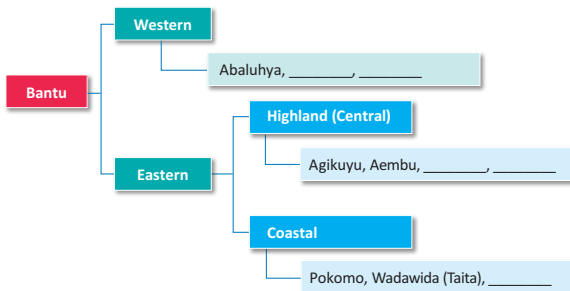


Activities

They are well researched, relevant, creative, fun, engaging and intended to bring out the various competencies, values and PCIs. They can be done individually, in pairs, in groups or as a whole class.

Activity 3 Group work

Grade 10 learners at Bute Senior School have created the following chart on the Bantu linguistic group in Kenya. Study the chart and do the tasks that follow.



1. Draw the chart above in your exercise books. Complete the chart by adding other communities that belong to each linguistic group. Pick the names of the communities from the bar below.

Abagusii, Ameru, Mijikenda, Akamba, Abakuria, Wataveta, Ambeere, Waswahili.

Remember

This is a platform for raising awareness about contemporary issues.

Remember

Any person, including you or I, can fall victim to human trafficking. Stay vigilant! Report suspected cases to the authorities.

Learning spot

These are summaries that reinforce the learner's understanding of the concept being learnt. They mostly appear after an activity.

Learning spot

The establishment of colonial rule in Kenya offers valuable lessons that help us build a fairer and more harmonious society. It teaches us the importance of inclusive leadership, where everyone's voice is heard and respected. The harsh treatment of Kenyans during colonial times underscores the need to protect human rights and ensure that everyone is treated with dignity. The unfair economic practices of the colonial era remind us that economic policies should benefit all members of society, not just a select few.

The colonial strategy of dividing communities to maintain control shows us the power of unity and collaboration in achieving common goals. By learning from history, we can avoid repeating past mistakes and work towards a more just and equal future.

Assessment

These are summative questions come at the end of each sub strand. They test the learner's acquisition and retention of knowledge.

Assessment

1. Explain three factors that may have influenced early humans to change from migratory hunting and gathering to domestication of crops and animals.
2. (a) Explain the meaning of the phrase, 'Neolithic Revolution.'
(b) Highlight three advancements that took place during the Neolithic Revolution.
3. Describe three contributions of the Neolithic Revolution to modern society.
4. State two differences between the pastoralism activities of the Maasai and Fulani communities.

Summary

This section provides key points on the concepts learnt. They appear at the end of every sub-strand.

Summary

- For centuries, European foreigners interacted with the Africans through trade, exploration and missionary work.
- The scramble for and partition of Africa happened between 1884 and 1910. This is when European nations invaded and colonised the African continent.
- Economic, political, social and strategic reasons influenced the scramble for and partition of Africa.
- Some of the key players who determined the colonisation of Africa were missionaries, traders, explorers, European leaders, African rulers and European chartered companies.
- The European nations that participated in the scramble for and partition of Africa included Germany, France, Britain, Spain, Portugal and Belgium.

Did you know?

This section contains insights that enhance a deeper understanding and appreciation of concepts to be learnt.



Did you know?

The people of Kenya hold all sovereign power, which can only be exercised according to the Constitution. They can exercise this power either directly or through their elected representatives (Article 1 of the Constitution of Kenya, 2010).

Key words

This is a section where the definition of new words or phrases that are critical to the sub strand/concept is done.



Key word

De facto – by fact and not by law.

Explore

This is a section where the learners get to explore contemporary issues related to the concepts in the sub strand. It gives provocative but safe topics for the learners to engage and give opinions.



Explore

The first phase of Industrial Revolution led to the underdevelopment of Africa. Do you agree with this statement? Justify your answer.

Out of class activities

This section covers **school, home or community activities** where the learners may research on something and apply knowledge learnt to solve a problem.



Out of class activity

Community activity

1. Come up with messages or slogans that promote peaceful co-existence in society.
2. Develop posters showing the messages or slogans that you have come up with. See the sample poster on the side.
3. Display your posters in public spaces such as markets or community halls. You may also ask your parents, guardians or older relatives to post the messages on their social media platforms.

Different roots, one tree:
Let us grow together in peace!



Digital space

This is a section where the learners get to interact with technology. They are guided to use digital devices such as computers, tablets, cameras, radio and TV to perform the various tasks.



Digital space

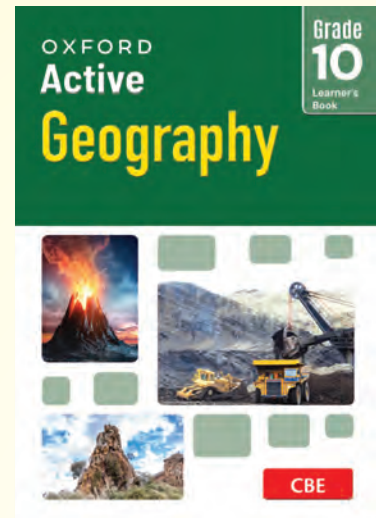
Work in groups

1. Watch a documentary on the establishment of colonial rule in Kenya. You may also read relevant print materials such as history books, magazines, journals and newspapers. Answer the following questions:
 - (a) Which European country colonised Kenya?
 - (b) When did the colonisation of Kenya begin?
 - (c) What were the reasons for the establishment of colonial rule in Kenya?
2. Prepare a summary of responses to the questions and share it in class.

Active Geography

Active Geography has:

1. All learning outcomes in the curriculum design are covered.
2. Concepts that are well developed to ensure a deep understanding.
3. Activities are presented in a creative and engaging manner.
4. Relevant and colourful illustrations to enhance learning.
5. Accompanying digital components are included to support interactive learning.
6. Concise notes that are tailored to deepen the understanding of Geography concepts.



Key features of the Learner's Book

The key features of the Learner's Book are explained below:

Summary

This section provides key points on the concepts learnt. Summaries appear at the end of every sub-strand.

Summary

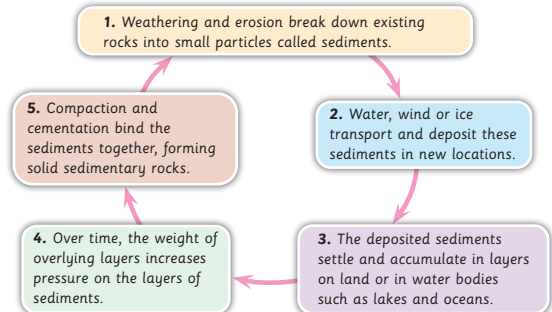
- A **rock** is naturally occurring solid substance composed of mineral particles. It forms the solid part of the earth's crust.
- The three main types of rocks according to their mode of formation are: Igneous rocks, sedimentary rocks and metamorphic rocks
- **Igneous rocks** are crystalline, have joints or cracks, lack fossils and are either acidic or basic.
- **Sedimentary rocks** have layers, contain fossils and are non-crystalline.
- **Metamorphic rocks** are often brightly coloured, resistant to weathering and may have layers.
- **Igneous rocks** are found in the Kenyan Highlands, the Rift Valley and around the Lake Victoria Basin.
- **Sedimentary rocks** are common along the Coast and in North-Eastern Kenya.

Activities

The activities are well researched, relevant, creative, fun and engaging. They are intended to bring out the various core competencies, values and PCIs.

Activity 4 Group work

1. The diagram below illustrates the stages involved in the formation of some sedimentary rocks. Study it and complete the tasks that follow.



A flow diagram showing the stages of sedimentary rocks formation

Discussion spot

This section prompts the learners to explore pertinent and emerging issues related to the respective sub-strands. It also encourages them to apply the skills and knowledge they have acquired.



Discussion spot

Debate this motion: **Kenya's economy would be more developed if the country's rock resources were fully utilised.**

Follow the instructions below.

1. Form two groups: Group A will argue in favour of the motion. Group B will present opposing arguments.
2. Debate and discuss: Each group will present its arguments and respond to counterarguments. Take notes on key points from both sides.
3. Summarise the lessons learned from the debate. Your summary should include arguments from both groups. State your personal opinion on the issue, with a supporting reason.



Case studies and maps

Case studies and thematic maps are aligned to specific topics in the Geography curriculum. They help the learners to relate theoretical concepts to real-life experiences.

3 Human and economic activities

Agriculture

Agriculture is the backbone of the society and one of the most common human activities worldwide. It involves growing crops and rearing animals to provide food, clothing and other essential products.

Types of agriculture in the world

Activity 1 Pair work

The pictures below show types of agriculture. Study them and do the tasks that follow.



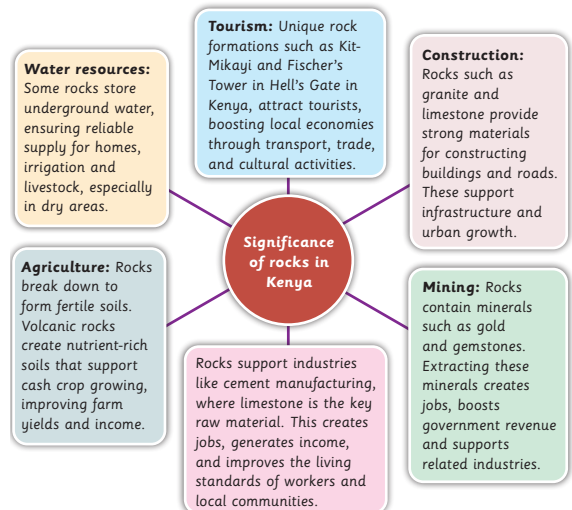
- Describe each type of agriculture based on:
 - The size of land used.
 - The level of technology needed.
 - The main purpose of the farming activity.
- Identify the type of agriculture shown in each picture.
- List other types of agriculture that you know.
- Share and discuss your answers with the class.

Learning point

- These notes reinforce the learner's knowledge on the various concepts covered in the book.
- They appear after an activity and are creatively presented across the book.

Learning point

The chart below gives a summary of the significance of rocks in Kenya.

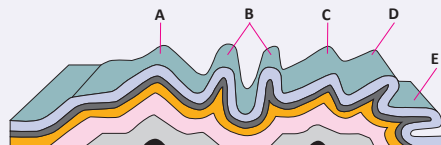


Assessment

These are summative assessments included in the book to test what the learners have understood throughout a sub-strand. They appear at the end of every sub strand.

Assessment

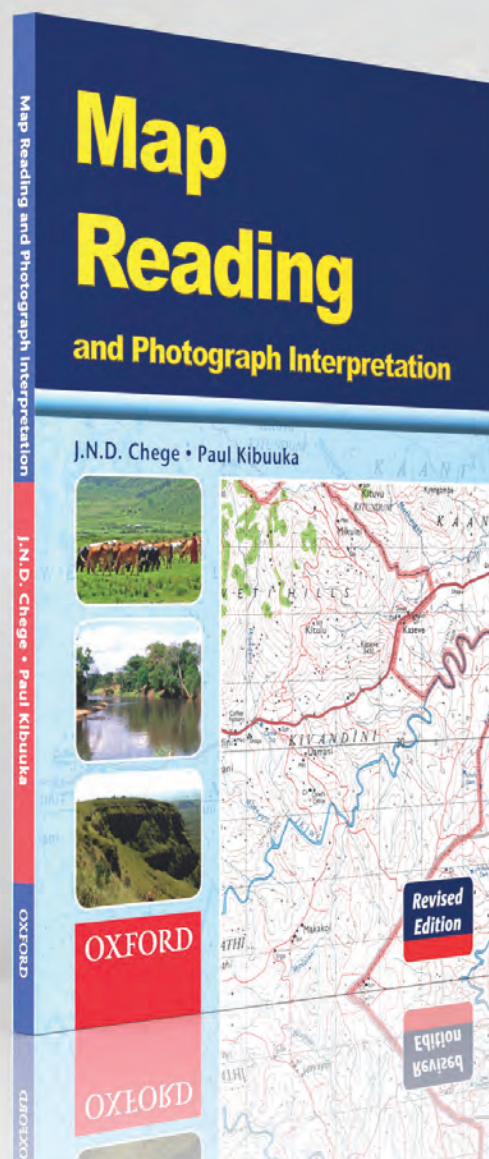
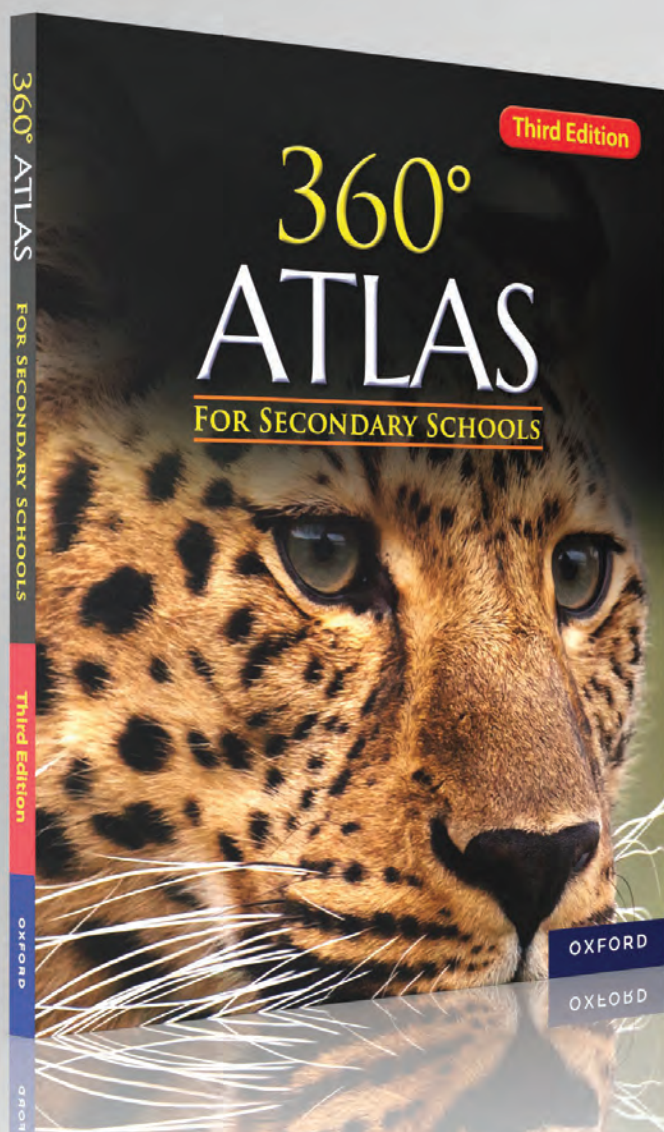
- Identify the types of folds labelled A to E.



- Explain the difference between a symmetrical fold and an asymmetrical fold.
- Describe how overthrust folds form.
- Explain the significance of folding and the resultant features in:
 - Economic importance
 - Climate
- Explain two ways in which fold mountains influence human settlement and population distribution.

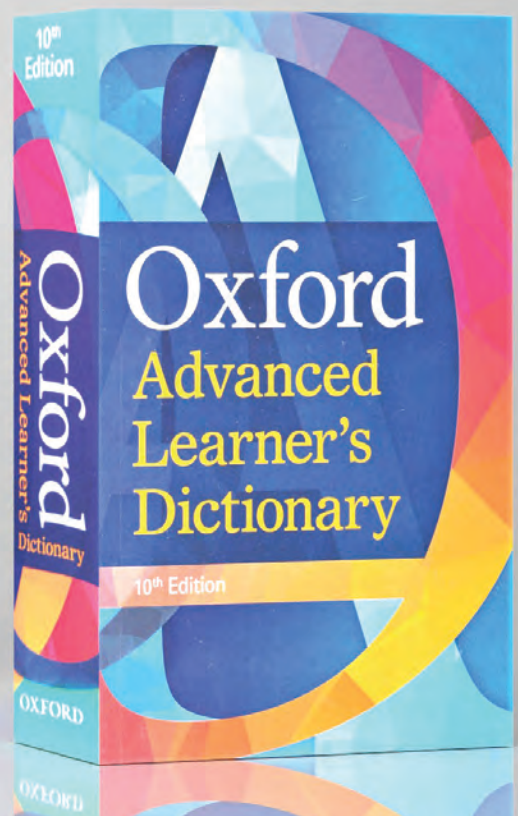
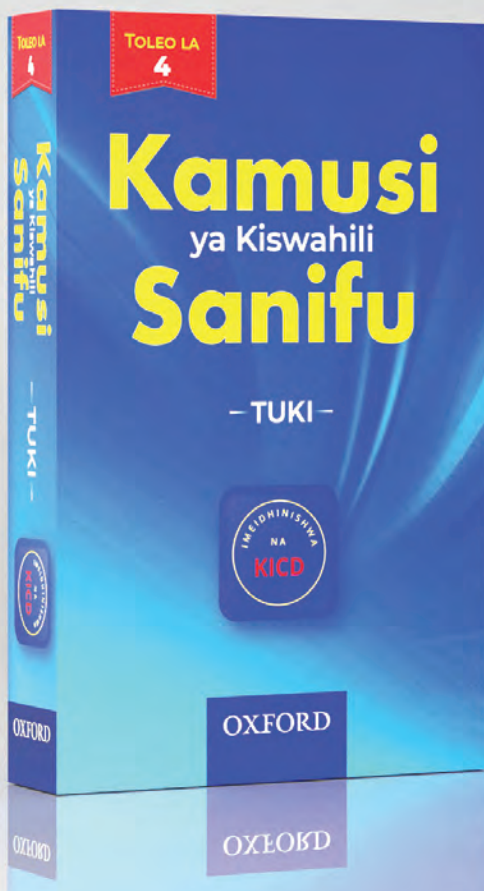
6

Must-haves and how to use them in CBE



Available in bookshops countrywide!

Senior School Must-Haves



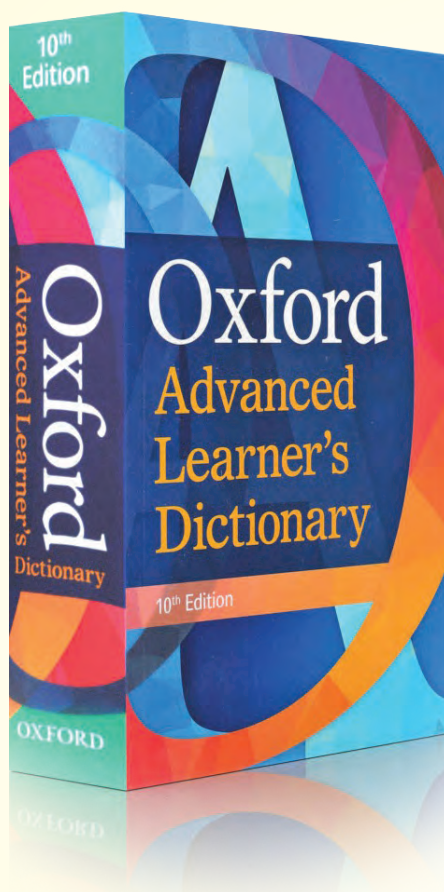
Oxford Advanced Learner's Dictionary

During English lessons in Senior School, learners will build their language skills; that is listening, speaking, reading and writing. They will also expand their vocabulary and their knowledge of grammar. The use of the dictionary in class is an integral part of this. The curriculum designs specify the use of the dictionary to achieve learning outcomes such as using digital and print dictionaries to check the meanings of words, confirming the meaning of words and phrases as inferred from the context, articulating sounds correctly in words, pronouncing words correctly, distinguishing classes of words on the basis of stress, spelling and pronouncing synonyms and antonyms correctly, searching for meaning and examples of phrasal verbs and confirming the meaning of idioms.

The *Oxford Advanced Learner's Dictionary*, the world's bestselling advanced-level dictionary for learners of English, during the lessons where outcomes such as those above are addressed.

For the learners, we recommend the print dictionary which is available in all major bookstores countrywide at a pocket friendly price. Teach the learners how to use the dictionary and guide them to understand how to find the information they need in the dictionary navigating to the right word, the right part of speech and the right section of the entry. The keywords of the Oxford 3 000 and Oxford 5 000 – clearly marked with key symbols in the entries – also help the learners understand which words they should spend time learning, because they are likely to encounter them again.

For the teacher, we recommend the online dictionary and app which are quick reference tools that you can use in class



from your mobile phone or laptop. If the learners have digital devices, they can also use the online dictionary or app.

Go to <https://www.oxfordlearnersdictionaries.com/definition/english/> for the online dictionary or download the Oxford Advanced Learner's App from Google Play or App Store.

Read more about the Oxford Advanced Learner's Dictionary from the brochure at <https://oxelt.gl/3f4TGSc>

Synonyms notes:

Also 190 synonyms comparing and contrasting groups of synonyms.

SYNONYMS

intelligent

smart • clever • brilliant • bright

These words all describe people who are good at learning, understanding and thinking about things, and the actions that show this ability.

intelligent good at learning, understanding and thinking in a logical way about things; showing this ability: *He's a highly intelligent man.* ◊ *She asked a lot of intelligent questions.*

smart (especially NAmE) quick at learning and understanding things; showing the ability to make good business or personal decisions: *She's smarter than her brother.* ◊ *That was a smart career move.*

clever (sometimes disapproving, especially BrE) quick at learning and understanding things; showing this ability: *How clever of you to work it out!* ◊ *He's too clever by half, if you ask me.* **NOTE** People use **clever** in the phrase: *Clever boy/girl!* to tell a young child that they have learnt or done sth well. When used to or about an adult **clever** can be disapproving.

brilliant extremely intelligent or showing a lot of skill: *He's a brilliant young scientist.*

bright intelligent; quick to learn: *She's probably the brightest student in the class.* **NOTE** **Bright** is used especially to talk about young people. Common collocations of **bright** include *girl, boy, kid, student, pupil.*

PATTERNS

- clever / brilliant at sth
- a(n) intelligent / smart / clever / brilliant / bright child / boy / girl / man / woman
- a(n) intelligent / smart / clever / brilliant thing to do

ob-liga-tory /ə'bligətri/; NAmE -tɔ:ri/ *adj.* 1 ~ (for sb) (to do sth) (formal) that you must do because of the law, rules, etc. **SYN** compulsory: ~ for sb to do sth *It is obligatory for all employees to wear protective clothing.* **OPP** optional 2 (often humorous) that you do because you always do it, or other people in the same situation always do it: *In the mid-60s he took the almost obligatory trip to India.*

antonym

synonym

Synonyms and antonyms: are clearly signalled throughout the dictionary using SYN and OPP signs – over 6 000 synonyms and over 2 000 antonyms.

limb ˈlɪm/ *noun* 1 ˈlɪm/ an arm or a leg; a similar part of an animal, such as a wing: *an artificial limb* ◊ *For a while, she lost the use of her limbs.* 2 -limbed (in adjectives) having the type of limbs mentioned: *long-limbed* ◊ *loose-limbed* 3 a large branch of a tree ◊ **VISUAL VOCAB** page V12

IDM out on a limb (informal) not supported by other people: *Are you prepared to go out on a limb (= risk doing sth that other people are not prepared to do) and make your suspicions public?* **tear/rip sb limb from limb** (often humorous) to attack sb very violently ◊ more at **RISK** v.

VOCABULARY BUILDING

Objects you can use

It is useful to know some general words to help you describe objects, especially if you do not know the name of a particular object.

- A **device** is something that has been designed to do a particular job: *There is a new device for cars that warns drivers of traffic jams ahead.*
- A **gadget** is a small object that does something useful, but is not really necessary: *His kitchen is full of gadgets he never uses.*
- An **instrument** is used especially for careful or scientific work: *'What do you call the instrument that measures temperature?'* 'A thermometer.'
- A **tool** is something that you use for making and repairing things: *'Have you got one of those tools for turning screws?'* 'Do you mean a screwdriver?'
- A **machine** has moving parts and is used for a particular job. It usually stands on its own: *'What's a blender?'* 'It's an electric machine for mixing soft food or liquid.'
- An **appliance** is a large machine that you use in the house, such as a washing machine.
- **Equipment** means all the things you need for a particular activity: *climbing equipment.*
- **Apparatus** means all the tools, machines or equipment that you need for something: *firefighters wearing breathing apparatus.*

Vocabulary building

Idioms – are in a clearly marked section at the end of the entry, introduced by an IDM symbol. In the online dictionary, type in the whole idiom and it takes you there. Grammar reference page R13 explains how idioms are presented in the dictionary.

Example sentences – over 100 000 of them in print, over 230 000 online – show learners how to use words in context.

Pronunciation: The IPA transcriptions of headwords. IPA transcriptions are supported by the audio pronunciations online and in the app.

Stress for the **noun**

Stress for the **verb**

Phrasal verbs are in clearly marked sections introduced by a PHRV sign at the end of verb entries. In the online dictionary each phrasal verb is a separate entry and can be found instantly by typing in verb and particle. Grammar reference pages R8-9 in the back of the dictionary clearly explain what phrasal verbs are and how the meanings and grammar work.

pro-hib-ition-ist /prəʊ'biʃənɪst/ *noun* a person who supports the act of making sth illegal, especially the sale of alcoholic drinks

pro-hib-i-tive /prə'hɪbətɪv/ *NAmE also* prəʊ'h-/ *adj.* **1** (of a price or a cost) so high that it prevents people from buying sth or doing sth **SYN** **exorbitant**: *prohibitive costs* ◇ *The price of property in the city is prohibitive.* **2** preventing people from doing sth by law: *prohibitive legislation* **3** (*NAmE*) (of a person taking part in an election or a competition) extremely likely to win: *Miami began the day a prohibitive Super Bowl favorite.* ► **pro-hib-i-tive-ly** *adv.*: *Car insurance can be prohibitively expensive for young drivers.*

pro-ject  **A1**  *noun, verb*

■ **noun** /'prɒdʒekt/; *NAmE* 'prɑ:dʒ-/

• **SCHOOL/COLLEGE WORK** **1**  **A1** a piece of work involving careful study of a subject over a period of time, done by school or college students: *a history project* ◇ *~on sth* *My class is doing a project on medieval towns.* ◇ *The final term will be devoted to project work.*

• **PLANNED WORK** **2**  **B1** a planned piece of work that is designed to find information about sth, to produce sth new, or to improve sth: *We worked on various projects together.* ◇ *to fund/finance a project* ◇ *to undertake/complete a project* ◇ *a building/construction project* ◇ *~to do sth* *They've set up a research project to investigate the harmful effects of air pollution.* ◇ *a project manager/team*

• **SET OF AIMS/ACTIVITIES** **3** a set of aims, ideas or activities that sb is interested in or wants to bring to people's attention: *The party attempted to assemble its aims into a focused political project.*

• **HOUSING** **4** (*NAmE*) = **HOUSING PROJECT**: *Going into the projects alone can be dangerous.*

■ **verb** /prə'dʒekt/

• **ESTIMATE** **1**  **B2** [T, usually passive] to estimate what the size, cost or amount of sth will be in the future based on what is happening now **SYN** **forecast**: *be projected* *A growth rate of 4 per cent is projected for next year.* ◇ *be projected at sth* *The overall cost is projected at \$11 billion.* ◇ *be projected to do sth* *Unemployment is projected to rise to over 5 per cent next year.* ◇ *it is projected that ...* *It is projected that the unemployment rate will fall.*

• **PLAN** **2**  **B2** [T, usually passive] to plan an activity, a project etc. for a time in the future: *be projected* *The next edition of the book is projected for publication in March.* ◇ *The projected housing development will go ahead next year.*

• **LIGHT/IMAGE** **3**  **B2** [T] *~sth (on/onto sth)* to make light, an image, etc. fall onto a flat surface or screen: *They projected the digital image onto the model's surface.*

• **STICK OUT** **4** [I] + *adv.* / *prep.* to stick out beyond an edge or a surface **SYN** **protrude**: *a building with balconies projecting out over the street*

• **PRESENT YOURSELF** **5** [T] to present sb/sth/yourself to other people in a particular way, especially one that gives a good impression: *~sth* *They sought advice on how to project a more positive image of their company.* ◇ *~sb/sth/yourself (as sb/sth)* *He projected himself as a man worth listening to.*

• **SEND/THROW UP OR AWAY** **6** [T] *~sth/sb (+ adv./prep.)* to send or throw sth up or away from yourself: *Actors must learn to project their voices.* ◇ (*figurative*) *the powerful men who would project him into the White House*

PHRV **pro-ject sth onto sb** (*psychology*) to imagine that other people have the same feelings, problems, etc. as you, especially when this is not true

Kamusi Ya Kiswahili Sanifu 4

Wanafunzi wanahitajika kutafuta **maana za maneno**, **nahau** na **misemo** katika kamusi. Pia, wanapaswa kutamka maneno ipasavyo, **kueleza maana za methali**, **kuchambua vipera vya fasihi**, **kujifunza ngeli na upatanisho wa kisarufi pamoja na mnyambuliko wa vitenzi**. **Kamusi ya Kiswahili Sanifu Toleo la 4** itawasaidia sana kupata taarifa hizi.

Aina ya neno: nomino

Kidhizo **mtu** /'mtu/ **nm** (wa-) [a-/wa-] kiumbe agh. anayetembea wima ambaye hutumia fikira zaidi kuliko silika kufanya mambo yake **VSW** mwanadamu, mahuluku, binadamu.

Visawe

Misemo

Kiambishi cha wingi wa nomino: watu

Ngeli ya nomino

Maana

Uelekezi wa kitenzi: kitenzi elekezi

Aina ya neno: kitenzi

Matamshi ya neno

Mnyambuliko wa kitenzi

Mfano wa matumizi

Nahau

Msemo

Methali



funik.a /fu'nika/ **kt** [ele] (~ia, ~iana, ~ika, ~iwa, ~iza, ~wa) weka kitu juu ya kingine ili kilichoko chini yake kisionekane **VSW** fichi, fuchama, sitiri, fidika: ~ chungu.

NH ~ maneno eleza maneno nusu kwa kufichaficha.

MS ~ gubigubi funika kabisa bila kuacha sehemu hata ndogo.

MT ~ kombe mwanaharamu apite (1) si vizuri kuzifichua siri zako kwa watu usiowaamini. (2) kubali yaishe. taz. kivuli.

360° Atlas

The 360° Atlas has unique features which makes it the atlas of choice. The features are explained below.

Fact Sheet

These give vital statistics for counties and other areas of study.

Counties fact sheet

County Headquarters: City Hall

Area: 695 km²

Population*: 3 138 369

Population density*: 4 515 people per km²

Major trading centres: The Central Business District (CBD) popularly referred to as 'Town', Eastleigh, Dandora, Muthurwa, Mutindwa, Adams Arcade, Westlands, City Park Market, Nairobi West, Kangemi, Dagoretti, Kawangware, Gikomba, Kariakor, Burma, Ngara, Githurai

Main economic activities: Information and communication technology, food and beverage processing, glass manufacturing, textile making, steel processing, leather processing, tourism, finance

Main attractions: Nairobi National Park, Nairobi National Museum, Bomas of Kenya, Karen Blixen Museum, Giraffe Centre

* 2009 National Census

Photographs

Relevant quality photographs that aid the learning of concepts.



↑ Cashew nuts are grown in Kwale, Kilifi, Tana River and Lamu counties. The raw nuts are exported mainly to India.

Digital accompaniment

The atlases have selected digital content to aid in the understanding of Social Studies concepts.

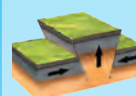
Block mountains

Due to compression



The earth's crust may experience compressional forces.

This leads to the formation of faults.



The compressional forces, combined with pressure from below, cause the middle block to rise over the side blocks. This forms an overhanging block mountain.

The edges are smoothed over time by erosion.



Scan the QR code to watch a video clip on the formation of block mountains.

Notes

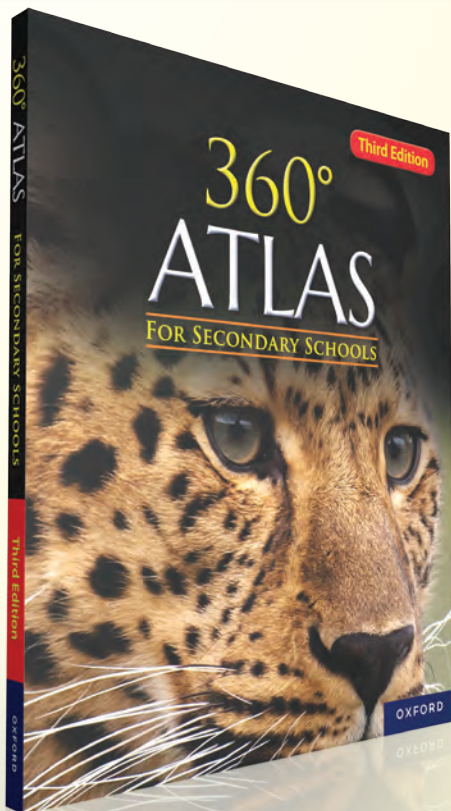
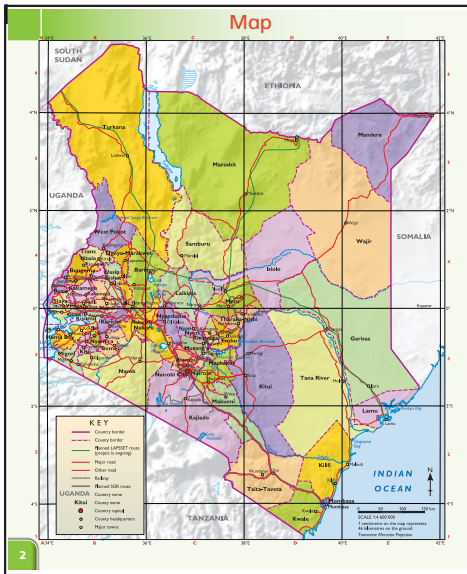
Brief notes that aid the understanding of Social Studies concepts. The notes also address topical concerns such as climate change and desertification.

Climatic zones of Eastern Africa and their characteristics

Climatic zone	Characteristics
Highland	- Temperatures are modified by altitude - Lower slopes are warmer than higher slopes - Rainfall varies with altitude, as well as position of the slope on the mountainside - Windward slopes receive more rainfall than leeward slopes
Tropical wet	- Average annual temperature of about 28° C - Heavy rainfall ranging between 1 100 and 1 600 mm experienced throughout the year

Maps

Many rich, varied and updated thematic maps aligned to specific topics in the Social Studies curriculum design. They have comprehensive keys that make it easy to relate features and draw linkages between various regions of study.



Worksheets

Have numerous test item to aid revision of Social Studies topics while interacting with the content in the atlas.

Part A Map reading

Worksheet 1

- Fill in the table below correctly. One has been done for you.

	Scale	Type of scale
(a)	1 cm represents 10 km	Statement scale
(b)	0 10 20 30 km	
(c)	1: 25 000	
(d)	1 cm to 100 km	
(e)	$\frac{1}{10\,000}$	

- Write if the sentence is **true** or **false**.
 - A small-scale map represents a big area with few details.
 - A large-scale map represents a small area of land. It shows a lot of information.
- Study the topographical map of Nkubu (Sheet 122/1) on **page 8** and its key on **page 9**. Do the following tasks.
 - State the distance, in kilometres, of the road from the southern end near Erimbene up to the junction near the Sub-Chief's Camp.
 - Calculate the area under forest cover in the western area of the map. State your answer in square kilometres.
 - What is the bearing of Kithangari School from Gaatia School?
 - Give the four-and six-figure grid references for the church at Igandene.

Self-check section

Numerous activities that aid the study of Social Studies concepts while interacting with the content in the atlases.

Self-check

- Using the points of the compass, give the direction of Nairobi from the following places:
 - Kajiado
 - Bura
 - Kapenguria
 - Voi.
- Using latitudes and longitudes, give the location of the following places:
 - Cairo
 - Accra
 - Johannesburg.

Map Reading and Photograph Interpretation

The map reading and photograph interpretation text book from Oxford offers an in-depth exploration of the various concepts in the School curriculum. The text book has:

1. Notes relevant to the outcomes of Senior School Geography as well as History and Citizenship Curriculum Designs,
2. 16 full-colour topographical maps,
3. A mix of coloured and black and white photographs,
4. Assessment activities questions relevant to Senior School,
5. Large and clear maps,
6. Accompanying digital component.

Key features of the resource book

The key features in the Map Reading and Photograph Interpretation text book are explained below:

Notes: Well researched, relevant to the concept and accompanied by diagrams and sketches to enable the learners understand the concepts. The notes are also written in simple language.

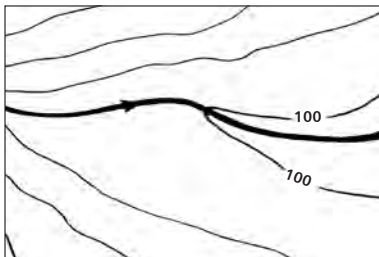


Fig. 6.9 (b) Mature valley



Fig. 6.9 (c) Old-age valley (flood plain)

Spurs

A spur is a prominent projection of raised land from higher ground, such as a hill or mountainside, into lower land.

Hills

A hill is an upland that rises above the general relatively low ground but is of less height compared to a mountain.

Their shape varies from regular to irregular.

Regular hills look evenly shaped and tend to be conical shaped. On topographical maps, they are depicted by concentric contours that give the hill a rounded shape.

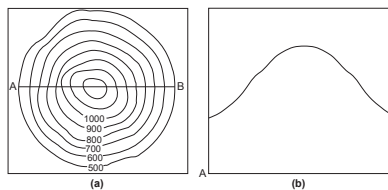


Fig. 6.11 A regular-shaped (conical) hill

Some hills are irregularly shaped due to erosion, the shape of the rock outcrop or due to other geomorphological processes.

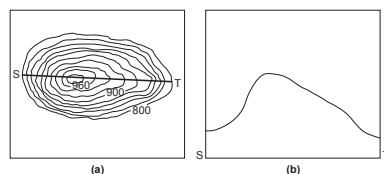
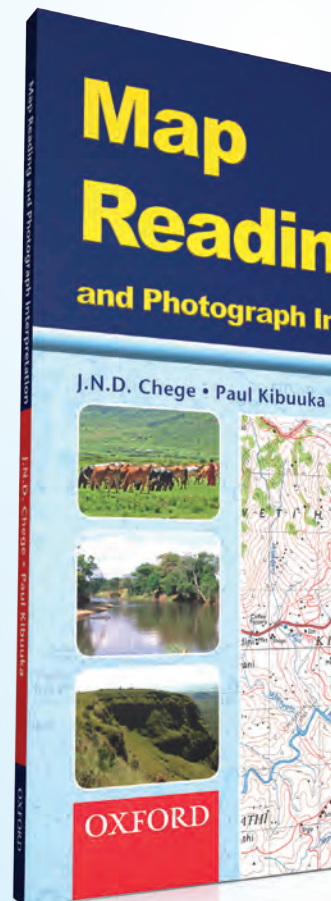


Fig. 6.12 An irregularly shaped hill



Practise questions which serve as summative assessment and are in two sets: one set for map reading and another set for photograph interpretation. Each set of questions appears at the end of its respective strand.

Photograph A

Using Photograph A alongside Map No. 1 of Nairobi, answer the following questions.

- What type of photograph is this one?
 - Draw a frame, the same size as that of the photograph. On the frame, mark and name:
 - Nairobi Dam
 - A railway line
 - The major roads
 - Kenyatta and Nairobi Dam estates
 - Kibera settlement
 - Wilson Airport.
- Compare the settlements in Kenyatta estate with those of Kibera area.

Photograph B

With reference to the photograph and Map No. 2 of Naivasha:

- Explain the distribution of settlements in the area shown on the photograph.
- Explain the relationship between relief and communication lines.
- Describe the distribution of vegetation in the area covered by the photograph.
- Give reasons why the area to the left hand side on the photograph is suitable for farming.

Photograph C

- Using photograph C and with reference to Map 3 of Machakos, draw a rectangle the same size as that of the photograph. On the rectangle, mark and name:

- the main all-weather roads
- three dry-weather roads

- Kyandani hill
- a sports track
- a hospital
- The D.C.'s office.

Photograph D

Photograph D shows part of the area shown on Map No. 4 of Nakuru.

- What type of photograph is this?
- Draw a rectangle the same size as the photograph. Using both the photograph and the map, mark on the rectangle and name the following features:
 - Lake Nakuru
 - A forest
 - A river
 - A railway line
 - Bondeni, Race Track and Freehold estates
 - The show ground
 - Menengai Forest.
- Describe the siting of Nakuru town.
- Photograph D was taken in January 1969 while Map No. 4 is 1997 edition. Identify on the map changes that have taken place on the actual ground since 1969.

Photograph F

Photograph F shows part of Mombasa island and the west mainland. Using this photograph and with reference to Map No. 5 of Mombasa, answer the following questions.

- Draw a rectangle to represent the photograph. On the rectangle, mark and name the following

Formative exercises that appear after every concept and give the learners a chance to gauge their understanding of the concepts.

Exercise 6

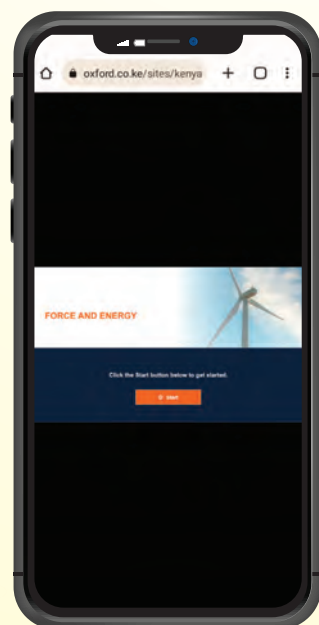
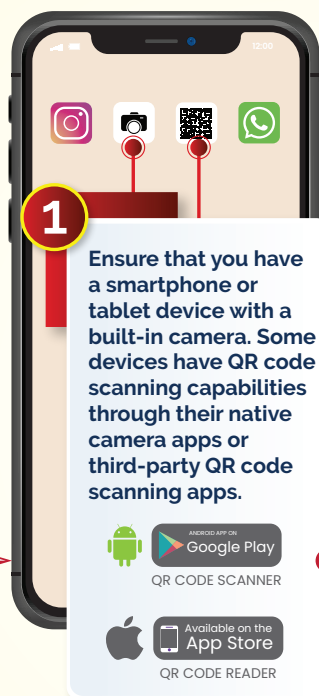
Study Map No. 10 of LUMBWA and answer the following questions.

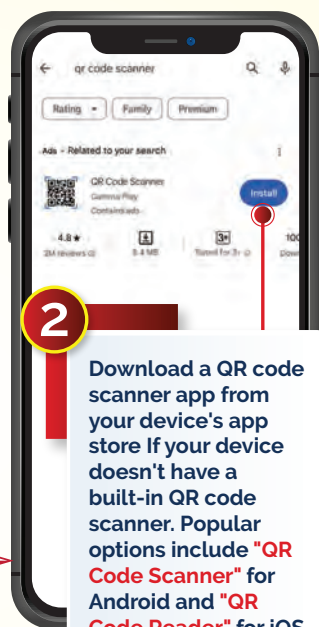
- Identify and state the methods used to represent relief on this map.
 - Describe the relief of the area shown by the map extract.
- Name three types of vegetation shown on the map.
 - Describe the distribution of vegetation in the area represented by the map extract.
- Describe the drainage of the area shown in the map extract.

Exercise 1

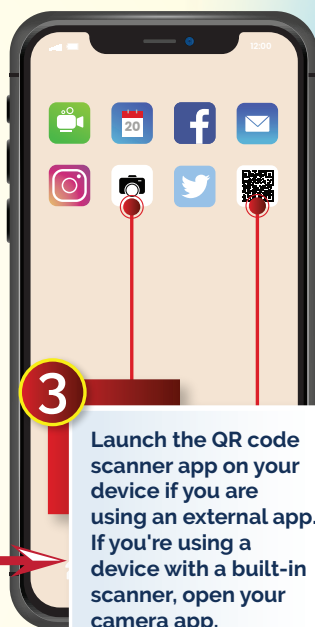
- Categorize all the photographs appearing in this book.
- Study Photograph 1.2. Describe the vegetation shown.
- Explain the uses of aerial photographs.

A step-by-step guide on how to efficiently use QR codes for seamless access to digital content.





2 Download a QR code scanner app from your device's app store. If your device doesn't have a built-in QR code scanner, popular options include "QR Code Scanner" for Android and "QR Code Reader" for iOS.

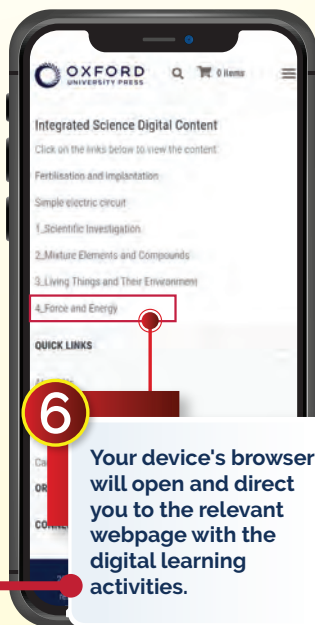


3 Launch the QR code scanner app on your device if you are using an external app. If you're using a device with a built-in scanner, open your camera app.



4 Hold your device's camera over the QR code so that it fits within the viewfinder. Make sure the QR code is in focus and not obscured.

The scanner app or built-in camera app will automatically detect the QR code.



6 Your device's browser will open and direct you to the relevant webpage with the digital learning activities.



5 Once the QR code is successfully scanned, your device will typically display a notification or prompt you to visit a link.

General Queries

	Email
Orders	Orders.KE@oup.com
Customer Service & General Enquiries	CustomerService.KE@oup.com
Marketing Enquiries	Marketing.KE@oup.com
Publishing Enquiries	Publishing.KE@oup.com
Compliance	Enquiries.KE@oup.com

Sales Queries

Contact your sales regional representative on the following numbers:

Eastern Block

Name	Region	Contact Details
Susan Mutonga Regional Sales Coordinator	Eastern Block (Coast/Eastern/Nairobi/Central)	0722 664 175
Wilson Mwenda Educational Business Advisor	Nairobi (Nairobi, Kajiado)	0720 987 899
Michael Mburu Educational Business Advisor	Nairobi (Nairobi, Kiambu)	0721 837 151
Simon Mwangi Educational Business Advisor	Central (Murang'a, Nyeri, Kirinyaga, Laikipia)	0743 566 102
Titus Kathuli Educational Business Advisor	Lower Eastern (Makueni, Machakos, Kitui)	0726 896 178
James Maorwe Educational Business Advisor	Upper Eastern (Embu, Isiolo, Tharaka-Nithi, Meru, Garissa)	0728 091 854
Romanus Ngare Educational Business Advisor	Coast (Mombasa, Kilifi, Kwale, Taita Taveta)	0710 858 605

Western Block

Name	Region	Contact Details
Paul Achungo Regional Sales Coordinator	Western Block (Rift Valley, Nyanza, Western, Upper Central)	0743 566 096
Vincent Okumu Educational Business Advisor	North Nyanza (Kisumu, Siaya, Homa Bay, Migori)	0743 794 071
Simon Magabe Educational Business Advisor	South Nyanza (Kisii, Nyamira, Bomet, Narok)	0743 566 101
Peter Miugo Educational Business Advisor	Central Rift (Nakuru, Kericho, Baringo, Nyandarua)	0743 566 097
Stephen Bor Educational Business Advisor	Western (Kakamega, Bungoma, Busia, Vihiga, Trans Nzoia, Uasin Gishu, Nandi, West Pokot, Elgeyo Marakwet)	0728 547 572

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